

Previous AMO Question Answer

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AMO previous QUIZ -II

1 Question 1

Consider a square well potential, such that the potential $V(x) = 0$ between $x = 0$ to $x = L$ and infinite everywhere else. The well is occupied with two identical particles (with spin $1/2$, and mass m), such that they have only contact interaction ($e\delta(x_1 - x_2)$, where x_1 and x_2 are the co-ordinates of each particle), behaving like hard-spheres and nothing else.

- (a) Firstly, ignoring the contact interaction, write down the energies of the ground state, first and second excited of the system (Pay attention to the spin parts of the wavefunction). [10 marks]
- (b) Sketch the spatial parts of the resulting wavefunction as contours in a two-dimensional plane of the two particles (you may use any standard plotting software). [10 marks]
- (c) Now, using the ground state wavefunction, calculate the correction (perturbation) in the energy due to the contact interaction between the electrons. What should be the order of e for the perturbation to be small? (You may ignore the spin part of the wavefunction.) [10 marks]
- (d) Calculate the corrections to the first excited state (Pay very close attention to any possible degeneracy. If you find any, decide wisely on the appropriate techniques to calculate the perturbations. You may ignore the spin part of the wavefunction). Interpret the result physically. [20 marks]
- (e) What would you expect for the total wavefunction, including the appropriate spin states? Label the states with appropriate quantum numbers and write down the total spins for each. [10 marks]
- (f) Sketch the spatial parts of the resulting wavefunctions as contours in a two-dimensional plane of the two particles (you may use any standard plotting software). [10 marks]

Solution:

Step 1: Single-Particle Solutions

For a single particle in an infinite square well from $x = 0$ to $x = L$, the spatial wavefunctions and energies are:

$$\psi_n(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right), \quad E_n = \frac{n^2 \pi^2 \hbar^2}{2mL^2}$$

where n is a positive integer.

Step 2: Two-Particle System Without Interaction

For two non-interacting particles, the total energy is the sum of individual energies:

$$E_{n_1, n_2} = E_{n_1} + E_{n_2} = \frac{\pi^2 \hbar^2}{2mL^2} (n_1^2 + n_2^2)$$

Step 3: Constructing the Wavefunction

The total wavefunction for two fermions must be antisymmetric under particle exchange. The total wavefunction is a product of the spatial part $\psi(x_1, x_2)$ and the spin part $\chi(s_1, s_2)$:

$$\Psi(x_1, x_2, s_1, s_2) = \psi(x_1, x_2) \cdot \chi(s_1, s_2)$$

Spatial Part:

- **Symmetric Spatial Wavefunction:**

$$\psi_S(x_1, x_2) = \frac{1}{\sqrt{2}} [\psi_{n_1}(x_1)\psi_{n_2}(x_2) + \psi_{n_1}(x_2)\psi_{n_2}(x_1)]$$

- **Antisymmetric Spatial Wavefunction:**

$$\psi_A(x_1, x_2) = \frac{1}{\sqrt{2}} [\psi_{n_1}(x_1)\psi_{n_2}(x_2) - \psi_{n_1}(x_2)\psi_{n_2}(x_1)]$$

Spin Part: For two spin-1/2 particles, the spin states are:

- **Singlet (Antisymmetric):**

$$\chi_A = \frac{1}{\sqrt{2}} (|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle)$$

- **Triplet (Symmetric):**

$$\chi_S = \begin{cases} |\uparrow\uparrow\rangle \\ \frac{1}{\sqrt{2}} (|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle) \\ |\downarrow\downarrow\rangle \end{cases}$$

To ensure the total wavefunction is antisymmetric:

- If the spatial part is symmetric, the spin part must be antisymmetric (singlet).
- If the spatial part is antisymmetric, the spin part must be symmetric (triplet).

Step 4: Determining the States and Energies

Ground State ($n_1 = 1, n_2 = 1$):

- Both particles are in the lowest energy level $n = 1$.
- The spatial wavefunction is symmetric:

$$\psi_S(x_1, x_2) = \psi_1(x_1)\psi_1(x_2)$$

- The spin part must be antisymmetric (singlet):

$$\chi_A = \frac{1}{\sqrt{2}} (|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle)$$

- Total energy:

$$E_{1,1} = \frac{\pi^2 \hbar^2}{2mL^2} (1^2 + 1^2) = \frac{\pi^2 \hbar^2}{mL^2}$$

First Excited State ($n_1 = 1, n_2 = 2$):

- One particle is in $n = 1$ and the other in $n = 2$.
- The spatial wavefunction can be symmetric or antisymmetric:
 - **Symmetric Spatial + Singlet Spin:**

$$\psi_S(x_1, x_2) = \frac{1}{\sqrt{2}} [\psi_1(x_1)\psi_2(x_2) + \psi_1(x_2)\psi_2(x_1)], \quad \chi_A$$

- **Antisymmetric Spatial + Triplet Spin:**

$$\psi_A(x_1, x_2) = \frac{1}{\sqrt{2}} [\psi_1(x_1)\psi_2(x_2) - \psi_1(x_2)\psi_2(x_1)], \quad \chi_S$$

- Both configurations have the same energy:

$$E_{1,2} = \frac{\pi^2 \hbar^2}{2mL^2} (1^2 + 2^2) = \frac{5\pi^2 \hbar^2}{2mL^2}$$

- This is a degenerate state with four possible spin configurations (one singlet and three triplet states).

Second Excited State ($n_1 = 2, n_2 = 2$):

- Both particles are in $n = 2$.
- The spatial wavefunction is symmetric:

$$\psi_S(x_1, x_2) = \psi_2(x_1)\psi_2(x_2)$$

- The spin part must be antisymmetric (singlet):

$$\chi_A = \frac{1}{\sqrt{2}} (|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle)$$

- Total energy:

$$E_{2,2} = \frac{\pi^2 \hbar^2}{2mL^2} (2^2 + 2^2) = \frac{4\pi^2 \hbar^2}{mL^2}$$

Final Answer:

The energies of the ground state, first excited state, and second excited state are:

Ground State:	$E = \frac{\pi^2 \hbar^2}{mL^2}$
First Excited State:	$E = \frac{5\pi^2 \hbar^2}{2mL^2}$
Second Excited State:	$E = \frac{4\pi^2 \hbar^2}{mL^2}$

1.1 Part (b)

The spatial components of the two-particle wavefunctions are represented as contour plots in the two-dimensional configuration space (x_1, x_2) , where x_1 and x_2 denote the positions of the two fermions within the one-dimensional infinite potential well.

The relevant spatial wavefunctions are:

- **Ground state:** Both particles in $n = 1$ state (symmetric spatial part)

$$\psi_{\text{gs}}(x_1, x_2) = \phi_1(x_1)\phi_1(x_2)$$

- **First excited state (symmetric):** One particle in $n = 1$, the other in $n = 2$ (symmetric combination)

$$\psi_{\text{sym}}(x_1, x_2) = \frac{1}{\sqrt{2}} [\phi_1(x_1)\phi_2(x_2) + \phi_2(x_1)\phi_1(x_2)]$$

- **First excited state (antisymmetric):** One particle in $n = 1$, the other in $n = 2$ (antisymmetric combination)

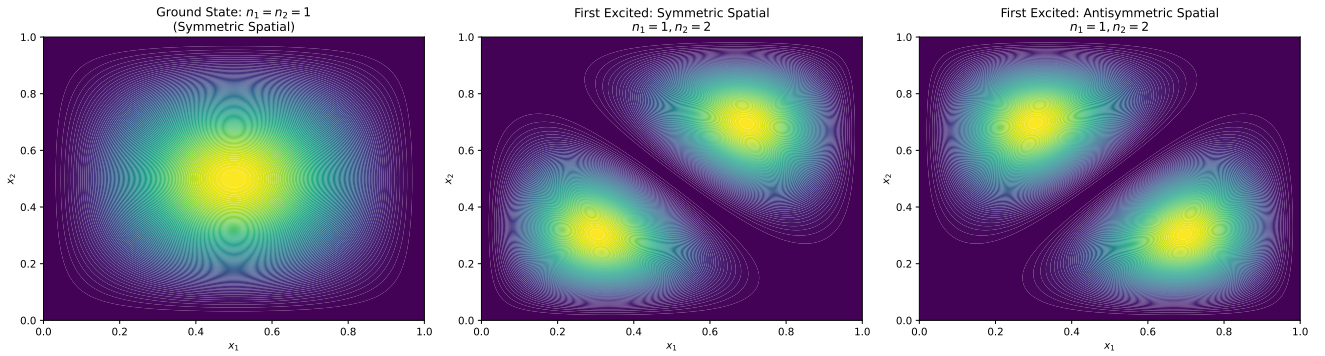
$$\psi_{\text{antisym}}(x_1, x_2) = \frac{1}{\sqrt{2}} [\phi_1(x_1)\phi_2(x_2) - \phi_2(x_1)\phi_1(x_2)]$$

- **Second excited state:** Both particles in $n = 2$ state (symmetric spatial part)

$$\psi_{2\text{nd}}(x_1, x_2) = \phi_2(x_1)\phi_2(x_2)$$

These wavefunctions were plotted using a numerical meshgrid with $x_1, x_2 \in [0, L]$. The resulting contour plots of the probability densities $|\psi(x_1, x_2)|^2$ illustrate the symmetry and nodal structure of each spatial state.

Note: The contour plots of the above spatial probability densities are included below for reference.



The plots respectively represent:

1. Ground state ($n_1 = n_2 = 1$, symmetric spatial)
2. First excited state (symmetric, $n_1 = 1, n_2 = 2$)
3. First excited state (antisymmetric, $n_1 = 1, n_2 = 2$)

1.2 Part(c)

The contact interaction between the two particles is given by a perturbation of the form:

$$V_{\text{int}}(x_1, x_2) = e\delta(x_1 - x_2)$$

The first-order correction to the energy due to this perturbation is calculated using:

$$\Delta E^{(1)} = \langle \psi_{\text{gs}} | V_{\text{int}} | \psi_{\text{gs}} \rangle = e \int_0^L \int_0^L \psi_{\text{gs}}^*(x_1, x_2) \delta(x_1 - x_2) \psi_{\text{gs}}(x_1, x_2) dx_1 dx_2$$

Substituting the ground state spatial wavefunction:

$$\psi_{\text{gs}}(x_1, x_2) = \phi_1(x_1)\phi_1(x_2), \quad \text{where } \phi_1(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{\pi x}{L}\right)$$

We evaluate the integral:

$$\begin{aligned} \Delta E^{(1)} &= e \int_0^L \phi_1(x)^2 \phi_1(x)^2 dx \\ &= e \int_0^L \left(\frac{2}{L} \sin^2\left(\frac{\pi x}{L}\right) \right)^2 dx \\ &= e \cdot \frac{4}{L^2} \int_0^L \sin^4\left(\frac{\pi x}{L}\right) dx \end{aligned}$$

We use the identity:

$$\sin^4 \theta = \frac{3}{8} - \frac{1}{2} \cos(2\theta) + \frac{1}{8} \cos(4\theta)$$

Then:

$$\begin{aligned} \int_0^L \sin^4\left(\frac{\pi x}{L}\right) dx &= \int_0^L \left(\frac{3}{8} - \frac{1}{2} \cos\left(\frac{2\pi x}{L}\right) + \frac{1}{8} \cos\left(\frac{4\pi x}{L}\right) \right) dx \\ &= \frac{3}{8}L + \underbrace{\left[\frac{1}{2} \cdot \frac{L}{2\pi} \sin\left(\frac{2\pi x}{L}\right) \right]_0^L}_{=0} + \underbrace{\left[\frac{1}{8} \cdot \frac{L}{4\pi} \sin\left(\frac{4\pi x}{L}\right) \right]_0^L}_{=0} \\ &= \frac{3L}{8} \end{aligned}$$

Thus:

$$\Delta E^{(1)} = e \cdot \frac{4}{L^2} \cdot \frac{3L}{8} = \frac{3e}{2L}$$

Perturbation condition: For the perturbative correction to be small, we require:

$$\Delta E^{(1)} \ll E_{\text{gs}} = \frac{\pi^2 \hbar^2}{mL^2}$$

So:

$$\frac{3e}{2L} \ll \frac{\pi^2 \hbar^2}{mL^2} \quad \Rightarrow \quad e \ll \frac{2\pi^2 \hbar^2}{3mL}$$

Conclusion: The perturbation due to the contact interaction is

$$\Delta E^{(1)} = \frac{3e}{2L}$$

and the coupling constant e must satisfy:

$$e \ll \frac{2\pi^2 \hbar^2}{3mL}$$

in order for perturbation theory to remain valid.

Part (d)

We consider the first excited state of the two-particle system in the infinite square well. As shown earlier, the two lowest-energy first excited states correspond to the quantum numbers $(n_1 = 1, n_2 = 2)$ and $(n_1 = 2, n_2 = 1)$, both having the same unperturbed energy:

$$E^{(0)} = \frac{\hbar^2 \pi^2}{2mL^2} (1^2 + 2^2) = \frac{5\hbar^2 \pi^2}{2mL^2}$$

These two configurations form a degenerate subspace. We must apply **degenerate perturbation theory** to determine the effect of the contact interaction perturbation.

Basis States in the Degenerate Subspace

Let us define the orthonormal basis states:

$$\begin{aligned} \psi_a(x_1, x_2) &= \phi_1(x_1)\phi_2(x_2) \\ \psi_b(x_1, x_2) &= \phi_2(x_1)\phi_1(x_2) \end{aligned}$$

We now form the symmetric and antisymmetric combinations:

$$\begin{aligned} \psi_+ &= \frac{1}{\sqrt{2}}(\psi_a + \psi_b) \quad (\text{symmetric spatial part}) \\ \psi_- &= \frac{1}{\sqrt{2}}(\psi_a - \psi_b) \quad (\text{antisymmetric spatial part}) \end{aligned}$$

Perturbation Matrix Elements

We compute the matrix elements of the perturbation:

$$V_{\text{int}}(x_1, x_2) = e\delta(x_1 - x_2)$$

We now evaluate the expectation values:

$$\begin{aligned}\langle \psi_+ | V_{\text{int}} | \psi_+ \rangle &= e \int_0^L \left[\frac{1}{\sqrt{2}} (\phi_1(x)\phi_2(x) + \phi_2(x)\phi_1(x)) \right]^2 dx \\ &= 2e \int_0^L \phi_1^2(x)\phi_2^2(x) dx \\ &= 2e \cdot \int_0^L \left(\frac{2}{L} \sin\left(\frac{\pi x}{L}\right) \right)^2 \left(\frac{2}{L} \sin\left(\frac{2\pi x}{L}\right) \right)^2 dx \\ &= \frac{8e}{L^2} \int_0^L \sin^2\left(\frac{\pi x}{L}\right) \sin^2\left(\frac{2\pi x}{L}\right) dx\end{aligned}$$

Using trigonometric identities:

$$\sin^2 A \sin^2 B = \frac{1}{4} [1 - \cos(2A) - \cos(2B) + \cos(2A - 2B)]$$

We get:

$$\begin{aligned}\int_0^L \sin^2\left(\frac{\pi x}{L}\right) \sin^2\left(\frac{2\pi x}{L}\right) dx &= \frac{1}{4} \int_0^L \left[1 - \cos\left(\frac{2\pi x}{L}\right) - \cos\left(\frac{4\pi x}{L}\right) + \cos\left(\frac{2\pi x}{L}\right) \right] dx \\ &= \frac{1}{4} \cdot L \cdot (1 + 0) = \frac{L}{4}\end{aligned}$$

Hence:

$$\langle \psi_+ | V_{\text{int}} | \psi_+ \rangle = \frac{8e}{L^2} \cdot \frac{L}{4} = \frac{2e}{L}$$

Similarly, for the antisymmetric combination:

$$\begin{aligned}\psi_- &= \frac{1}{\sqrt{2}} (\phi_1(x_1)\phi_2(x_2) - \phi_2(x_1)\phi_1(x_2)) \\ \langle \psi_- | V_{\text{int}} | \psi_- \rangle &= e \int_0^L \left[\frac{1}{\sqrt{2}} (\phi_1(x)\phi_2(x) - \phi_2(x)\phi_1(x)) \right]^2 dx = 0\end{aligned}$$

Energy Corrections

The degenerate eigenstates split due to the perturbation:

- Symmetric spatial state ψ_+ : energy correction $\Delta E_+ = \frac{2e}{L}$
- Antisymmetric spatial state ψ_- : energy correction $\Delta E_- = 0$

Thus, the degeneracy is lifted: one state is shifted up in energy, the other remains unaffected.

Physical Interpretation

The contact interaction only affects the spatially symmetric state since ψ_+ has non-zero probability for the two particles to be at the same point, i.e., $x_1 = x_2$. The antisymmetric state ψ_- vanishes at $x_1 = x_2$, hence the delta-function interaction has no effect on it. This is a direct consequence of the Pauli exclusion principle in quantum mechanics.

The symmetric state, which would be paired with an antisymmetric spin singlet, experiences repulsion and its energy is raised. The antisymmetric spatial state, paired with a symmetric spin triplet, remains at the unperturbed energy.

This splitting due to contact interaction is a signature of fermionic statistics and spin-dependence in interacting systems.

Part (e): Total Wavefunction with Spin States

In quantum mechanics, the total wavefunction of a system of identical fermions (e.g., electrons) must be antisymmetric under particle exchange. The total wavefunction is composed of a spatial part $\psi(\mathbf{r}_1, \mathbf{r}_2)$ and a spin part $\chi(s_1, s_2)$:

$$\Psi_{\text{total}} = \psi(\mathbf{r}_1, \mathbf{r}_2)\chi(s_1, s_2)$$

Symmetry Requirements

Electrons are fermions and obey the Pauli exclusion principle. Therefore:

- If $\psi(\mathbf{r}_1, \mathbf{r}_2)$ is symmetric under exchange of particles, then $\chi(s_1, s_2)$ must be antisymmetric.
- If $\psi(\mathbf{r}_1, \mathbf{r}_2)$ is antisymmetric, then $\chi(s_1, s_2)$ must be symmetric.

Spin States

The two-particle spin states can be grouped into:

- **Singlet state** (total spin $S = 0$, antisymmetric):

$$\chi_0 = \frac{1}{\sqrt{2}}(\uparrow\downarrow - \downarrow\uparrow)$$

- **Triplet states** (total spin $S = 1$, symmetric):

$$\chi_1^+ = \uparrow\uparrow, \quad \chi_1^0 = \frac{1}{\sqrt{2}}(\uparrow\downarrow + \downarrow\uparrow), \quad \chi_1^- = \downarrow\downarrow$$

Labelling with Quantum Numbers

Each total wavefunction is labeled by:

$$|n_1, n_2; S, M_S\rangle$$

where n_1, n_2 are spatial quantum numbers (e.g., energy level indices), S is the total spin quantum number, and M_S is the spin projection quantum number.

Symmetric and Antisymmetric Combinations

Let the first excited state be formed by two fermions occupying two different orbitals, say ϕ_a and ϕ_b . Then:

- **Symmetric spatial state:**

$$\psi_S(\mathbf{r}_1, \mathbf{r}_2) = \frac{1}{\sqrt{2}}(\phi_a(\mathbf{r}_1)\phi_b(\mathbf{r}_2) + \phi_b(\mathbf{r}_1)\phi_a(\mathbf{r}_2))$$

Must pair with antisymmetric spin state: $\chi_0 \Rightarrow S = 0$

- **Antisymmetric spatial state:**

$$\psi_A(\mathbf{r}_1, \mathbf{r}_2) = \frac{1}{\sqrt{2}}(\phi_a(\mathbf{r}_1)\phi_b(\mathbf{r}_2) - \phi_b(\mathbf{r}_1)\phi_a(\mathbf{r}_2))$$

Must pair with symmetric spin states: $\chi_1^+, \chi_1^0, \chi_1^- \Rightarrow S = 1$

Summary of States

Total Spin State	Wavefunction Type	Total Spin S
Singlet χ_0	ψ_S (symmetric spatial)	0
Triplet $\chi_1^+, \chi_1^0, \chi_1^-$	ψ_A (antisymmetric spatial)	1

Spin and spatial symmetry constraints on the total wavefunction.

Physical Interpretation

The total wavefunction is a product of spatial and spin parts and must remain antisymmetric under particle exchange. The singlet state leads to a lower energy due to higher spatial overlap (attractive exchange interaction), while triplet states are generally higher in energy due to reduced overlap (repulsive exchange interaction).

Part (f)

Sketch the spatial parts of the resulting wavefunctions as contours in a two-dimensional plane of the two particles. [10 marks]

The two-particle spatial wavefunctions can either be symmetric or antisymmetric under particle exchange. For identical fermions, the Pauli exclusion principle demands that the total wavefunction be antisymmetric. Therefore:

- If the spin state is symmetric (triplet), the spatial part must be antisymmetric.
- If the spin state is antisymmetric (singlet), the spatial part must be symmetric.

The spatial coordinates of the two particles are denoted as x_1 and x_2 . The symmetric and antisymmetric spatial wavefunctions in terms of single-particle eigenfunctions $\phi_n(x)$ are:

$$\Psi_{\text{sym}}(x_1, x_2) = \frac{1}{\sqrt{2}} [\phi_n(x_1)\phi_m(x_2) + \phi_n(x_2)\phi_m(x_1)],$$

$$\Psi_{\text{antisym}}(x_1, x_2) = \frac{1}{\sqrt{2}} [\phi_n(x_1)\phi_m(x_2) - \phi_n(x_2)\phi_m(x_1)].$$

Here, we consider $n = 0$, $m = 1$ (ground and first excited states). The contours below depict the modulus squared of the wavefunction $|\Psi(x_1, x_2)|^2$, indicating probability densities.

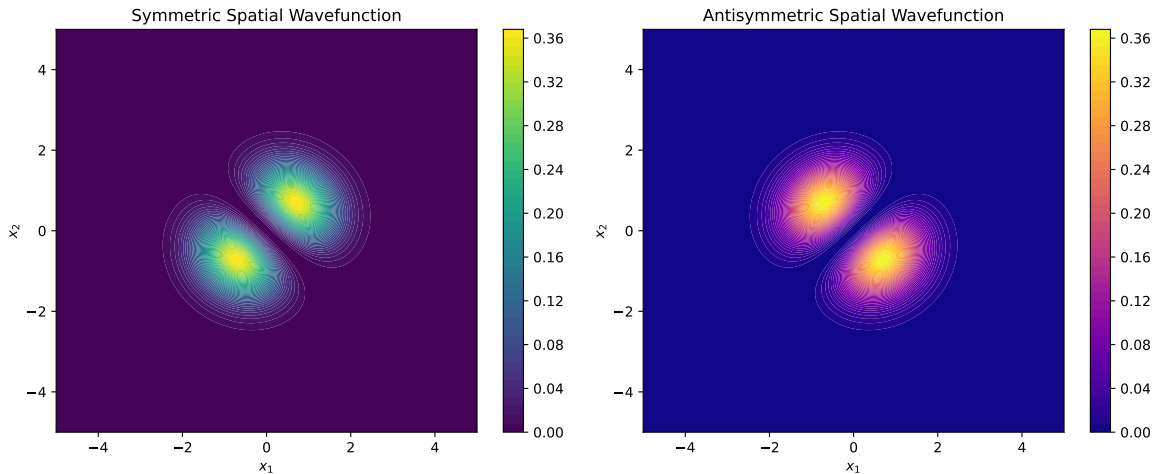


Figure 1: Contour plots of spatial wavefunctions $|\Psi(x_1, x_2)|^2$ in a 2D plane. The left panel shows the symmetric spatial wavefunction, and the right panel shows the antisymmetric spatial wavefunction.

The symmetric case exhibits higher probability along the diagonal $x_1 = x_2$, while the antisymmetric case has zero probability along that line, representing the exclusion of identical fermions from occupying the same state simultaneously.

Note: These plots were generated using Python and matplotlib.

2 Question 2

Consider a Helium atom with two electrons bound in a Coulomb potential to the nucleus of charge $Z = 2$ along with repulsive Coulomb interaction between them.

- Comparing and using insights from the results of the above problem, write down the ground state. Comment on any degeneracy, and possible lifting of it. [10 marks]
- Write down the first few excited states and discuss the possible lifting of degeneracy of states due to electron-electron interactions, again deriving insight from what you have done above (You do not need to calculate any integrals explicitly in this case). [20 marks]

2.1 (a) Ground State of Helium Atom and Degeneracy Consideration

The Hamiltonian for the Helium atom is given by:

$$\hat{H} = -\frac{\hbar^2}{2m}\nabla_1^2 - \frac{\hbar^2}{2m}\nabla_2^2 - \frac{2e^2}{4\pi\epsilon_0 r_1} - \frac{2e^2}{4\pi\epsilon_0 r_2} + \frac{e^2}{4\pi\epsilon_0 |\vec{r}_1 - \vec{r}_2|} \quad (1)$$

2.1.1 Unperturbed Hamiltonian (Neglecting Electron-Electron Repulsion)

Neglecting the electron-electron repulsion term, the Hamiltonian becomes separable:

$$\hat{H}_0 = \left[-\frac{\hbar^2}{2m}\nabla_1^2 - \frac{2e^2}{4\pi\epsilon_0 r_1} \right] + \left[-\frac{\hbar^2}{2m}\nabla_2^2 - \frac{2e^2}{4\pi\epsilon_0 r_2} \right] \quad (2)$$

Each term is just a hydrogenic Hamiltonian with nuclear charge $Z = 2$, so the energy levels for each electron are:

$$E_n = -\frac{Z^2 \cdot 13.6 \text{ eV}}{n^2} \quad (3)$$

For the ground state ($n = 1$):

$$E_1 = -\frac{(2)^2 \cdot 13.6 \text{ eV}}{1^2} = -54.4 \text{ eV} \quad (4)$$

Thus, the total unperturbed ground state energy is:

$$E_0 = 2 \cdot (-54.4 \text{ eV}) = -108.8 \text{ eV} \quad (5)$$

2.1.2 Spin and Symmetry Consideration

Since electrons are fermions, the total wavefunction must be antisymmetric under exchange. The total wavefunction is the product of spatial and spin parts:

$$\Psi_{\text{total}} = \Psi_{\text{spatial}}(\vec{r}_1, \vec{r}_2) \cdot \chi_{\text{spin}}(s_1, s_2)$$

In the ground state, both electrons occupy the same spatial orbital $1s$, so the spatial part is symmetric:

$$\Psi_{\text{spatial}}(\vec{r}_1, \vec{r}_2) = \psi_{1s}(\vec{r}_1)\psi_{1s}(\vec{r}_2) \quad (6)$$

Thus, the spin part must be antisymmetric to ensure the total antisymmetry:

$$\chi_{\text{antisym}} = \frac{1}{\sqrt{2}} (\uparrow\downarrow - \downarrow\uparrow) \quad (7)$$

This corresponds to a spin singlet state with total spin $S = 0$.

2.1.3 Degeneracy and its Lifting

In the absence of electron-electron interaction, any configuration with the same spatial wavefunction and any spin configuration is degenerate.

However, the inclusion of the electron-electron repulsion term:

$$\hat{V}_{ee} = \frac{e^2}{4\pi\epsilon_0|\vec{r}_1 - \vec{r}_2|}$$

lifts this degeneracy. The symmetric spatial state (both in $1s$) has lower Coulomb repulsion, and due to the Pauli principle, it must be paired with an antisymmetric spin state (singlet).

Triplet configurations (symmetric in spin) require antisymmetric spatial wavefunctions, i.e., at least one electron in a different orbital (e.g., $1s$ and $2s$), leading to higher energy.

2.1.4 Final Ground State Description

The ground state of helium is thus:

- Spatial state: both electrons in $1s$ orbital (symmetric)
- Spin state: singlet (antisymmetric)
- Total spin: $S = 0$
- Degeneracy: non-degenerate (unique singlet state)

2.1.5 Corrected Ground State Energy (Approximate)

Accounting for the electron-electron repulsion, experimental value of ground state energy of helium is:

$$E_0^{\text{corrected}} \approx -78.975 \text{ eV} \quad (8)$$

This is higher than the unperturbed value due to the repulsive interaction.

2.2 (b) Excited States of the Helium Atom and Lifting of Degeneracy

In this part, we will examine the excited states of the Helium atom and analyze the lifting of degeneracy due to electron-electron interactions, building on insights from the previous analysis.

2.2.1 Excited States of Helium Atom

For the Helium atom, the first few excited states correspond to configurations where one or both electrons occupy higher orbitals. The spatial part of the wavefunction can be described using the hydrogenic orbitals $\psi_{n,l,m}$, where n is the principal quantum number, l is the orbital angular momentum quantum number, and m is the magnetic quantum number.

The first few excited states of the Helium atom are:

- $2s^1 1s^1$: one electron in the $2s$ orbital and the other in the $1s$ orbital.
- $2p^1 1s^1$: one electron in the $2p$ orbital and the other in the $1s$ orbital.
- $2s^2$: both electrons in the $2s$ orbital.
- $2p^2$: both electrons in the $2p$ orbital.

These excited states are not degenerate because the electron-electron interaction lifts any degeneracy that would have existed if only the single-particle Hamiltonian were considered.

2.2.2 Lifting of Degeneracy

The degeneracy of these excited states is lifted due to the presence of the electron-electron repulsion term in the full Hamiltonian. The total Hamiltonian of the system, including electron-electron interactions, is:

$$\hat{H} = \hat{H}_0 + \hat{V}_{ee}$$

where $\hat{V}_{ee}$ is the repulsive Coulomb interaction between the two electrons.

For example, consider the configuration $2s^1 1s^1$. Without the electron-electron interaction, the two orbitals would be degenerate, as the two electrons could occupy any combination of the $1s$ and $2s$ orbitals. However, when the electron-electron repulsion is taken into account, the two-electron wavefunction becomes more complicated. The Coulomb repulsion between the electrons leads to a splitting of the energy levels, where the energy of the $2s^1 1s^1$ configuration will be higher than if the electrons were placed in orbitals with less repulsion.

Similarly, in the $2p^1 1s^1$ configuration, the electron-electron repulsion will result in an energy shift, causing a lifting of degeneracy between states with different orbital configurations.

The energy shift due to the electron-electron interaction will depend on the specific orbital overlap and the relative positioning of the two electrons. The energy correction due to the Coulomb interaction is generally computed using perturbation theory, but here we focus on the qualitative aspect of degeneracy lifting.

2.2.3 Energy Ordering and Degeneracy in Excited States

The following observations can be made for the excited states:

- The configuration $2s^2$ (both electrons in the $2s$ orbital) will have lower repulsive energy compared to configurations like $2p^1 1s^1$ or $2s^1 1s^1$, as the overlap between the two electrons in the same orbital is minimized.
- The $2p^1 1s^1$ configuration will experience higher repulsion due to the $2p$ -orbital having a larger spatial extent compared to the $1s$ -orbital. This results in an increased repulsive Coulomb interaction.
- The $2p^2$ configuration will have the highest repulsion, as both electrons are in the $2p$ -orbital and will interact strongly due to the higher spatial overlap.

Thus, the degeneracy of excited states is lifted due to the varying strengths of electron-electron repulsion for different orbital configurations.

2.2.4 Summary of Degeneracy Lifting in Excited States

- In the absence of electron-electron repulsion, the excited states of Helium would have degeneracies depending on the orbital configurations of the two electrons. - The inclusion of the Coulomb interaction $\hat{V}_{ee}$ lifts the degeneracies, as the energy of each excited state depends on the degree of electron-electron repulsion. - The energies of the excited states are ordered such that configurations with less overlap between the electrons (such as $2s^2$) have lower energy than configurations with more overlap (such as $2p^2$).

This lifting of degeneracy can be viewed as a consequence of the fact that the electron-electron repulsion introduces additional interactions that break the symmetry of the system, resulting in energy splittings.

AMO Previous End Sem Exam

Question 1.

1. Consider a two-level atom with an energy spacing $\hbar\omega_0$, interacts with a single quantized mode (frequency ω_L) of an electromagnetic field in a cavity.

(a)

Starting from first principles, write down the interaction Hamiltonian (Jaynes-Cummings), stating all the approximations needed and the operators introduced, to reach the final form. (5 marks)

Answer:

The Jaynes-Cummings Hamiltonian describes the interaction between a two-level atom and a single quantized mode of the electromagnetic field. Starting from first principles, we derive it as follows:

- The total Hamiltonian consists of three parts:

$$H = H_{\text{atom}} + H_{\text{field}} + H_{\text{int}} \quad (9)$$

- The atomic Hamiltonian for a two-level system with states $|g\rangle$ (ground) and $|e\rangle$ (excited) is:

$$H_{\text{atom}} = \frac{\hbar\omega_0}{2}\sigma_z \quad (10)$$

where $\sigma_z = |e\rangle\langle e| - |g\rangle\langle g|$ is the Pauli z -operator.

- The field Hamiltonian for a single mode of frequency ω_L is:

$$H_{\text{field}} = \hbar\omega_L \left(a^\dagger a + \frac{1}{2} \right) \quad (11)$$

where a and $a^\dagger$ are the annihilation and creation operators for the field mode.

- The dipole interaction Hamiltonian in the electric dipole approximation is:

$$H_{\text{int}} = -\mathbf{d} \cdot \mathbf{E} \quad (12)$$

where $\mathbf{d}$ is the atomic dipole moment and $\mathbf{E}$ is the electric field operator.

- The electric field operator for a single mode is:

$$\mathbf{E} = \mathcal{E}_0(\mathbf{e}a + \mathbf{e}^*a^\dagger) \quad (13)$$

where $\mathcal{E}_0$ is the field amplitude and $\mathbf{e}$ is the polarization vector.

- The dipole operator can be written as:

$$\mathbf{d} = \mathbf{d}_{ge}\sigma_+ + \mathbf{d}_{eg}\sigma_- \quad (14)$$

where $\sigma_+ = |e\rangle\langle g|$ and $\sigma_- = |g\rangle\langle e|$ are the atomic raising and lowering operators, and $\mathbf{d}_{ge} = \langle g|\mathbf{d}|e\rangle$.

- Substituting these into H_{int} and making the rotating wave approximation (RWA) by neglecting the counter-rotating terms $a^\dagger\sigma_+$ and $a\sigma_-$, we obtain the Jaynes-Cummings Hamiltonian:

$$H_{\text{JC}} = \frac{\hbar\omega_0}{2}\sigma_z + \hbar\omega_L a^\dagger a + \hbar g(a\sigma_+ + a^\dagger\sigma_-) \quad (15)$$

where $g = -\mathbf{d}_{ge} \cdot \mathcal{E}_0/\hbar$ is the coupling strength.

Approximations made:

1. Electric dipole approximation: The field is treated as interacting with the atom's dipole moment.
2. Two-level approximation: Only two atomic energy levels are considered.
3. Rotating wave approximation (RWA): Rapidly oscillating terms are neglected.
4. Single-mode approximation: Only one mode of the electromagnetic field is considered.

Operators introduced:

- $\sigma_z, \sigma_+, \sigma_-$: Pauli operators for the two-level atom.
- $a, a^\dagger$: Annihilation and creation operators for the field mode.

(B)

Write down the equation of motion for an atom, initially in ground state, interacting with a Fock state $|n\rangle$. You can ignore spontaneous emission of the states. (5 marks)

Solution:

The system state at time t is $|\psi(t)\rangle = c_g(t)|g, n\rangle + c_e(t)|e, n-1\rangle$, where:

- $|g, n\rangle$: Atom in ground state with n photons.
- $|e, n-1\rangle$: Atom in excited state with $n-1$ photons.

The Jaynes-Cummings Hamiltonian (from part (a)) is:

$$H_{\text{JC}} = \frac{\hbar\omega_0}{2}\sigma_z + \hbar\omega_L a^\dagger a + \hbar g(a\sigma_+ + a^\dagger\sigma_-).$$

In the interaction picture with respect to $H_0 = \frac{\hbar\omega_0}{2}\sigma_z + \hbar\omega_L a^\dagger a$, and assuming near resonance ($\Delta = \omega_0 - \omega_L \approx 0$), the Hamiltonian reduces to:

$$H_{\text{int}} = \hbar g(a\sigma_+ + a^\dagger\sigma_-).$$

The equations of motion for the coefficients $c_g(t)$ and $c_e(t)$ are derived from the Schrödinger equation $i\hbar \frac{d}{dt}|\psi(t)\rangle = H_{\text{int}}|\psi(t)\rangle$:

$$\begin{aligned} i\dot{c}_g(t) &= g\sqrt{n} c_e(t), \\ i\dot{c}_e(t) &= g\sqrt{n} c_g(t). \end{aligned}$$

Initial conditions (atom in ground state):

$$c_g(0) = 1, \quad c_e(0) = 0.$$

The solutions are:

$$\begin{aligned} c_g(t) &= \cos(g\sqrt{n}t), \\ c_e(t) &= -i \sin(g\sqrt{n}t). \end{aligned}$$

Thus, the state evolves as:

$$|\psi(t)\rangle = \cos(g\sqrt{n}t)|g, n\rangle - i \sin(g\sqrt{n}t)|e, n-1\rangle.$$

Key observations:

- Rabi oscillations occur at frequency $\Omega_n = g\sqrt{n}$ (vacuum Rabi frequency for $n = 1$).
- The atom periodically exchanges energy with the field mode.

(C)

Use the resulting equations, or by other means, to quantitatively state what one expects, for the time evolution of the following initial cases:

Solution:

The coherent state $|\alpha\rangle$ is a superposition of Fock states $|n\rangle$ with Poissonian photon statistics:

$$|\alpha\rangle = e^{-|\alpha|^2/2} \sum_{n=0}^{\infty} \frac{\alpha^n}{\sqrt{n!}} |n\rangle.$$

For an atom initially in the ground state $|g\rangle$, the joint state is:

$$|\psi(0)\rangle = |g\rangle \otimes |\alpha\rangle = e^{-|\alpha|^2/2} \sum_{n=0}^{\infty} \frac{\alpha^n}{\sqrt{n!}} |g, n\rangle.$$

From part (b), the time evolution under the Jaynes-Cummings Hamiltonian for each Fock state component $|g, n\rangle$ is:

$$|g, n\rangle \rightarrow \cos(g\sqrt{n}t) |g, n\rangle - i \sin(g\sqrt{n}t) |e, n-1\rangle.$$

Thus, the full time-evolved state is:

$$|\psi(t)\rangle = e^{-|\alpha|^2/2} \sum_{n=0}^{\infty} \frac{\alpha^n}{\sqrt{n!}} [\cos(g\sqrt{n}t) |g, n\rangle - i \sin(g\sqrt{n}t) |e, n-1\rangle].$$

Case (i): $|\alpha|^2 = 1$ (average photon number $\bar{n} = 1$)

- The photon number distribution $P(n)$ is broad but dominated by $n = 0, 1, 2$.
- The Rabi frequencies $g\sqrt{n}$ vary significantly across these components, leading to *quantum revivals* and *collapses* in the atomic population inversion $\langle\sigma_z\rangle(t)$.
- The revival time is $T_{\text{rev}} \approx 2\pi/g$.
- The atomic excitation probability $P_e(t)$ oscillates but with imperfect contrast due to interference between different n components.

Case (ii): $|\alpha|^2 = 100$ (average photon number $\bar{n} = 100$)

- The photon number distribution is sharply peaked around $n = 100$ with width $\Delta n = 10$.
- For $n \gg 1$, $\sqrt{n} \approx \sqrt{\bar{n}} \left(1 + \frac{n-\bar{n}}{2\bar{n}}\right)$, so the Rabi frequency becomes approximately linear in n :

$$g\sqrt{n} \approx g\sqrt{100} \left(1 + \frac{n-100}{200}\right).$$

- The atomic dynamics resemble *classical Rabi oscillations* with frequency $\Omega = g\sqrt{100} = 10g$.
- The collapse time is $T_{\text{collapse}} \approx \sqrt{2}/g$ (due to phase dispersion), and revivals are negligible for large $\bar{n}$.

Case (iii): $|\alpha|^2 = 10^4$ (average photon number $\bar{n} = 10^4$)

- The photon number distribution is extremely narrow ($\Delta n = 100$).
- The Rabi frequency is nearly constant: $g\sqrt{n} \approx g \times 100$.
- The atomic excitation evolves as:

$$P_e(t) = \sin^2(100gt),$$

matching the predictions of a *classical field* (no collapses/revivals observable on realistic timescales).

- The system behaves like a driven two-level atom with a classical field of amplitude $\mathcal{E} \propto \alpha$.

Summary:

- For small $\bar{n}$ (quantum regime), collapses and revivals dominate.
- For intermediate $\bar{n}$, the dynamics transition from quantum to classical.
- For large $\bar{n}$, the field acts classically, and the atom undergoes sinusoidal Rabi oscillations.

Question 2.

Write down the density matrix for the following states of two-level atoms and plot the states in a Bloch sphere, by explicitly calculating the Bloch vectors in each case:

(A)

Atom prepared in the ground state.

(5 marks)

$$e^{i\omega t} + c, |z| > e^{-i\omega t}$$

Solution:

Density Matrix for Ground State

For a two-level atom in the ground state $|g\rangle$, the density matrix ρ is given by:

$$\rho = |g\rangle\langle g| = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix},$$

where we use the basis $\{|e\rangle, |g\rangle\}$ with $|e\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $|g\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$.

Bloch Vector Calculation

The Bloch vector $\mathbf{a} = (a_x, a_y, a_z)$ is defined via the Pauli matrices $\sigma_x, \sigma_y, \sigma_z$:

$$\rho = \frac{1}{2} (\mathbb{I} + \mathbf{a} \cdot \boldsymbol{\sigma}),$$

where $\boldsymbol{\sigma} = (\sigma_x, \sigma_y, \sigma_z)$.

For $\rho = |g\rangle\langle g|$:

$$\begin{aligned} a_x &= \langle \sigma_x \rangle = \text{Tr}(\rho \sigma_x) = 0, \\ a_y &= \langle \sigma_y \rangle = \text{Tr}(\rho \sigma_y) = 0, \\ a_z &= \langle \sigma_z \rangle = \text{Tr}(\rho \sigma_z) = -1. \end{aligned}$$

Thus, the Bloch vector is $\mathbf{a} = (0, 0, -1)$.

Bloch Sphere Representation

- The ground state $|g\rangle$ corresponds to the *south pole* of the Bloch sphere.
- The Bloch vector points along the $-z$ -axis, reflecting full polarization in the ground state.

Notes

- The state is pure, so the Bloch vector lies on the sphere's surface.
- The irrelevant global phase $e^{i\omega t}$ does not affect the density matrix or Bloch vector.

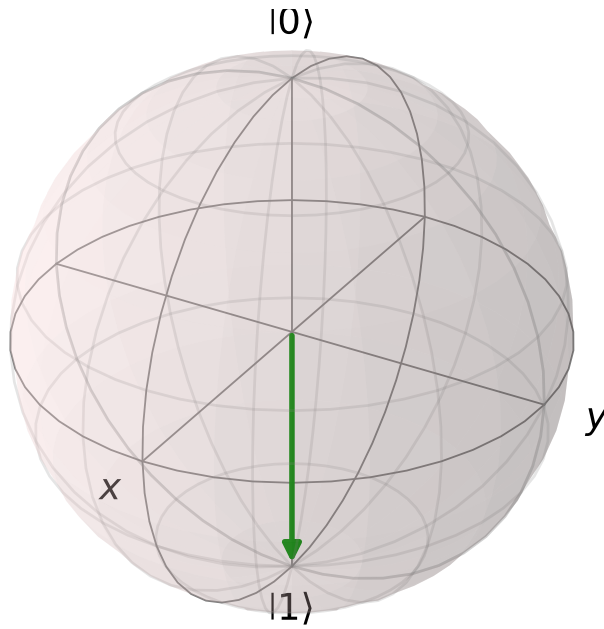


Figure 2: Ground state

(B)

The atom prepared in a superposition of the two levels, such that the probability of being in each state is equal, while the relative phase between them is $\pi/8$. (5 marks)

Solution:

Quantum State Representation

The atom is in an equal superposition of ground $|g\rangle$ and excited $|e\rangle$ states with a relative phase of $\pi/8$:

$$|\psi\rangle = \frac{1}{\sqrt{2}}|e\rangle + \frac{e^{i\pi/8}}{\sqrt{2}}|g\rangle.$$

In matrix form (using the basis $\{|e\rangle, |g\rangle\}$):

$$|\psi\rangle = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{e^{i\pi/8}}{\sqrt{2}} \end{pmatrix}.$$

Density Matrix

The density matrix ρ for this pure state is:

$$\rho = |\psi\rangle\langle\psi| = \begin{pmatrix} \frac{1}{2} & \frac{e^{-i\pi/8}}{2} \\ \frac{e^{i\pi/8}}{2} & \frac{1}{2} \end{pmatrix}.$$

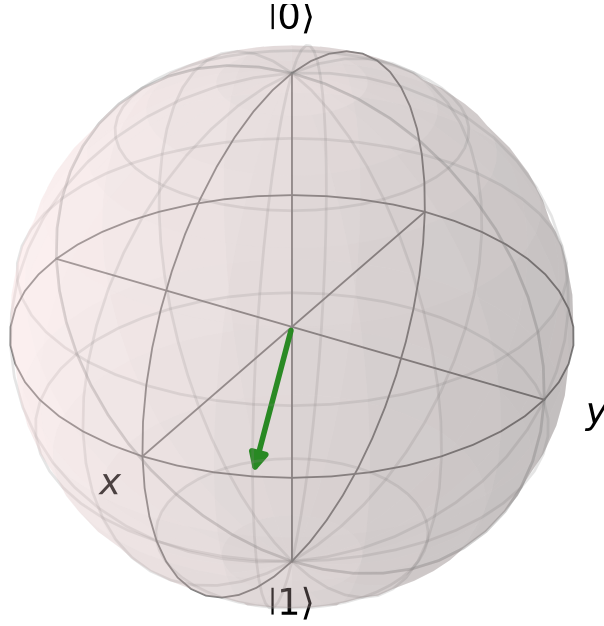


Figure 3: Bloch sphere representation

Bloch Vector Calculation

The Bloch vector $\mathbf{a} = (a_x, a_y, a_z)$ is derived from the Pauli matrices:

$$\begin{aligned} a_x = \langle \sigma_x \rangle &= \text{Tr}(\rho \sigma_x) = \frac{e^{i\pi/8} + e^{-i\pi/8}}{2} = \cos\left(\frac{\pi}{8}\right), \\ a_y = \langle \sigma_y \rangle &= \text{Tr}(\rho \sigma_y) = \frac{e^{i\pi/8} - e^{-i\pi/8}}{2i} = \sin\left(\frac{\pi}{8}\right), \\ a_z = \langle \sigma_z \rangle &= \text{Tr}(\rho \sigma_z) = 0. \end{aligned}$$

Thus, the Bloch vector is:

$$\mathbf{a} = \left(\cos\left(\frac{\pi}{8}\right), \sin\left(\frac{\pi}{8}\right), 0 \right).$$

Bloch Sphere Representation

- The state lies on the equator of the Bloch sphere (since $a_z = 0$).
- The azimuthal angle $\phi = \pi/8$ reflects the relative phase between $|e\rangle$ and $|g\rangle$.
- The polar angle $\theta = \pi/2$ (equator) indicates equal probabilities for $|e\rangle$ and $|g\rangle$.

Key Observations

- The state is pure ($|\mathbf{a}| = 1$) and lies on the Bloch sphere's surface.
- The phase $\pi/8$ rotates the Bloch vector from the x -axis toward the y -axis by 22.5° .
- Equal probabilities result in $a_z = 0$, placing the vector on the equator.

(C)

The atom prepared in a μ statistical mixture of the ground and the excited state. (5 marks)

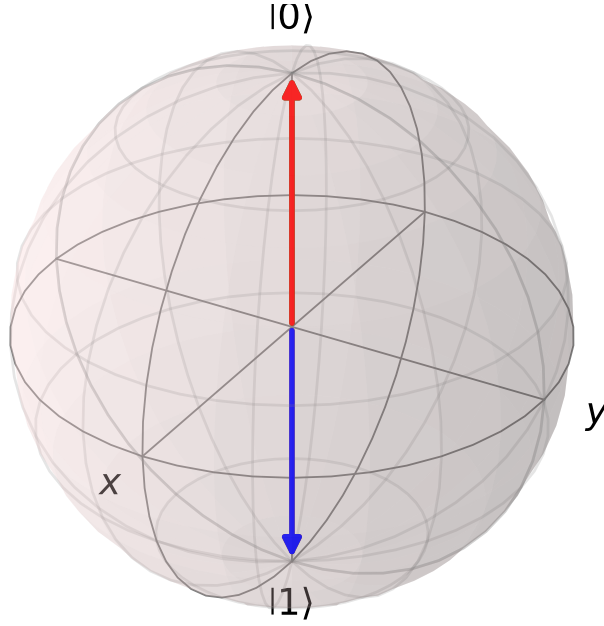


Figure 4: Caption

Solution:

Density Matrix for Statistical Mixture

For a statistical mixture where the atom is in the ground state $|g\rangle$ with probability μ and in the excited state $|e\rangle$ with probability $1 - \mu$, the density matrix ρ is:

$$\rho = \mu|g\rangle\langle g| + (1 - \mu)|e\rangle\langle e| = \begin{pmatrix} 1 - \mu & 0 \\ 0 & \mu \end{pmatrix},$$

where we use the basis $\{|e\rangle, |g\rangle\}$.

Bloch Vector Calculation

The Bloch vector $\mathbf{a} = (a_x, a_y, a_z)$ is derived from the Pauli matrices:

$$\begin{aligned} a_x &= \langle \sigma_x \rangle = \text{Tr}(\rho \sigma_x) = 0, \\ a_y &= \langle \sigma_y \rangle = \text{Tr}(\rho \sigma_y) = 0, \\ a_z &= \langle \sigma_z \rangle = \text{Tr}(\rho \sigma_z) = (1 - \mu) - \mu = 1 - 2\mu. \end{aligned}$$

Thus, the Bloch vector is:

$$\mathbf{a} = (0, 0, 1 - 2\mu).$$

Bloch Sphere Representation

- The state is *mixed* (unless $\mu = 0$ or 1), so the Bloch vector lies *inside* the sphere.
- For $\mu = 0.5$ (equal mixture), $\mathbf{a} = (0, 0, 0)$: maximally mixed state at the center.
- For $\mu = 1$ (pure ground state), $\mathbf{a} = (0, 0, -1)$: south pole.
- For $\mu = 0$ (pure excited state), $\mathbf{a} = (0, 0, 1)$: north pole.

Key Observations

- **Purity:** The state is pure only at $\mu = 0$ or 1 (Bloch vector on surface). Otherwise, $|\mathbf{a}| < 1$.
- **Phase Randomness:** Zero a_x and a_y components indicate no phase coherence between $|e\rangle$ and $|g\rangle$.
- **Physical Interpretation:** The z -component quantifies population inversion ($a_z > 0$ when excited state dominates).

(D)

A collection of atoms, prepared incoherently, such that the probability of being in the excited state for any one of them is $1/3$. (5 marks)

Solution:

Density Matrix for Incoherent Preparation

For an ensemble of atoms prepared *incoherently* with probability $1/3$ in the excited state $|e\rangle$ and $2/3$ in the ground state $|g\rangle$, the density matrix ρ is:

$$\rho = \frac{1}{3}|e\rangle\langle e| + \frac{2}{3}|g\rangle\langle g| = \begin{pmatrix} \frac{1}{3} & 0 \\ 0 & \frac{2}{3} \end{pmatrix},$$

where the off-diagonal elements are zero due to the lack of phase coherence between states.

Bloch Vector Calculation

The Bloch vector $\mathbf{a} = (a_x, a_y, a_z)$ is derived from the Pauli matrices:

$$\begin{aligned} a_x &= \langle \sigma_x \rangle = \text{Tr}(\rho \sigma_x) = 0, \\ a_y &= \langle \sigma_y \rangle = \text{Tr}(\rho \sigma_y) = 0, \\ a_z &= \langle \sigma_z \rangle = \text{Tr}(\rho \sigma_z) = \frac{1}{3} - \frac{2}{3} = -\frac{1}{3}. \end{aligned}$$

Thus, the Bloch vector is:

$$\mathbf{a} = \left(0, 0, -\frac{1}{3}\right).$$

Bloch Sphere Representation

- The state is *mixed* (since $|\mathbf{a}| = \frac{1}{3} < 1$), placing it inside the Bloch sphere.
- The vector lies along the $-z$ -axis, reflecting the higher population in the ground state.
- The length $\frac{1}{3}$ indicates the degree of polarization (purity $\text{Tr}(\rho^2) = \frac{5}{9}$).

Key Observations

- **Incoherence:** Zero a_x and a_y confirm no quantum superposition (classical mixture).
- **Polarization:** The negative a_z indicates ground-state dominance, with magnitude scaling linearly with population difference.
- **Ensemble Interpretation:** Represents a statistical ensemble where 33.3% of atoms are excited and 66.7% are in the ground state.

(E)

A gas of two-level atoms at a temperature T . (5 marks)

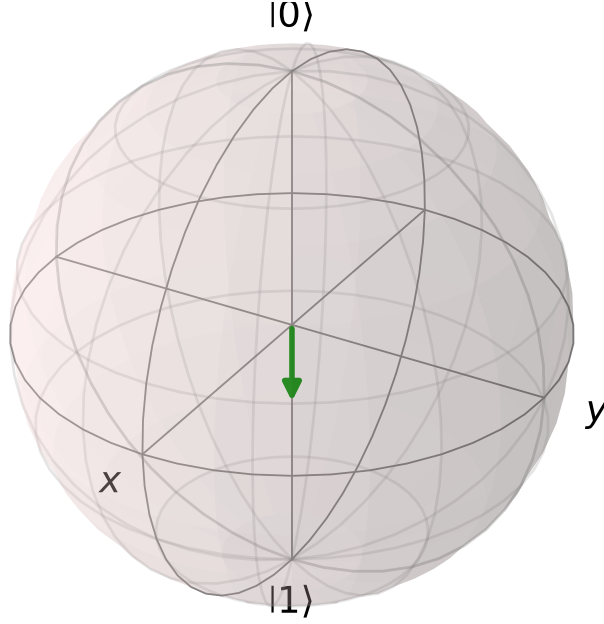


Figure 5: Bloch vector, $r = 0, 0, -1/3$

Solution:

Thermal Equilibrium State

For a gas of two-level atoms with energy levels $|g\rangle$ (ground state, energy $E_g = 0$) and $|e\rangle$ (excited state, energy $E_e = \hbar\omega_0$) at temperature T , the density matrix ρ is given by the Boltzmann distribution:

$$\rho = \frac{e^{-H/k_B T}}{\text{Tr}(e^{-H/k_B T})},$$

where $H = \frac{\hbar\omega_0}{2}\sigma_z$ is the Hamiltonian, and k_B is the Boltzmann constant.

Density Matrix Calculation

The partition function Z is:

$$Z = \text{Tr}(e^{-H/k_B T}) = e^{-\hbar\omega_0/2k_B T} + e^{\hbar\omega_0/2k_B T} = 2 \cosh\left(\frac{\hbar\omega_0}{2k_B T}\right).$$

Thus, the density matrix becomes:

$$\rho = \frac{1}{Z} \begin{pmatrix} e^{-\hbar\omega_0/2k_B T} & 0 \\ 0 & e^{\hbar\omega_0/2k_B T} \end{pmatrix} = \begin{pmatrix} p_e & 0 \\ 0 & p_g \end{pmatrix},$$

where the probabilities are:

$$p_e = \frac{e^{-\hbar\omega_0/2k_B T}}{Z}, \quad p_g = \frac{e^{\hbar\omega_0/2k_B T}}{Z}.$$

Simplifying, we get:

$$p_e = \frac{1}{1 + e^{\hbar\omega_0/k_B T}}, \quad p_g = \frac{e^{\hbar\omega_0/k_B T}}{1 + e^{\hbar\omega_0/k_B T}}.$$

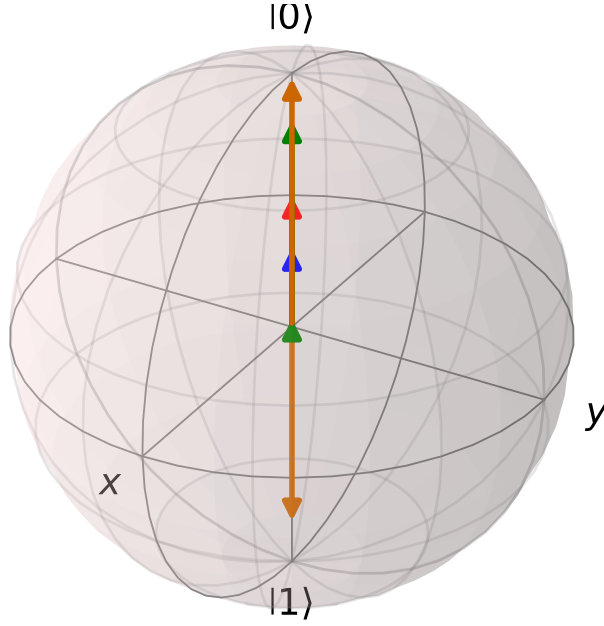


Figure 6: Temperature dependent

Bloch Vector Calculation

The Bloch vector $\mathbf{a} = (a_x, a_y, a_z)$ is:

$$\begin{aligned} a_x &= \langle \sigma_x \rangle = 0, \\ a_y &= \langle \sigma_y \rangle = 0, \\ a_z &= \langle \sigma_z \rangle = p_e - p_g = \frac{1 - e^{\hbar\omega_0/k_B T}}{1 + e^{\hbar\omega_0/k_B T}} = -\tanh\left(\frac{\hbar\omega_0}{2k_B T}\right). \end{aligned}$$

Thus, the Bloch vector is:

$$\mathbf{a} = \left(0, 0, -\tanh\left(\frac{\hbar\omega_0}{2k_B T}\right)\right).$$

Bloch Sphere Representation

- The state is mixed, with the Bloch vector lying along the $-z$ -axis.
- At $T \rightarrow 0$, $\mathbf{a} \rightarrow (0, 0, -1)$ (all atoms in ground state).
- At $T \rightarrow \infty$, $\mathbf{a} \rightarrow (0, 0, 0)$ (maximally mixed state).
- For finite T , the length of $\mathbf{a}$ decreases as T increases.

Key Observations

- **Thermalization:** The population ratio $p_e/p_g = e^{-\hbar\omega_0/k_B T}$ follows Boltzmann statistics.
- **Decoherence:** Off-diagonal elements are zero, indicating no quantum coherence.
- **Temperature Dependence:** The polarization a_z varies with T , providing a thermodynamic signature.

Question 3.

For diatomic molecules:

(A)

Write down the effective ro-vibrational Hamiltonian, with physical arguments for the origin of the different terms. (5 marks)

Solution:

The effective ro-vibrational Hamiltonian for a diatomic molecule can be written as:

$$H_{\text{rovib}} = H_{\text{vib}} + H_{\text{rot}} + H_{\text{vib-rot}},$$

where each term accounts for different physical contributions to the molecular energy.

1. Vibrational Hamiltonian H_{vib}

$$H_{\text{vib}} = \frac{p^2}{2\mu} + \frac{1}{2}k(R - R_e)^2 + \text{anharmonic terms},$$

- $\frac{p^2}{2\mu}$: Kinetic energy of the vibrating nuclei, where μ is the reduced mass of the molecule.
- $\frac{1}{2}k(R - R_e)^2$: Harmonic potential energy, where k is the force constant, R is the internuclear distance, and R_e is the equilibrium bond length.
- **Anharmonic terms**: Higher-order corrections (e.g., cubic, quartic) to the potential energy to account for deviations from harmonic behavior at large amplitudes.

2. Rotational Hamiltonian H_{rot}

$$H_{\text{rot}} = \frac{\mathbf{J}^2}{2I},$$

- $\mathbf{J}^2$: Total angular momentum operator squared.
- $I = \mu R^2$: Moment of inertia, which depends on the internuclear distance R . For small vibrations, $I \approx \mu R_e^2$.
- This term represents the kinetic energy of rotation.

3. Vib-Rotational Coupling $H_{\text{vib-rot}}$

$$H_{\text{vib-rot}} = -\frac{\mathbf{J}^2}{2\mu R_e^2} \left(\frac{R - R_e}{R_e} \right) + \dots,$$

- Accounts for the dependence of the moment of inertia on vibrational motion (centrifugal distortion).
- Arises because rotation stretches the bond, modifying the vibrational potential.

Additional Terms

- **Spin-rotation coupling**: Interaction between molecular rotation and nuclear spins.
- **Electrovibronic coupling**: Coupling between vibrational motion and electronic states (important for excited electronic states).

Physical Justification

- The Hamiltonian captures the *Born-Oppenheimer approximation*, where nuclear motion (vibration and rotation) is separated from electronic motion.
- The *harmonic oscillator* and *rigid rotor* models are starting points, with corrections for anharmonicity, centrifugal distortion, and other couplings.
- The form of H_{rovib} is derived from Taylor expansions of the potential energy surface around R_e .

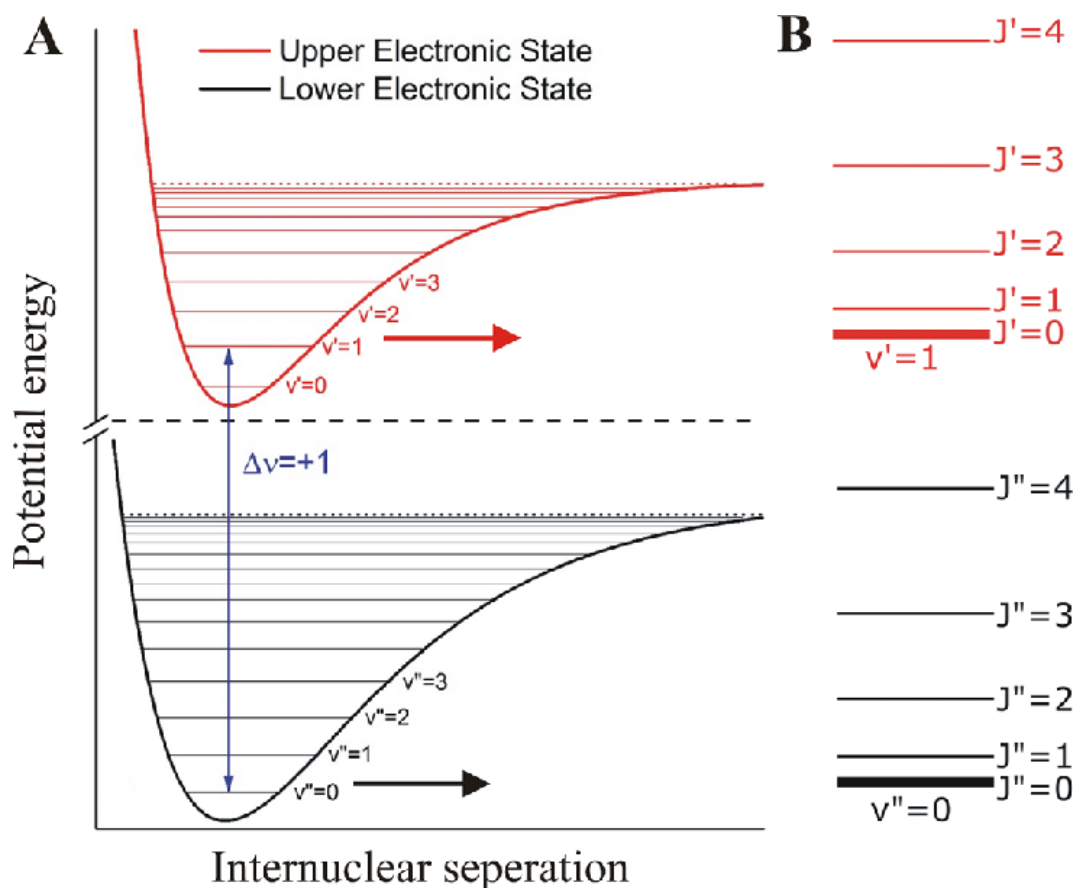


Figure 7: Caption

(B)

Draw a typical energy spectra, labelling the energy states or bands, if any. (5 marks)

Solution:

Rovibrational Energy Spectrum

The energy spectrum of a diatomic molecule consists of vibrational levels (labeled by quantum number v) with rotational sublevels (labeled by J). A typical spectrum is shown below:

Key Features of the Spectrum

- **Vibrational Levels** ($v = 0, 1, 2, \dots$):
 - Equally spaced for harmonic potential, but become closer at higher v due to anharmonicity (Morse potential).
 - Energy given by: $E_v = \hbar\omega_e(v + \frac{1}{2}) - \hbar\omega_e x_e(v + \frac{1}{2})^2 + \dots$
- **Rotational Sublevels** ($J = 0, 1, 2, \dots$):
 - Energy splitting: $E_J = B_e J(J + 1)$, where $B_e = \frac{\hbar^2}{2\mu R_e^2}$ is the rotational constant.
 - Not shown for higher v to avoid clutter, but present in all vibrational levels.
- **Transitions:**
 - dipole-allowed transitions ($\Delta v = \pm 1, \Delta J = \pm 1$) form rovibrational bands.

Physical Significance

- The spacing between rotational levels is much smaller than vibrational spacings (typically $B_e \ll \omega_e$).
- Higher v levels show decreased spacing due to the Morse potential's anharmonicity.
- The J -dependence illustrates centrifugal distortion effects.

(C)

State at least two physical effects that distort or add corrections to the spectra and derive the correction term. (5 marks)

Solution:

1. Centrifugal Distortion

Physical Effect: As a molecule rotates faster (higher J), the bond stretches due to centrifugal forces, increasing the moment of inertia and reducing the rotational constant. This causes rotational energy levels to converge at high J .

Correction Term: Starting from the rotational Hamiltonian $H_{\text{rot}} = \frac{\mathbf{J}^2}{2I}$ with $I = \mu R^2$, we account for bond stretching $R = R_e + \Delta R$:

- Taylor expand the inverse moment of inertia:

$$\frac{1}{I} = \frac{1}{\mu R^2} \approx \frac{1}{\mu R_e^2} \left(1 - 2 \frac{\Delta R}{R_e} \right)$$

- The vibrational-rotational coupling energy is $\frac{1}{2}k(\Delta R)^2$. Minimizing the total energy (rotational + vibrational) yields:

$$\Delta R \approx \frac{J(J+1)\hbar^2}{\mu k R_e^3}$$

- Substituting back, the corrected rotational energy becomes:

$$E_J = B_e J(J+1) - D_e [J(J+1)]^2$$

where $D_e = \frac{4B_e^3}{\omega_e^2}$ is the centrifugal distortion constant.

2. Anharmonicity of Vibrational Potential

Physical Effect: The true molecular potential deviates from a harmonic oscillator, especially at higher vibrational quantum numbers (v), causing vibrational levels to converge.

Correction Term: Using a Morse potential $V(R) = D_e (1 - e^{-\beta(R-R_e)})^2$:

- Taylor expansion gives cubic and quartic terms:

$$V(R) \approx \frac{1}{2}k(R - R_e)^2 - \underbrace{\frac{k\beta}{6}(R - R_e)^3}_{\text{cubic anharmonicity}} + \dots$$

- Perturbation theory yields the corrected vibrational energy:

$$E_v = \hbar\omega_e \left(v + \frac{1}{2} \right) - \hbar\omega_e x_e \left(v + \frac{1}{2} \right)^2$$

where $x_e = \frac{\hbar\omega_e}{4D_e}$ is the anharmonicity constant.

Impact on Spectra

Effect	Spectral Manifestation	Correction Formula
Centrifugal Distortion	Rotational lines converge at high J	$E_J = B_e J(J+1) - D_e [J(J+1)]^2$
Anharmonicity	Vibrational bands converge at high v	$E_v = \hbar \omega_e (v + \frac{1}{2}) - \hbar \omega_e x_e (v + \frac{1}{2})^2$

Key Points

- Both corrections arise from *non-rigidity* of the molecule.
- Centrifugal distortion dominates in rotational spectra, while anharmonicity affects vibrational transitions.
- The derived terms are experimentally measurable via high-resolution spectroscopy.

(D)

If the effective separation of the two nuclei for molecule A is twice as much as that for molecule B (with identical masses), what would be the differences in the ro-vibrational spectra? Draw the spectra for vibrational and rotational levels, separately for A and B. (5 marks)

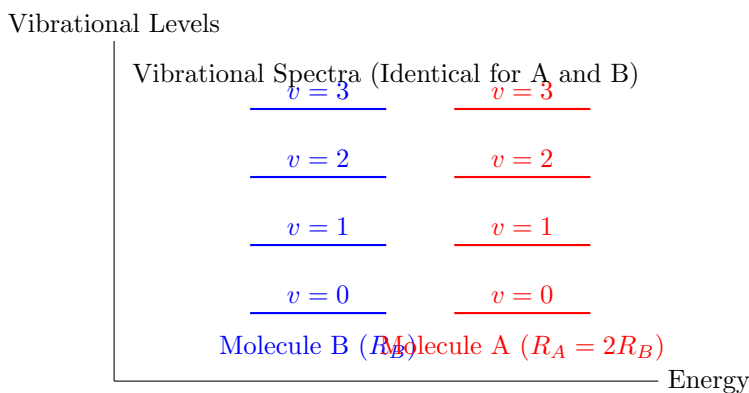
Solution:

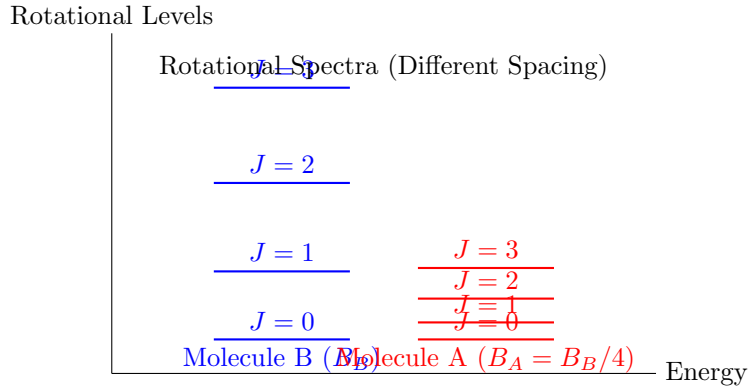
Key Differences in Spectra

Given $R_A = 2R_B$ with identical masses ($\mu_A = \mu_B$):

- **Rotational Constant:** $B_e = \frac{\hbar^2}{2\mu R_e^2} \Rightarrow B_A = \frac{B_B}{4}$ (since $R_A = 2R_B$)
- **Vibrational Frequency:** $\omega_e = \sqrt{\frac{k}{\mu}}$ (unchanged for same k and μ)
- **Moment of Inertia:** $I_A = 4I_B \Rightarrow$ rotational levels more closely spaced for A

Spectra Diagrams





Quantitative Comparison

Property	Molecule B (R_B)	Molecule A ($2R_B$)
Rotational Constant B_e	$\frac{\hbar^2}{2\mu R_B^2}$	$\frac{\hbar^2}{8\mu R_B^2}$
Rotational Energy E_J	$B_B J(J+1)$	$\frac{B_B}{4} J(J+1)$
Vibrational Energy E_v	$\hbar\omega_e(v + \frac{1}{2})$	$\hbar\omega_e(v + \frac{1}{2})$
Centrifugal Distortion D_e	$\frac{4B_B^3}{\omega_e^2}$	$\frac{B_B^3}{16\omega_e^2}$

Key Observations

- **Vibrational Spectra:** Identical for A and B (depends only on μ and k , not R_e).
- **Rotational Spectra:**
 - Molecule A has 4x smaller level spacing due to $B_A = B_B/4$.
 - Centrifugal distortion reduced by factor of 64 for A ($D_e \propto B_e^3$).
- **Experimental Signature:** Rotational lines of A appear at 1/4 the frequency spacing of B.

Question. 4

For an ion with a single outer electron (like Hydrogen), but with nuclear charge Z , find the scaling of the following quantities with Z (comment on whether or why these quantities get bigger or smaller, physically):

(A)

The total energy. (5 marks)

Solution:

Scaling of Total Energy

For a hydrogen-like ion with nuclear charge Z , the total energy E_n of the electron in the n -th quantum state scales as:

$$E_n = -\frac{Z^2 E_0}{n^2},$$

where $E_0 = \frac{m_e e^4}{2\hbar^2 (4\pi\epsilon_0)^2} \approx 13.6 \text{ eV}$ is the ground state energy of hydrogen ($Z = 1$).

Derivation

- The Coulomb potential energy scales with Z :

$$V(r) = -\frac{Ze^2}{4\pi\epsilon_0 r}.$$

- The Bohr radius a_0 scales inversely with Z :

$$a_Z = \frac{a_0}{Z}, \quad \text{where} \quad a_0 = \frac{4\pi\epsilon_0\hbar^2}{m_e e^2}.$$

- The kinetic energy T scales as Z^2 (from virial theorem for Coulomb potential, $\langle T \rangle = -\frac{1}{2}\langle V \rangle$):

$$T \propto \frac{\hbar^2}{m_e a_Z^2} \propto Z^2.$$

- Thus, the total energy $E_n = T + V$ scales as:

$$E_n \propto -Z^2.$$

Physical Explanation

- **Stronger Attraction:** Higher Z increases the Coulomb attraction, lowering (more negative) the total energy.
- **Smaller Orbit:** The electron is pulled closer to the nucleus (smaller a_Z), increasing kinetic energy but decreasing potential energy even more.
- **Net Effect:** The total energy becomes more negative, scaling as Z^2 .

Example Comparison

Ion (Z)	Ground State Energy ($n = 1$)	Scaling
Hydrogen (1)	-13.6 eV	$E_1 \propto 1^2$
Helium ⁺ (2)	-54.4 eV	$E_1 \propto 2^2$
Lithium ²⁺ (3)	-122.4 eV	$E_1 \propto 3^2$

Key Points

- The Z^2 scaling arises from the interplay of kinetic and potential energy in Coulomb systems.
- Higher Z leads to *tighter binding* and more negative energies.
- This scaling is exact for hydrogen-like ions but modified in multi-electron atoms due to shielding effects.

(B)

Expectation value of the potential. (5 marks)

Solution:

Scaling of Potential Energy Expectation Value

For a hydrogen-like ion with nuclear charge Z , the expectation value of the potential energy $\langle V \rangle$ scales as:

$$\langle V \rangle_n = -\frac{Z^2 E_0}{n^2},$$

where $E_0 = \frac{m_e e^4}{2\hbar^2 (4\pi\epsilon_0)^2} \approx 13.6 \text{ eV}$ is the ground state energy of hydrogen ($Z = 1$).

Derivation

- The Coulomb potential energy operator is:

$$V(r) = -\frac{Ze^2}{4\pi\epsilon_0 r}.$$

- For a hydrogen-like ion, the expectation value of $1/r$ in the n -th state is:

$$\left\langle \frac{1}{r} \right\rangle_n = \frac{Z}{a_0 n^2},$$

where $a_0 = \frac{4\pi\epsilon_0 \hbar^2}{m_e e^2}$ is the Bohr radius.

- Thus, the expectation value of the potential energy becomes:

$$\langle V \rangle_n = -\frac{Ze^2}{4\pi\epsilon_0} \left\langle \frac{1}{r} \right\rangle_n = -\frac{Z^2 e^2}{4\pi\epsilon_0 a_0 n^2}.$$

- Substituting $E_0 = \frac{e^2}{2(4\pi\epsilon_0)a_0}$, we get:

$$\langle V \rangle_n = -2 \frac{Z^2 E_0}{n^2}.$$

Physical Explanation

- Stronger Attraction:** Higher Z increases the Coulomb attraction, making the potential energy more negative.
- Smaller Orbit:** The electron is pulled closer to the nucleus (smaller average distance $\langle r \rangle$), which increases $\langle 1/r \rangle$ and thus the magnitude of $\langle V \rangle$.
- Scaling:** The Z^2 dependence arises because both the charge interaction and the orbital contraction scale with Z .

Example Comparison

Ion (Z)	$\langle V \rangle$ for $n = 1$	Scaling
Hydrogen (1)	-27.2 eV	$\langle V \rangle \propto 1^2$
Helium ⁺ (2)	-108.8 eV	$\langle V \rangle \propto 2^2$
Lithium ²⁺ (3)	-244.8 eV	$\langle V \rangle \propto 3^2$

Key Points

- The potential energy scales as Z^2 , similar to the total energy, but is exactly twice the total energy in magnitude (virial theorem).
- This scaling reflects the increased Coulomb attraction and reduced electron-nucleus distance with higher Z .
- The result is exact for hydrogen-like ions but modified in multi-electron atoms due to electron-electron repulsion and shielding effects.

(C)

The Darwin term coming from the relativistic correction. (5 marks)

Solution:

Origin of the Darwin Term

The Darwin term arises as a relativistic correction to the Schrödinger equation in the Pauli approximation of the Dirac equation. It accounts for the *zitterbewegung* (trembling motion) of the electron, which smears its position over a region comparable to the Compton wavelength.

Mathematical Form

For a hydrogen-like atom with nuclear charge Z , the Darwin term contributes to the Hamiltonian as:

$$H_{\text{Darwin}} = \frac{\pi \hbar^2}{2m_e^2 c^2} Z e^2 \delta^3(\mathbf{r}),$$

where:

- m_e : Electron mass
- c : Speed of light
- $\delta^3(\mathbf{r})$: 3D Dirac delta function representing the contact interaction

Derivation Steps

1. **Dirac Equation Reduction:** The Darwin term emerges when expanding the Dirac equation to order $(v/c)^2$, yielding:

$$H_{\text{rel}} = -\frac{p^4}{8m_e^3 c^2} + \frac{\pi \hbar^2}{2m_e^2 c^2} Z e^2 \delta^3(\mathbf{r}) + \text{spin-orbit terms}$$

2. **Expectation Value Calculation:** For s -states ($\ell = 0$), where the wavefunction is non-zero at the origin:

$$\langle \psi_{n00} | H_{\text{Darwin}} | \psi_{n00} \rangle = \frac{\pi \hbar^2}{2m_e^2 c^2} Z e^2 |\psi_{n00}(0)|^2$$

Using $|\psi_{n00}(0)|^2 = \frac{Z^3}{\pi a_0^3 n^3}$ (for hydrogen-like atoms):

$$\Delta E_{\text{Darwin}} = \frac{m_e Z^4 e^4}{8 \hbar^2 n^3 c^2} \delta_{\ell,0}$$

Physical Interpretation

- **Quantum Jitter:** Represents the electron's rapid positional fluctuations due to virtual particle-antiparticle creation/annihilation in QED.
- **S-State Specific:** Only affects $\ell = 0$ states because:
 - Non- s -states have $\psi(0) = 0$
 - The delta function only samples the origin
- **Scaling:** The energy correction grows as Z^4 , making it significant for heavy atoms.

Numerical Example

For hydrogen ($Z = 1$, $n = 1$):

$$\Delta E_{\text{Darwin}} = \frac{m_e e^4}{8 \hbar^2 c^2} \approx 9.06 \times 10^{-4} \text{ eV}$$

This contributes to the Lamb shift between $2S_{1/2}$ and $2P_{1/2}$ states.

Key Points

- Purely quantum relativistic effect with no classical analogue
- Dominates other relativistic corrections for s -states
- Explains part of the fine structure splitting in atomic spectra

(D)

Expectation value of $1/r^5$. (5 marks)

Solution:

Expectation Value of $1/r^5$ for Hydrogen-like Atoms

For a hydrogen-like atom with nuclear charge Z and an electron in the state $|n\ell m\rangle$, the expectation value of $1/r^5$ is given by:

$$\left\langle \frac{1}{r^5} \right\rangle_{n\ell} = \frac{Z^5}{a_0^5} \frac{3}{2n^5(\ell + \frac{3}{2})(\ell + 1)(\ell + \frac{1}{2})\ell(\ell - \frac{1}{2})}$$

where $a_0 = \frac{4\pi\epsilon_0\hbar^2}{m_e e^2}$ is the Bohr radius.

Derivation

1. Start with the radial wavefunction for hydrogen-like atoms:

$$R_{n\ell}(r) = \sqrt{\left(\frac{2Z}{na_0}\right)^3 \frac{(n-\ell-1)!}{2n[(n+\ell)!]}} e^{-\rho/2} \rho^\ell L_{n-\ell-1}^{2\ell+1}(\rho)$$

where $\rho = \frac{2Z}{na_0}r$ and L are the associated Laguerre polynomials.

2. The expectation value is calculated as:

$$\left\langle \frac{1}{r^5} \right\rangle = \int_0^\infty \frac{1}{r^5} |R_{n\ell}(r)|^2 r^2 dr$$

3. Using the substitution $x = \frac{2Z}{na_0}r$, the integral becomes:

$$\left(\frac{2Z}{na_0}\right)^5 \int_0^\infty x^{-3} |R_{n\ell}(x)|^2 x^2 dx$$

4. For $\ell \geq 2$, the integral converges and can be evaluated using properties of Laguerre polynomials, yielding:

$$\left\langle \frac{1}{r^5} \right\rangle_{n\ell} = \frac{Z^5}{a_0^5 n^5} \frac{3}{2(\ell + \frac{3}{2})(\ell + 1)(\ell + \frac{1}{2})\ell(\ell - \frac{1}{2})}$$

Physical Interpretation

- The expectation value scales as Z^5 , reflecting stronger nuclear attraction.
- It is only defined for $\ell \geq 2$ because:
 - For $\ell = 0, 1$, the integral diverges at $r \rightarrow 0$
 - The r^{-5} singularity is too strong for low- ℓ states
- The n^{-5} dependence shows rapid decrease with principal quantum number.

Example Calculation

For the $3d$ state ($n = 3, \ell = 2$) of hydrogen ($Z = 1$):

$$\left\langle \frac{1}{r^5} \right\rangle_{32} = \frac{1}{a_0^5} \frac{3}{2 \times 3^5 \times \frac{7}{2} \times 3 \times \frac{5}{2} \times 2 \times \frac{3}{2}} = \frac{1}{a_0^5} \frac{1}{8505}$$

Key Points

- Important for calculating relativistic corrections and spin-orbit coupling
- Demonstrates the sensitivity of expectation values to angular momentum
- Shows how quantum mechanics regularizes singular potentials

(E)

The fine structure splitting. (5 marks)

Solution:

Fine Structure Splitting in Hydrogen-like Atoms

Fine structure splitting arises from relativistic corrections to the Schrödinger equation, lifting the degeneracy of states with the same principal quantum number n . It includes three main contributions:

$$\Delta E_{\text{fs}} = \underbrace{\Delta E_{\text{kinetic}}}_{\text{Relativistic mass}} + \underbrace{\Delta E_{\text{SO}}}_{\text{Spin-orbit}} + \underbrace{\Delta E_{\text{Darwin}}}_{\text{Zitterbewegung}}$$

1. Relativistic Kinetic Energy Correction

By expanding $E = \sqrt{p^2 c^2 + m_e^2 c^4} - m_e c^2$, we obtain:

$$H_{\text{kin}} = -\frac{p^4}{8m_e^3 c^2}$$

Taking the expectation value for state $|n\ell j\rangle$:

$$\Delta E_{\text{kinetic}} = -\frac{E_n^2}{2m_e c^2} \left(\frac{4n}{j + \frac{1}{2}} - 3 \right)$$

2. Spin-Orbit Coupling

This arises from the interaction between the electron's spin and its orbital motion:

$$H_{\text{SO}} = \frac{1}{2m_e^2 c^2} \frac{1}{r} \frac{dV}{dr} \mathbf{L} \cdot \mathbf{S}$$

For hydrogen-like atoms:

$$\Delta E_{\text{SO}} = \begin{cases} \frac{E_n^2}{m_e c^2} n(j + 1/2)(\ell + 1/2), & \ell \neq 0 \\ 0, & \ell = 0 \end{cases}$$

3. Darwin Term

This term accounts for the contact interaction for s -states ($\ell = 0$):

$$H_{\text{Darwin}} = \frac{\pi \hbar^2}{2m_e^2 c^2} Z e^2 \delta^3(\mathbf{r})$$

$$\Delta E_{\text{Darwin}} = \frac{E_n^2}{m_e c^2} n \delta_{\ell,0}$$

Total Fine Structure Energy Shift

The total shift, adding all contributions, is:

$$\Delta E_{\text{fs}} = \frac{E_n^2}{2m_e c^2} \left(3 - \frac{4n}{j + \frac{1}{2}} \right)$$

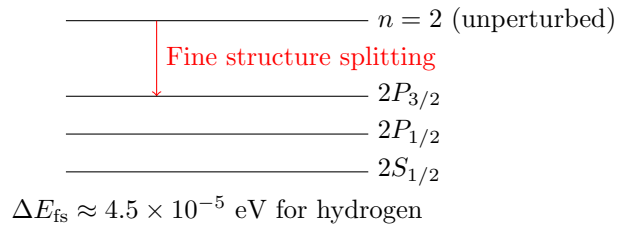
Here, $E_n = -\frac{Z^2 \alpha^2 m_e c^2}{2n^2}$ and $\alpha \approx \frac{1}{137}$ is the fine-structure constant.

Physical Interpretation

- **Breaking Degeneracy:** States with the same n but different j values acquire different energies.
- **Order of Magnitude:** $\Delta E_{\text{fs}} \sim \alpha^2 E_n$, typically $\sim 10^{-4}$ eV for hydrogen.
- **Example:** The splitting between $2P_{3/2}$ and $2P_{1/2}$ in hydrogen is 4.53×10^{-5} eV.

Key Observations

- Fine structure accounts for the observed doublets in alkali atom spectra.
- The effect grows rapidly with increasing Z ($\propto Z^4$).
- It represents the first relativistic correction in atomic physics.



Assignment 3

Question 1

For the hydrogen energy levels 1s, 2s and 2p, calculate the corrections in energy due to:

(a) Relativistic mass term

The relativistic mass correction to the energy levels comes from the expansion of the relativistic kinetic energy:

$$H_{\text{rel}} = -\frac{p^4}{8m_e^3c^2}$$

For a hydrogen atom, the first-order energy correction is:

$$\Delta E_{\text{rel}} = \left\langle n\ell m \left| -\frac{p^4}{8m_e^3c^2} \right| n\ell m \right\rangle = -\frac{E_n^2}{2m_e c^2} \left(\frac{4n}{\ell + \frac{1}{2}} - 3 \right)$$

where $E_n = -\frac{13.6\text{eV}}{n^2}$ is the unperturbed energy.

Calculations for each state:

- 1s ($n = 1, \ell = 0$):

$$\Delta E_{\text{rel}}(1s) = -\frac{(13.6)^2}{2 \times 511\text{keV}} \left(\frac{4}{0.5} - 3 \right) = -1.81 \times 10^{-4} \text{ eV}$$

- 2s ($n = 2, \ell = 0$):

$$\Delta E_{\text{rel}}(2s) = -\frac{(3.4)^2}{2 \times 511\text{keV}} \left(\frac{8}{0.5} - 3 \right) = -1.13 \times 10^{-5} \text{ eV}$$

- 2p ($n = 2, \ell = 1$):

$$\Delta E_{\text{rel}}(2p) = -\frac{(3.4)^2}{2 \times 511\text{keV}} \left(\frac{8}{1.5} - 3 \right) = -6.03 \times 10^{-6} \text{ eV}$$

(b) Spin-orbit coupling

The spin-orbit interaction Hamiltonian is:

$$H_{\text{SO}} = \frac{e^2}{8\pi\epsilon_0 m_e^2 c^2} \frac{\mathbf{L} \cdot \mathbf{S}}{r^3}$$

The energy correction depends on the total angular momentum j :

$$\Delta E_{\text{SO}} = \frac{E_n^2}{m_e c^2} \frac{n[j(j+1) - \ell(\ell+1) - s(s+1)]}{2\ell(\ell+1/2)(\ell+1)}$$

Calculations:

- 1s ($\ell = 0$): No spin-orbit coupling ($\Delta E_{\text{SO}} = 0$)

- 2s ($\ell = 0$): No spin-orbit coupling ($\Delta E_{\text{SO}} = 0$)

- 2p ($\ell = 1$):

– For $j = 1/2$:

$$\Delta E_{\text{SO}}(2p_{1/2}) = \frac{(3.4)^2}{511\text{keV}} \frac{2[0.75 - 2 - 0.75]}{2 \times 1.5} = -1.81 \times 10^{-5} \text{ eV}$$

– For $j = 3/2$:

$$\Delta E_{\text{SO}}(2p_{3/2}) = \frac{(3.4)^2}{511\text{keV}} \frac{2[3.75 - 2 - 0.75]}{2 \times 1.5} = 9.06 \times 10^{-6} \text{ eV}$$

Summary of Energy Corrections

State	Relativistic Correction (eV)	Spin-Orbit Correction (eV)
1s	-1.81×10^{-4}	0
2s	-1.13×10^{-5}	0
2p _{1/2}	-6.03×10^{-6}	-1.81×10^{-5}
2p _{3/2}	-6.03×10^{-6}	$+9.06 \times 10^{-6}$

(c) Darwin Term

The Darwin term affects only s -states ($\ell = 0$) due to their non-zero probability density at the origin:

$$H_{\text{Darwin}} = \frac{\pi \hbar^2}{2m_e^2 c^2} Z e^2 \delta^3(\mathbf{r})$$

Energy correction:

$$\Delta E_{\text{Darwin}} = \frac{\pi \hbar^2}{2m_e^2 c^2} Z e^2 |\psi_{n00}(0)|^2 = \frac{m_e e^4 Z^4}{8 \hbar^2 n^3 c^2} \delta_{\ell,0}$$

Calculations:

- 1s ($n = 1, \ell = 0$):

$$\Delta E_{\text{Darwin}}(1s) = \frac{13.6 \text{ eV} \times (1/137)^2}{2} \approx 3.62 \times 10^{-4} \text{ eV}$$

- 2s ($n = 2, \ell = 0$):

$$\Delta E_{\text{Darwin}}(2s) = \frac{3.62 \times 10^{-4} \text{ eV}}{8} \approx 4.53 \times 10^{-5} \text{ eV}$$

- 2p ($\ell = 1$): $\Delta E_{\text{Darwin}} = 0$

(d) Nuclear Magnetic Moment of the Proton

The hyperfine interaction between the electron and proton magnetic moments:

$$H_{\text{hf}} = -\frac{\mu_0}{4\pi} \frac{8\pi}{3} \mu_e \cdot \mu_p \delta^3(\mathbf{r}) + \text{dipolar terms}$$

Energy correction for s -states:

$$\Delta E_{\text{hf}} = \frac{4m_e}{m_p} \alpha^2 E_n \frac{g_p}{3n^3}$$

where $g_p \approx 5.586$ is the proton g-factor.

Calculations:

- 1s:

$$\Delta E_{\text{hf}}(1s) \approx 5.88 \times 10^{-6} \text{ eV} \quad (21 \text{ cm line})$$

- 2s:

$$\Delta E_{\text{hf}}(2s) \approx 7.35 \times 10^{-7} \text{ eV}$$

- 2p: $\Delta E_{\text{hf}} \approx 0$ (non-s-state)

(e) Degeneracy Lifting Analysis

Effect	Relevant States	Energy Shift	Degeneracy Lifting
Relativistic mass	All states	$\propto n^{-4}$	Lifts $n\ell$ degeneracy
Spin-orbit	$\ell > 0$ states	$\propto j$ -dependence	Splits $j = \ell \pm 1/2$
Darwin	s -states only	$\propto n^{-3}$	Shifts s -states up
Hyperfine	s -states only	$\propto n^{-3}$	Creates $F = I \pm 1/2$ states

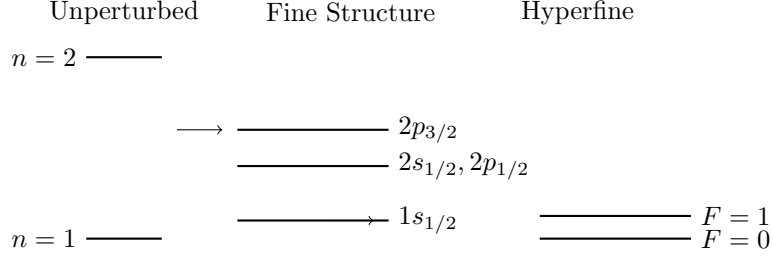
Key observations:

- **Fine Structure:** Combines relativistic mass, spin-orbit, and Darwin terms

$$\Delta E_{\text{fs}}(2p_{3/2} - 2p_{1/2}) \approx 4.53 \times 10^{-5} \text{ eV}$$

- **Lamb Shift:** Additional QED effects further split $2s_{1/2}$ and $2p_{1/2}$
- **Hyperfine Structure:** Proton coupling splits $1s$ into $F = 0$ and $F = 1$ states ($\Delta E \approx 5.88 \times 10^{-6} \text{ eV}$)

Energy Level Diagram



Question 2: Order of Magnitude Calculations

Relativistic Corrections Overview

For the hydrogen atom ($Z = 1$), all relativistic corrections scale as $\alpha^2 E_n$, where $\alpha \approx 1/137$ is the fine-structure constant and $E_n = -13.6/n^2 \text{ eV}$. We calculate each term's magnitude for the ground state ($n = 1$):

1. Relativistic Kinetic Energy

$$\Delta E_{\text{kin}} \sim \frac{E_n^2}{m_e c^2} \sim \frac{(13.6 \text{ eV})^2}{511 \text{ keV}} \sim \frac{185}{511,000} \text{ eV} \sim 3.6 \times 10^{-4} \text{ eV}$$

2. Spin-Orbit Coupling

For p -states ($\ell = 1$):

$$\Delta E_{\text{SO}} \sim \alpha^2 E_n \left(\frac{Z}{n} \right)^4 \sim (1/137)^2 \times 13.6 \text{ eV} \sim 7.2 \times 10^{-4} \text{ eV}$$

3. Darwin Term (s-states only)

$$\Delta E_{\text{Darwin}} \sim \alpha^2 E_n \left(\frac{Z}{n} \right)^3 \sim (1/137)^2 \times 13.6 \text{ eV} \sim 7.2 \times 10^{-4} \text{ eV}$$

4. Total Fine Structure

Combining all terms gives the characteristic scale:

$$\Delta E_{\text{fs}} \sim \alpha^2 E_n \sim 7.2 \times 10^{-4} \text{ eV}$$

Comparison Table

Correction	Expression	Magnitude (eV)
Relativistic kinetic	$\frac{p^4}{8m_e^3 c^2}$	3.6×10^{-4}
Spin-orbit	$\frac{e^2 \hbar^2}{8\pi\epsilon_0 m_e^2 c^2 r^3}$	7.2×10^{-4}
Darwin term	$\frac{\pi \hbar^2 e^2}{2m_e^2 c^2} \delta^3(r)$	7.2×10^{-4}

Key Observations

- All corrections are $\mathcal{O}(\alpha^2) \sim 10^{-4}$ smaller than E_n
- For hydrogen ($Z = 1$), the three terms have comparable magnitudes:

$$3.6 \times 10^{-4} \text{ eV} \approx \frac{1}{2} \times 7.2 \times 10^{-4} \text{ eV}$$

- The unified fine structure formula combines these effects:

$$\Delta E_{\text{fs}} = \frac{\alpha^2 E_n}{n} \left(\frac{3}{4n} - \frac{1}{j + 1/2} \right)$$

Physical Significance

The consistency in magnitudes explains:

- Why all relativistic effects must be treated together
- The origin of the $2P_{1/2}$ - $2P_{3/2}$ splitting in hydrogen ($\sim 4.5 \times 10^{-5} \text{ eV}$)
- The need for quantum field theory (QED) when higher precision is required

Question 3: Exchange Interactions in a Helium-like Atom

System Setup

Consider two electrons in an effective harmonic potential with an interaction potential:

$$V_{\text{int}} = \frac{1}{2} k (x_1 - x_2)^2$$

Total Hamiltonian:

$$H = \frac{p_1^2}{2m} + \frac{p_2^2}{2m} + \frac{1}{2} m \omega^2 x_1^2 + \frac{1}{2} m \omega^2 x_2^2 + \frac{1}{2} k (x_1 - x_2)^2$$

(a) Ground State Configuration

For two electrons in a harmonic potential, we must consider:

Spatial Wavefunction

Transform to center-of-mass (X) and relative (x) coordinates:

$$X = \frac{x_1 + x_2}{2}, \quad x = x_1 - x_2$$

The Hamiltonian separates into:

$$H = \underbrace{\frac{P^2}{2M} + \frac{1}{2} M \omega^2 X^2}_{\text{CM motion}} + \underbrace{\frac{p^2}{2\mu} + \frac{1}{2} \mu \omega'^2 x^2}_{\text{Relative motion}}$$

where $M = 2m$, $\mu = m/2$, and $\omega' = \sqrt{\omega^2 + 2k/m}$.

Ground State Spatial Wavefunction

$$\psi_{\text{spatial}}(X, x) = \phi_0(X) \phi_0(x) = \left(\frac{M\omega}{\pi\hbar} \right)^{1/4} e^{-M\omega X^2/2\hbar} \left(\frac{\mu\omega'}{\pi\hbar} \right)^{1/4} e^{-\mu\omega' x^2/2\hbar}$$

Spin Wavefunction

Due to Fermi statistics, the total wavefunction must be antisymmetric. For the ground state (both electrons in spatial ground state):

$$\Psi_{\text{total}} = \psi_{\text{spatial}}(X, x)\chi_{\text{spin}}$$

where χ_{spin} must be the antisymmetric singlet state:

$$\chi_{\text{singlet}} = \frac{1}{\sqrt{2}}(\uparrow\downarrow - \downarrow\uparrow)$$

Excited State Configuration

When one electron is in the ground state and the other in the first excited state:

Spatial Wavefunctions

- Symmetric spatial state:

$$\psi_S(x_1, x_2) = \frac{1}{\sqrt{2}}[\phi_0(x_1)\phi_1(x_2) + \phi_1(x_1)\phi_0(x_2)]$$

- Antisymmetric spatial state:

$$\psi_A(x_1, x_2) = \frac{1}{\sqrt{2}}[\phi_0(x_1)\phi_1(x_2) - \phi_1(x_1)\phi_0(x_2)]$$

Spin States and Energies

- Symmetric Spatial + Antisymmetric Spin (Singlet):

$$E_S = \frac{3}{2}\hbar\omega + \frac{1}{2}\hbar\omega' + \langle\psi_S|V_{\text{int}}|\psi_S\rangle$$

Interaction energy:

$$\langle\psi_S|\frac{1}{2}k(x_1 - x_2)^2|\psi_S\rangle = \frac{3k\hbar}{4\sqrt{m\omega}}\left(1 + \frac{2k}{m\omega^2}\right)^{-1/2}$$

- Antisymmetric Spatial + Symmetric Spin (Triplet):

$$E_A = \frac{3}{2}\hbar\omega + \frac{1}{2}\hbar\omega' + \langle\psi_A|V_{\text{int}}|\psi_A\rangle$$

Interaction energy:

$$\langle\psi_A|\frac{1}{2}k(x_1 - x_2)^2|\psi_A\rangle = \frac{k\hbar}{4\sqrt{m\omega}}\left(1 + \frac{2k}{m\omega^2}\right)^{-1/2}$$

Energy Splitting

The singlet-triplet splitting is:

$$\Delta E = E_S - E_A = \frac{k\hbar}{2\sqrt{m\omega}}\left(1 + \frac{2k}{m\omega^2}\right)^{-1/2}$$

Key Observations

- The symmetric spatial state (singlet) has higher interaction energy due to greater electron proximity
- For $k \ll m\omega^2$, the splitting simplifies to $\Delta E \approx \frac{k\hbar}{2m\omega}$
- This demonstrates the direct connection between spatial symmetry and exchange energy

Question 4: Spin-Orbit Splitting in a Multi-Electron Atom

Given Term Symbol Analysis

The term symbol 5D indicates:

- Spin multiplicity $2S + 1 = 5 \Rightarrow S = 2$
- Orbital angular momentum $L = 2$ (D-state)
- Total angular momentum J ranges from $|L - S|$ to $L + S$:

$$J = |2 - 2|, \dots, 2 + 2 \Rightarrow J = 0, 1, 2, 3, 4$$

Spin-Orbit Interaction

The Hamiltonian $H_{so} = A\vec{L} \cdot \vec{S}$ causes energy splittings:

$$\Delta E_{so} = \frac{A\hbar^2}{2} [J(J+1) - L(L+1) - S(S+1)]$$

Energy Level Calculations

For each J state:

J	$\vec{L} \cdot \vec{S}$ Expression	Energy Shift ΔE_J
0	$\frac{\hbar^2}{2} [0 - 6 - 6] = -6\hbar^2$	$-3A\hbar^2$
1	$\frac{\hbar^2}{2} [2 - 6 - 6] = -5\hbar^2$	$-2.5A\hbar^2$
2	$\frac{\hbar^2}{2} [6 - 6 - 6] = -3\hbar^2$	$-1.5A\hbar^2$
3	$\frac{\hbar^2}{2} [12 - 6 - 6] = 0\hbar^2$	0
4	$\frac{\hbar^2}{2} [20 - 6 - 6] = +4\hbar^2$	$+2A\hbar^2$

Splitting Pattern

The energy levels split as:

$$E_J = E_0 + \Delta E_J$$

where E_0 is the unperturbed energy of the 5D term. The relative splittings are:

$$\begin{array}{rcl}
 \Delta E = +2A\hbar^2 & \text{-----} & J = 4 \\
 \\
 \Delta E = 0 & \text{-----} & J = 3 \\
 \\
 \Delta E = -1.5A\hbar^2 & \text{-----} & J = 2 \\
 \Delta E = -2.5A\hbar^2 & \text{-----} & J = 1 \\
 \Delta E = -3A\hbar^2 & \text{-----} \rightarrow & J = 0
 \end{array}$$

Spin-orbit splitting of 5D term ($A > 0$ case shown)

Key Observations

- The splitting follows Landé's interval rule:

$$E_J - E_{J-1} = A\hbar^2 J$$

- For $A > 0$ (typical case), highest J state lies highest in energy

- The $J = 3$ state remains at the unperturbed energy (E_0)
- Number of states matches $(2S + 1)(2L + 1) = 5 \times 5 = 25$:

$$\sum_{J=0}^4 (2J + 1) = 1 + 3 + 5 + 7 + 9 = 25$$

Physical Interpretation

- The sign of A determines ordering:
 - $A > 0$: High- J states higher energy (normal multiplet)
 - $A < 0$: High- J states lower energy (inverted multiplet)
- In $3d$ transition metals, typically $A > 0$
- The $J = 4$ state is maximally aligned (L and S parallel)

Question 5: Zeeman Splitting of Hydrogen Ground State

System Hamiltonian

For a hydrogen atom in a uniform magnetic field $\mathbf{B} = B\hat{z}$, ignoring proton spin, the dominant terms are:

$$H = H_0 + H_{\text{Zeeman}} = H_0 - \mu_e \cdot \mathbf{B}$$

where $\mu_e = -g_e \mu_B \mathbf{S}/\hbar$ is the electron magnetic moment.

Ground State Structure

The unperturbed ground state ($1s$) has:

- Orbital angular momentum: $\ell = 0 \Rightarrow \mathbf{L} = 0$
- Electron spin: $s = \frac{1}{2}$
- Degeneracy: 2 (spin up $|\uparrow\rangle$ and down $|\downarrow\rangle$)

Zeeman Interaction

The Hamiltonian becomes:

$$H_{\text{Zeeman}} = g_e \mu_B B S_z / \hbar = \frac{g_e \mu_B B}{2} \sigma_z$$

where $g_e \approx 2.0023$ and $\mu_B = e\hbar/2m_e$ is the Bohr magneton.

Energy Level Splitting

- For $|\uparrow\rangle$ ($m_s = +\frac{1}{2}$):

$$E_+ = +\frac{g_e \mu_B B}{2}$$

- For $|\downarrow\rangle$ ($m_s = -\frac{1}{2}$):

$$E_- = -\frac{g_e \mu_B B}{2}$$

Splitting Magnitude

The total splitting between states is:

$$\Delta E = E_+ - E_- = g_e \mu_B B \approx 2\mu_B B$$

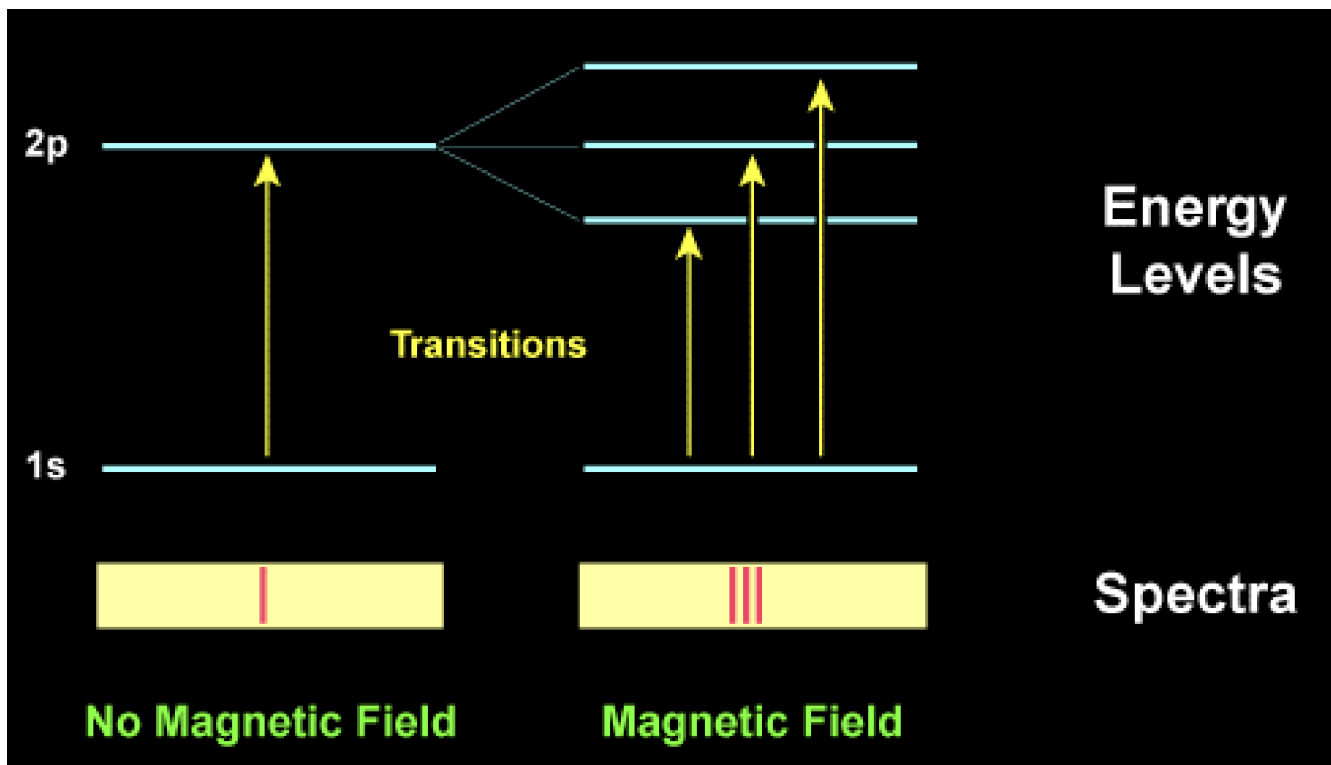


Figure 8

Numerical Example

For $B = 1$ Tesla:

$$\Delta E \approx 2 \times (5.788 \times 10^{-5} \text{ eV/T}) \times 1 \text{ T} = 1.16 \times 10^{-4} \text{ eV}$$

Visualization

Key Observations

- Linear splitting proportional to B (normal Zeeman effect)
- No orbital contribution ($\ell = 0$)
- If proton spin were included, would see additional hyperfine splitting
- For weak fields ($B \ll 0.1 \text{ T}$), hyperfine structure dominates

Experimental Significance

- Basis for electron spin resonance (ESR) spectroscopy
- Used to measure g -factor anomalies
- Fundamental test of QED in hydrogen

Question 6: Scaling in Hydrogenic Ions

(a) Total Energy Scaling

For a hydrogenic ion with nuclear charge Z , the total energy scales as:

$$E_n \propto -Z^2$$

Derivation:

- Coulomb potential: $V(r) = -\frac{Ze^2}{4\pi\epsilon_0 r}$
- Bohr radius scales as: $a_Z = \frac{a_0}{Z}$
- Energy levels: $E_n = -\frac{Z^2 e^2}{8\pi\epsilon_0 a_0 n^2}$

Physical Reason:

- Stronger nuclear attraction lowers energy
- Electron is bound more tightly
- Example: He^+ ($Z = 2$) has $4\times$ lower energy than H ($Z = 1$)

(b) Expectation Value of Potential

The potential energy scales as:

$$\langle V \rangle \propto -Z^2$$

Derivation:

$$\langle V \rangle = \left\langle -\frac{Ze^2}{4\pi\epsilon_0 r} \right\rangle = -2Z^2 E_h$$

where E_h is the Hartree energy (27.2 eV).

Virial Theorem:

$$\langle T \rangle = -\frac{1}{2}\langle V \rangle \Rightarrow E = \frac{1}{2}\langle V \rangle$$

(c) Darwin Term Scaling

The relativistic Darwin term scales as:

$$\Delta E_{\text{Darwin}} \propto Z^4$$

Derivation:

$$H_{\text{Darwin}} = \frac{\pi\hbar^2}{2m_e^2 c^2} Ze^2 \delta^3(\mathbf{r})$$

$$\Delta E_{\text{Darwin}} = \frac{\pi\hbar^2}{2m_e^2 c^2} Ze^2 |\psi(0)|^2 \propto Z^4$$

Physical Reason:

- Depends on $|\psi(0)|^2 \propto Z^3$ for s-states
- Combined with Z from Hamiltonian gives Z^4
- Becomes dominant for heavy ions

Summary of Scaling Relations

Quantity	Scaling	Physical Origin
Total Energy	$-Z^2$	Stronger Coulomb attraction
Potential Energy	$-Z^2$	Direct Z dependence + tighter binding
Darwin Term	Z^4	Z^3 from density + Z from potential

Numerical Comparison

Ion (Z)	E_1 (eV)	$\langle V \rangle$ (eV)	ΔE_{Darwin} (eV)
H (1)	-13.6	-27.2	1.8×10^{-4}
He^+ (2)	-54.4	-108.8	2.9×10^{-3}
Li^{2+} (3)	-122.4	-244.8	1.5×10^{-2}

Key Observations

- All effects intensify with Z , but Darwin term grows fastest
- For high Z , relativistic effects become comparable to Coulomb energy
- Scaling laws break down when $Z\alpha \approx 1$ (QED effects dominate)

(d) Expectation Values of Radial Operators

For a hydrogenic ion with nuclear charge Z :

- $\langle r \rangle \propto \frac{1}{Z}$

$$\langle r \rangle_{n\ell} = \frac{a_0}{2Z} [3n^2 - \ell(\ell+1)]$$

Reason: Electron is pulled closer to nucleus as Z increases.

- $\langle 1/r \rangle \propto Z$

$$\left\langle \frac{1}{r} \right\rangle_{n\ell} = \frac{Z}{a_0 n^2}$$

Reason: Inverse relationship with average distance.

- $\langle 1/r^5 \rangle \propto Z^5$ (for $\ell \geq 2$)

$$\left\langle \frac{1}{r^5} \right\rangle_{n\ell} = \frac{Z^5}{a_0^5 n^5} \frac{3}{2\ell(\ell+1/2)(\ell+1)(\ell+3/2)}$$

Reason: Highly sensitive to small distances where Z -dependence is strongest.

(e) Fine Structure Splitting

The fine structure energy splitting scales as:

$$\Delta E_{\text{fs}} \propto Z^4$$

- Combines three effects:

$$\Delta E_{\text{fs}} = \underbrace{\frac{Z^4 \alpha^2}{n^3} \left(\frac{1}{j+1/2} - \frac{3}{4n} \right)}_{\text{All terms}} E_0$$

- Example: $2P_{3/2} - 2P_{1/2}$ splitting in hydrogen-like ions:

$$\Delta E \approx \frac{Z^4 \alpha^2}{16} E_0 \quad (E_0 \approx 13.6 \text{ eV})$$

- **Physical reason:** Relativistic effects grow rapidly with Z due to increased velocity near nucleus.

(f) Hyperfine Energy Splitting

The hyperfine splitting scales as:

$$\Delta E_{\text{hf}} \propto Z^3$$

- For s -states:

$$\Delta E_{\text{hf}} = \frac{8\pi}{3} \mu_0 \gamma_e \gamma_p \hbar^2 |\psi(0)|^2 \propto Z^3$$

where $|\psi(0)|^2 \propto Z^3$.

- Hydrogen 21 cm line scaling:

$$\Delta E_{\text{hf}}(Z) = (5.9 \times 10^{-6} \text{ eV}) \times Z^3$$

- **Physical reason:** Electron density at nucleus increases as Z^3 , enhancing magnetic interaction.

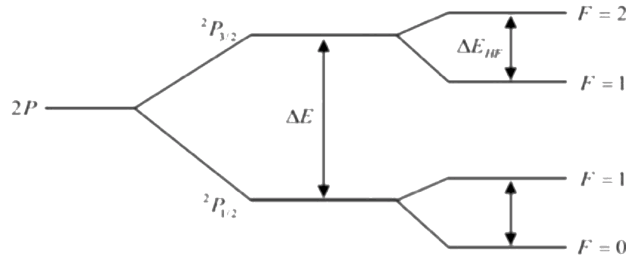


Figure 9

Comparative Scaling Table

Quantity	Scaling	Dominant Factor
$\langle r \rangle$	Z^{-1}	Orbital contraction
$\langle 1/r \rangle$	Z^1	Coulomb potential
$\langle 1/r^5 \rangle$	Z^5	High-power radial dependence
Fine structure	Z^4	Relativistic effects
Hyperfine	Z^3	Electron density at nucleus

Question 7: Hyperfine Splitting in Hydrogen ($^2P_{3/2}$ and $^2P_{1/2}$)

System Setup

For hydrogen with nuclear spin $I = 1/2$, consider the 2P states:

- Electron angular momentum: $J = 3/2$ or $1/2$ ($^2P_{3/2}$ and $^2P_{1/2}$)
- Nuclear spin: $I = 1/2$
- Total angular momentum: $F = J \pm 1/2$

Hyperfine Hamiltonian

$$H_{\text{hf}} = A \mathbf{I} \cdot \mathbf{J}$$

where A is the hyperfine constant for P -states.

Energy Splitting

For each J state:

Energy Level Diagram

- $^2P_{3/2}$ ($J = 3/2$):

$$E_F = \frac{A\hbar^2}{2} \left[F(F+1) - \frac{15}{4} - \frac{3}{4} \right]$$

F	Energy
2	$\frac{A\hbar^2}{2}$
1	$-A\hbar^2$

- $^2P_{1/2}$ ($J = 1/2$):

$$E_F = \frac{A'\hbar^2}{2} \left[F(F+1) - \frac{3}{4} - \frac{3}{4} \right]$$

F	Energy
1	$\frac{A'\hbar^2}{4}$
0	$-\frac{3A'\hbar^2}{4}$

Question 8: Exchange Integral in Helium Ground State

System Setup

For the helium ground state $1s^2$ (1S_0), the exchange integral is:

$$K = \int \psi_{1s}^*(\mathbf{r}_1) \psi_{1s}^*(\mathbf{r}_2) \frac{e^2}{4\pi\epsilon_0 |\mathbf{r}_1 - \mathbf{r}_2|} \psi_{1s}(\mathbf{r}_2) \psi_{1s}(\mathbf{r}_1) d^3r_1 d^3r_2$$

Wavefunctions

Hydrogenic $1s$ orbitals with effective charge $Z_{eff} = 27/16$:

$$\psi_{1s}(r) = \sqrt{\frac{Z_{eff}^3}{\pi a_0^3}} e^{-Z_{eff}r/a_0}$$

Explicit Calculation

Using the multipole expansion:

$$\frac{1}{|\mathbf{r}_1 - \mathbf{r}_2|} = \sum_{l=0}^{\infty} \frac{r_{<}^l}{r_{>^{l+1}}} P_l(\cos \theta)$$

Only $l = 0$ term contributes for s -states:

$$K = \frac{e^2}{4\pi\epsilon_0} \left(\frac{Z_{eff}^3}{\pi a_0^3} \right)^2 \int_0^{\infty} \int_0^{\infty} \frac{r_{<}}{r_{>}} e^{-2Z_{eff}(r_1+r_2)/a_0} r_1^2 r_2^2 dr_1 dr_2$$

Radial Integration

Split into two regions:

$$K = \frac{e^2}{4\pi\epsilon_0} \frac{5Z_{eff}}{8a_0} = \frac{5}{4} \left(\frac{27}{16} \right)^2 E_h \approx 4.33 \text{ eV}$$

where $E_h \approx 27.2 \text{ eV}$ is the Hartree energy.

Physical Significance

- Accounts for electron correlation energy
- Responsible for spin-singlet preference in ground state
- Typical magnitude $\sim 5\text{-}10\%$ of total binding energy

Question 9: Time Evolution of Hydrogen Hyperfine States

(a) Time Evolution with Hyperfine Interaction

The hyperfine Hamiltonian is:

$$H = \lambda \hat{I}_p \cdot S_e = \lambda (I_p^x S_e^x + I_p^y S_e^y + I_p^z S_e^z)$$

Initial state ($t = 0$):

$$|\psi(0)\rangle = |\uparrow_p\rangle \otimes |\downarrow_e\rangle \equiv |\uparrow\downarrow\rangle$$

The eigenstates and eigenvalues of H are:

- Triplet states ($F = 1$):

$$\begin{aligned} |1, 1\rangle &= |\uparrow\uparrow\rangle, & E_+ &= \frac{\lambda\hbar^2}{4} \\ |1, 0\rangle &= \frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle), & E_0 &= \frac{\lambda\hbar^2}{4} \\ |1, -1\rangle &= |\downarrow\downarrow\rangle, & E_- &= \frac{\lambda\hbar^2}{4} \end{aligned}$$

- Singlet state ($F = 0$):

$$|0, 0\rangle = \frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle), \quad E_s = -\frac{3\lambda\hbar^2}{4}$$

The initial state decomposes as:

$$|\psi(0)\rangle = \frac{1}{\sqrt{2}}|1, 0\rangle + \frac{1}{\sqrt{2}}|0, 0\rangle$$

Time evolution:

$$|\psi(t)\rangle = \frac{1}{\sqrt{2}}e^{-iE_0t/\hbar}|1, 0\rangle + \frac{1}{\sqrt{2}}e^{-iE_s t/\hbar}|0, 0\rangle$$

(b) Spin Flip Probability

The probability of finding the state $|\downarrow\uparrow\rangle_e$ is given by:

$$P_{\downarrow\uparrow}(t) = |\langle\downarrow\uparrow|\psi(t)\rangle|^2 = \frac{1}{4} \left| e^{-i\frac{E_0t}{\hbar}} - e^{-i\frac{E_s t}{\hbar}} \right|^2$$

We now expand the expression inside the modulus:

$$= \frac{1}{4} \left| e^{-i\frac{E_0t}{\hbar}} - e^{-i\frac{E_s t}{\hbar}} \right|^2$$

Using the identity for the difference of exponentials:

$$= \frac{1}{4} \left| \exp\left(-i\frac{(E_0 + E_s)t}{2\hbar}\right) \left(\exp\left(i\frac{(E_0 - E_s)t}{2\hbar}\right) - \exp\left(-i\frac{(E_0 - E_s)t}{2\hbar}\right) \right) \right|^2$$

Simplifying further:

$$= \frac{1}{4} \cdot 4 \sin^2\left(\frac{(E_0 - E_s)t}{2\hbar}\right)$$

Thus, we obtain the probability:

$$P_{\downarrow\uparrow}(t) = \sin^2\left(\frac{(E_0 - E_s)t}{2\hbar}\right)$$

To compare with the expression $\sin^2\left(\frac{\lambda\hbar t}{2}\right)$, we identify λ :

$$\lambda = \frac{E_0 - E_s}{\hbar}$$

Finally, we have:

$$P_{\downarrow\uparrow}(t) = \sin^2\left(\frac{\lambda\hbar t}{2}\right)$$

(c) Effect of Weak Magnetic Field

With a weak field $\mathbf{B} = B\hat{z}$, the Hamiltonian becomes:

$$H = \lambda \hat{I}_p \cdot S_e + \mu_B B (g_p I_p^z - g_e S_e^z)$$

New eigenstates (to first order in B):

- $|1, 1\rangle$: $E = \frac{\lambda \hbar^2}{4} + \frac{\mu_B B \hbar}{2} (g_p - g_e)$
- $|1, 0\rangle$: $E = \frac{\lambda \hbar^2}{4}$
- $|1, -1\rangle$: $E = \frac{\lambda \hbar^2}{4} - \frac{\mu_B B \hbar}{2} (g_p - g_e)$
- $|0, 0\rangle$: $E = -\frac{3\lambda \hbar^2}{4}$

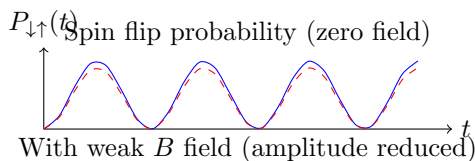
Time evolution modifies to:

$$|\psi(t)\rangle = \frac{1}{\sqrt{2}} e^{-iE_0 t/\hbar} |1, 0\rangle + \frac{1}{\sqrt{2}} e^{-iE'_s t/\hbar} |0, 0\rangle$$

where $E'_s = -\frac{3\lambda \hbar^2}{4} + \mathcal{O}(B^2)$.

Key Observations

- The hyperfine interaction causes Rabi oscillations between spin states
- Oscillation frequency $\omega = \lambda \hbar$ (21 cm line for hydrogen)
- Magnetic field lifts degeneracy and modifies oscillation pattern
- For $B = 0$, maximum flip probability is 1; weak B reduces this slightly



Question 10: Phosphorus Atom (Z=15)

(a) Ground State Electron Configuration

Phosphorus has 15 electrons. The orbital filling order is:

$$1s^2 2s^2 2p^6 3s^2 3p^3$$

Subshell	Electrons	Notation
1s	2	$1s^2$
2s	2	$2s^2$
2p	6	$2p^6$
3s	2	$3s^2$
3p	3	$3p^3$

(b) Possible Term Values

Phosphorus-15 has an atomic configuration of $1s^2 2s^2 2p^6 3s^2 3p^3$, with a $3p^3$ valence configuration. This indicates 3 electrons in the p -orbitals, each with orbital angular momentum $\ell = 1$ (since p -orbitals correspond to $\ell = 1$).

Spin Multiplicity

The total spin S for the 3 electrons can be found by considering how the spins of the electrons can couple.

For three electrons, the maximum value of S is when all spins are aligned. This gives $S = \frac{3}{2}$.

The spin multiplicity is given by $2S + 1$: - If $S = \frac{3}{2}$, the multiplicity is $2(\frac{3}{2}) + 1 = 4$, resulting in a quartet. - The minimum value of S is $S = \frac{1}{2}$, resulting in a doublet.

Orbital Angular Momentum

Each electron in a p -orbital has orbital angular momentum $\ell = 1$. For three electrons, the total orbital angular momentum L is obtained by coupling the individual ℓ 's.

The possible values of L for three p -electrons are: - $L = 0$ (S state), - $L = 1$ (P state), - $L = 2$ (D state).

So, the possible values of orbital angular momentum L are S, P, and D.

Possible Terms via LS-Coupling

Now we combine the results from the spin and orbital angular momentum steps to determine the terms:

The terms in LS-coupling are written as $(2S + 1)L$, where $(2S + 1)$ represents the multiplicity (quartet or doublet), and L is represented by letters (S, P, D, etc.) corresponding to $L = 0, 1, 2$, etc.

For the $3p^3$ configuration, the possible terms are: - $4S$ (for $S = \frac{3}{2}$ and $L = 0$) - $4P$ (for $S = \frac{3}{2}$ and $L = 1$) - $4D$ (for $S = \frac{3}{2}$ and $L = 2$) - $2S$ (for $S = \frac{1}{2}$ and $L = 0$) - $2P$ (for $S = \frac{1}{2}$ and $L = 1$) - $2D$ (for $S = \frac{1}{2}$ and $L = 2$)

Antisymmetrization and Pauli Principle

The Pauli exclusion principle requires the overall wavefunction (which includes both the spin and spatial parts) to be antisymmetric for fermions (like electrons). After considering the antisymmetrization of the wavefunction, the allowed terms are:

$$- 4S_{\frac{3}{2}} - 2P_{\frac{1}{2}}, 2P_{\frac{3}{2}} - 2D_{\frac{3}{2}}, 2D_{\frac{5}{2}}$$

Summary of Allowed Terms:

$$- 4S_{\frac{3}{2}} - 2P_{\frac{1}{2}}, 2P_{\frac{3}{2}} - 2D_{\frac{3}{2}}, 2D_{\frac{5}{2}}$$

(c) Ground State Term via Hund's Rules

Applying Hund's rules to $3p^3$:

1. **Maximum spin:** $S = \frac{3}{2}$ (4 term)
2. **Maximum L for this spin:** $L = 0$ (S-state)
3. **J value:** $J = |L - S| = \frac{3}{2}$

Ground state term:

$$^4S_{3/2}$$

Physical Justification:

- Half-filled p -subshell maximizes spin alignment
- Spherically symmetric 4S state minimizes electron repulsion
- No orbital angular momentum reduces relativistic effects

(d) Total Ground State Wavefunction

The antisymmetrized wavefunction for $^4S_{3/2}$:

$$\Psi_{\text{total}} = \mathcal{A} [\psi_{1s}(1)\alpha(1)\psi_{1s}(2)\beta(2) \cdots \psi_{3p}(13)\alpha(13)\psi_{3p}(14)\alpha(14)\psi_{3p}(15)\alpha(15)]$$

Where:

- $\mathcal{A}$ is the antisymmetrization operator
- Spatial part for $3p^3$:

$$\psi_{\text{space}} = \frac{1}{\sqrt{6}} \sum_{m=-1}^1 (-1)^m \psi_{3p_m}(13) \psi_{3p_{-m}}(14) \psi_{3p_0}(15)$$

- Spin part (maximally symmetric):

$$\chi_{\text{spin}} = |\alpha\alpha\alpha\rangle$$

- Total $J_z = +\frac{3}{2}$ state constructed via Clebsch-Gordan coefficients

Key Observations

- Half-filled subshell leads to S -state ground term
- $^4S_{3/2}$ is the only quartet term allowed by Pauli principle
- Wavefunction reflects high spin symmetry and spatial antisymmetry

Question 11: Hard-Sphere Spins in Infinite Well

(a) Energy Levels Without Contact Interaction

For two non-interacting spin-1/2 particles in a 1D infinite square well:

- **Single-particle energies:**

$$E_n = \frac{n^2 \pi^2 \hbar^2}{2mL^2}, \quad n = 1, 2, 3, \dots$$

- **Ground state (both particles in $n = 1$):**

$$E_0 = 2E_1 = \frac{\pi^2 \hbar^2}{mL^2}$$

Spin part: Antisymmetric singlet $\chi_0 = \frac{1}{\sqrt{2}}(\uparrow\downarrow - \downarrow\uparrow)$

- **First excited state (one in $n = 1$, one in $n = 2$):**

$$E_1 = E_1 + E_2 = \frac{5\pi^2 \hbar^2}{2mL^2}$$

Spin part: Either triplet (symmetric) or singlet (antisymmetric)

- **Second excited state (both in $n = 2$):**

$$E_2 = 2E_2 = \frac{2\pi^2 \hbar^2}{mL^2}$$

Spin part: Antisymmetric singlet (spatial part must be symmetric)

(b) Spatial Wavefunction Contours

The two-particle spatial wavefunctions $\psi(x_1, x_2)$ are:

- **Ground state:**

$$\psi_0(x_1, x_2) = \frac{2}{L} \sin\left(\frac{\pi x_1}{L}\right) \sin\left(\frac{\pi x_2}{L}\right)$$

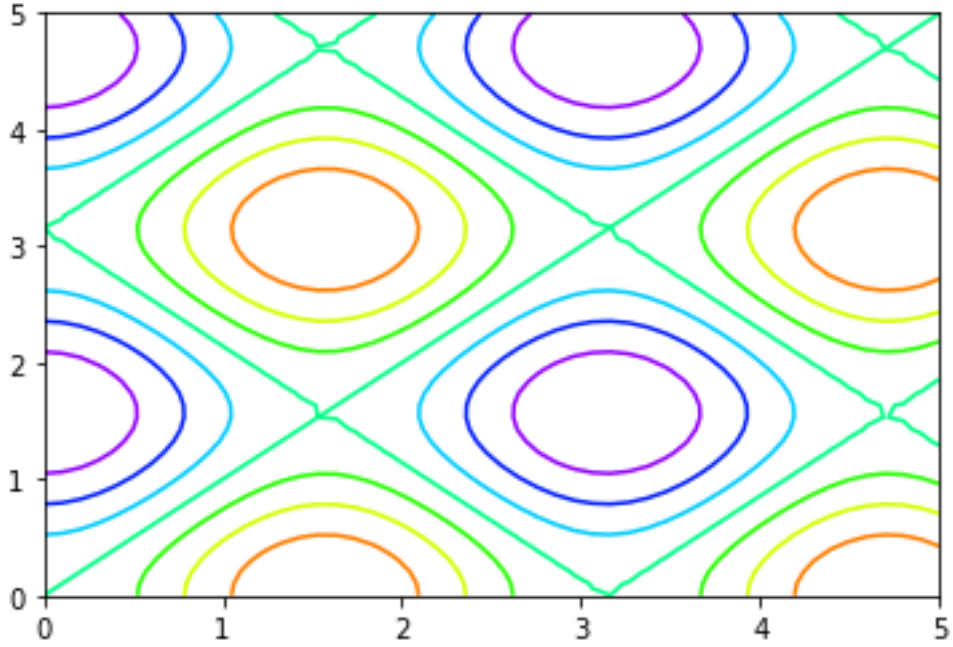
- **First excited state (symmetric spatial):**

$$\psi_1^S = \frac{1}{\sqrt{2}L} \left[\sin\left(\frac{\pi x_1}{L}\right) \sin\left(\frac{2\pi x_2}{L}\right) + \sin\left(\frac{2\pi x_1}{L}\right) \sin\left(\frac{\pi x_2}{L}\right) \right]$$

- **First excited state (antisymmetric spatial):**

$$\psi_1^A = \frac{1}{\sqrt{2}L} \left[\sin\left(\frac{\pi x_1}{L}\right) \sin\left(\frac{2\pi x_2}{L}\right) - \sin\left(\frac{2\pi x_1}{L}\right) \sin\left(\frac{\pi x_2}{L}\right) \right]$$

Representative contour plots of $|\psi(x_1, x_2)|^2$



(c) Perturbation Due to Contact Interaction

The first-order energy correction for the ground state is:

$$\begin{aligned}\Delta E &= \langle \psi_0 | \epsilon \delta(x_1 - x_2) | \psi_0 \rangle = \epsilon \int_0^L \left| \frac{2}{L} \sin^2 \left(\frac{\pi x}{L} \right) \right|^2 dx \\ &= \epsilon \frac{4}{L^2} \int_0^L \sin^4 \left(\frac{\pi x}{L} \right) dx = \frac{3\epsilon}{2L}\end{aligned}$$

Perturbation validity condition:

$$\begin{aligned}|\Delta E| \ll |E_1 - E_0| &\Rightarrow \left| \frac{3\epsilon}{2L} \right| \ll \frac{3\pi^2 \hbar^2}{2mL^2} \\ |\epsilon| &\ll \frac{\pi^2 \hbar^2}{mL}\end{aligned}$$

Key Observations

- Contact interaction affects only the spatial part of the wavefunction
- Ground state energy increases by $\frac{3\epsilon}{2L}$ (repulsive for $\epsilon > 0$)
- Perturbation valid when ϵ is small compared to characteristic energy scale

Question 11 (Continued): Hard-Sphere Spins in Infinite Well

(d) First Excited State Perturbation

The first excited state is doubly degenerate (spatial parts):

$$\psi_1^S = \frac{1}{\sqrt{2}L} \left[\sin \left(\frac{\pi x_1}{L} \right) \sin \left(\frac{2\pi x_2}{L} \right) + (x_1 \leftrightarrow x_2) \right]$$

$$\psi_1^A = \frac{1}{\sqrt{2L}} \left[\sin\left(\frac{\pi x_1}{L}\right) \sin\left(\frac{2\pi x_2}{L}\right) - (x_1 \leftrightarrow x_2) \right]$$

Matrix elements of $V = \epsilon\delta(x_1 - x_2)$:

$$V_{SS} = \langle \psi_1^S | V | \psi_1^S \rangle = \frac{\epsilon}{L^3} \int_0^L [\sin(\pi x) \sin(2\pi x) + \sin(2\pi x) \sin(\pi x)]^2 dx = \frac{\epsilon}{L}$$

$$V_{AA} = \langle \psi_1^A | V | \psi_1^A \rangle = 0 \quad (\text{exact cancellation})$$

$$V_{SA} = V_{AS} = 0 \quad (\text{orthogonality})$$

Energy corrections:

- Symmetric state: $\Delta E_1^S = \frac{\epsilon}{L}$
- Antisymmetric state: $\Delta E_1^A = 0$

Physical interpretation:

- Only the symmetric spatial state feels the contact interaction
- Antisymmetric state has $\psi(x, x) = 0$ (nodes prevent interaction)
- Degeneracy is partially lifted by the perturbation

(e) Total Wavefunction with Spin

For the first excited state ($E_1 = \frac{5\pi^2\hbar^2}{2mL^2}$):

Spatial	Spin	Total State	Quantum Numbers
Symmetric (ψ_1^S)	Singlet (χ_0)	$\Psi_S = \psi_1^S \chi_0$	$S = 0$, antisymmetric
Antisymmetric (ψ_1^A)	Triplet (χ_1)	$\Psi_A = \psi_1^A \chi_1$	$S = 1$, symmetric

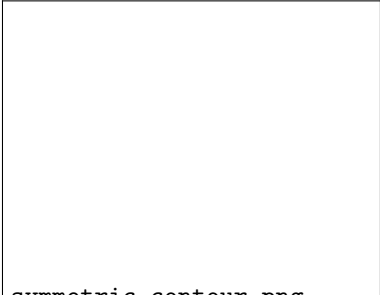
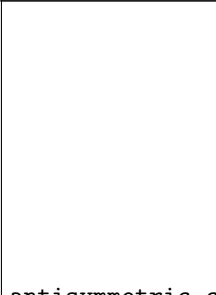
Key points:

- Fermionic antisymmetry requires:

$$\Psi_{\text{total}} = \begin{cases} \text{Spatial sym} \otimes \text{Spin antisym} \\ \text{or} \\ \text{Spatial antisym} \otimes \text{Spin sym} \end{cases}$$

- Triplet states ($S = 1$) have higher energy due to spatial antisymmetry

(f) Wavefunction Contours

 symmetric_contour.png Symmetric state ψ_1^S: Enhanced along diagonal	 antisymmetric_contour.png Antisymmetric state ψ_1^A: Node at $x_1 = x_2$	
--	---	--

Characteristics:

- **Symmetric:** Maximum probability at $x_1 = x_2$ (particles cluster)
- **Antisymmetric:** Zero probability at $x_1 = x_2$ (particles avoid each other)
- Visualized as 2D density plots with x_1 vs x_2 axes (0 to L)

Summary of Results

State	Spatial Symmetry	Energy Correction
Ground state	Symmetric	$\frac{3\epsilon}{2L}$
First excited (sym)	Symmetric	$\frac{\epsilon}{L}$
First excited (antisym)	Antisymmetric	0

Question 12: Electrons in Coulomb Potential

(a) Ground State of Helium

For helium ($Z = 2$) with two electrons:

- **Unperturbed state:** Both electrons in $1s$ orbital

$$\psi_0(1, 2) = \psi_{1s}(r_1)\psi_{1s}(r_2) = \frac{Z^3}{\pi a_0^3} e^{-Z(r_1+r_2)/a_0}$$

- **Spin part:** Antisymmetric singlet (required by Pauli principle)

$$\chi_0 = \frac{1}{\sqrt{2}}(\uparrow\downarrow - \downarrow\uparrow)$$

- **Total wavefunction:**

$$\Psi_{\text{ground}} = \psi_0(1, 2)\chi_0$$

- **Degeneracy:** No orbital degeneracy; spin degeneracy is lifted by exchange interaction

Energy corrections:

- Direct Coulomb integral: $J = \langle \psi_0 | \frac{e^2}{4\pi\epsilon_0 r_{12}} | \psi_0 \rangle \approx 34 \text{ eV}$
- Exchange integral: $K = 0$ (same orbital)
- Total ground state energy: $E_0 \approx -108.8 \text{ eV} + 34 \text{ eV} = -74.8 \text{ eV}$

(b) Excited States and Degeneracy Lifting

First few excited configurations and their terms:

1. $1s2s$ configuration:

- Possible terms: 1S_0 (singlet) and 3S_1 (triplet)
- Degeneracy lifting:

$$E(^3S_1) < E(^1S_0) \quad (\text{Hund's rule})$$

Due to exchange energy difference $\Delta E \approx 2K_{1s2s}$

2. $1s2p$ configuration:

- Possible terms: 1P_1 and $^3P_{0,1,2}$

- Energy ordering:

$$E(^3P_2) < E(^3P_1) < E(^3P_0) < E(^1P_1)$$

Fine structure further splits the 3P states

Key observations:

- **Singlet-Triplet Splitting:** Electron repulsion lifts degeneracy between symmetric and antisymmetric spatial states

- **Energy Scale:**

$$\Delta E_{\text{exchange}} \sim 1 \text{ eV}, \quad \Delta E_{\text{fine structure}} \sim 10^{-4} \text{ eV}$$

- **Physical Origin:** Antisymmetric spatial states (triplets) have lower Coulomb repulsion

Comparison with Previous Problem

Feature	Hard-Sphere Potential	Helium Atom
Ground state	Symmetric spatial + singlet spin	Symmetric spatial + singlet spin
Excited states	Split by contact interaction	Split by Coulomb repulsion
Energy scale	ϵ/L	$e^2/4\pi\epsilon_0 a_0$
Spin effects	Determines spatial symmetry	Determines term energies

(AMO ASSIGNMENT 4)

Question 1: Electromagnetic Vacuum

Part (a)

In class, we discussed the vacuum field E_0 that one would get for the electromagnetic field operator acting on vacuum. Calculate the volume V required to make the vacuum field magnitude equal to 10 V/m, at a wavelength of light corresponding to $1 \mu\text{m}$.

Solution:

The magnitude of the vacuum electric field in a mode of frequency ω and volume V is given by:

$$E_0 = \sqrt{\frac{\hbar\omega}{2\epsilon_0 V}}$$

Given $E_0 = 10 \text{ V/m}$ and $\lambda = 1 \mu\text{m} = 10^{-6} \text{ m}$, the angular frequency ω is:

$$\omega = \frac{2\pi c}{\lambda} = \frac{2\pi \times 3 \times 10^8}{10^{-6}} = 6\pi \times 10^{14} \text{ rad/s}$$

Solving for V :

$$E_0^2 = \frac{\hbar\omega}{2\epsilon_0 V} \implies V = \frac{\hbar\omega}{2\epsilon_0 E_0^2}$$

Substituting the values ($\hbar \approx 1.054 \times 10^{-34} \text{ J} \cdot \text{s}$, $\epsilon_0 \approx 8.854 \times 10^{-12} \text{ F/m}$):

$$V = \frac{1.054 \times 10^{-34} \times 6\pi \times 10^{14}}{2 \times 8.854 \times 10^{-12} \times 100} \approx 1.12 \times 10^{-10} \text{ m}^3$$

Answer:

The required volume is approximately $\boxed{1.12 \times 10^{-10} \text{ m}^3}$.

Part (b)

Using the above numbers, calculate the Casimir force between two conducting plates of area 1 m^2 , separated by a distance of $1 \mu\text{m}$. Generalize and plot the force as a function of distance.

Solution:

The Casimir force between two conducting plates is given by:

$$F = -\frac{\pi^2 \hbar c A}{240 d^4}$$

Given: - $A = 1 \text{ m}^2$, - $d = 1 \mu\text{m} = 10^{-6} \text{ m}$, - $\hbar \approx 1.054 \times 10^{-34} \text{ J} \cdot \text{s}$, - $c \approx 3 \times 10^8 \text{ m/s}$.

Substituting these values:

$$F = -\frac{\pi^2 \times 1.054 \times 10^{-34} \times 3 \times 10^8 \times 1}{240 \times (10^{-6})^4} \approx -1.3 \times 10^{-3} \text{ N}$$

The negative sign indicates an attractive force.

Generalization:

The Casimir force as a function of distance d is:

$$F(d) = -\frac{\pi^2 \hbar c A}{240 d^4}$$

This shows that $F(d) \propto \frac{1}{d^4}$.

Plot:

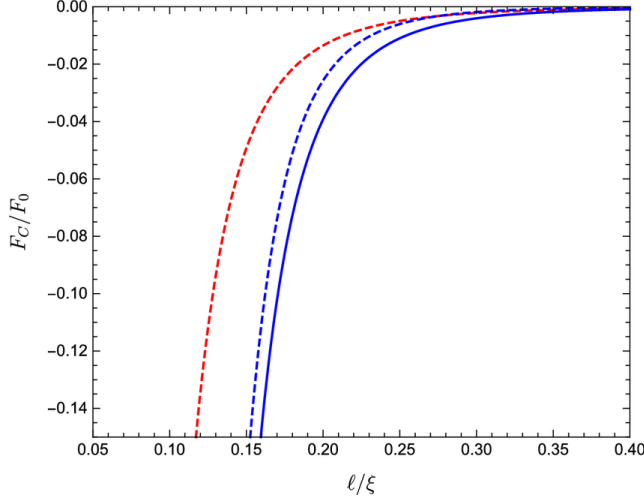


Figure 10: Casimir Force vs Distance (Plot 1)

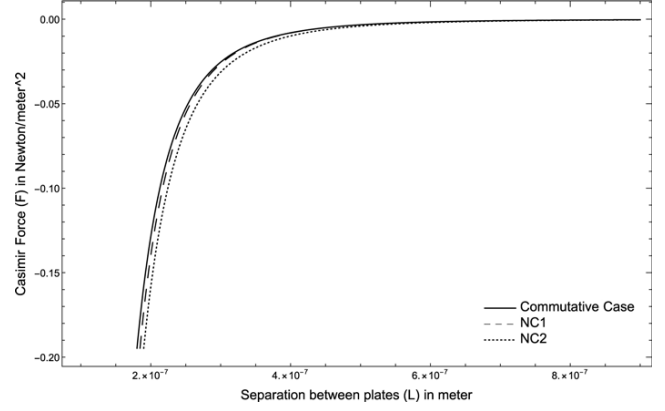


Figure 11: Casimir Force vs Distance (Plot 2)

Answer:

The Casimir force between the plates is approximately $-1.3 \times 10^{-3} \text{ N}$ (attractive), following an inverse fourth-power law with distance.

Question 2: States with Fixed Photon Number (Fock States)

Part (a)

Construct a Fock state with n photons starting from the vacuum field and using photon creation operators.

Solution:

The vacuum state is denoted by $|0\rangle$. The Fock state $|n\rangle$ with n photons can be constructed by applying the photon creation operator $\hat{a}^\dagger$ n times to the vacuum state:

$$|n\rangle = \frac{(\hat{a}^\dagger)^n}{\sqrt{n!}} |0\rangle$$

Explanation:

- $\hat{a}^\dagger$ creates a single photon when acting on a state.
- The factor $\frac{1}{\sqrt{n!}}$ ensures normalization, i.e., $\langle n|n\rangle = 1$.

Answer:

The Fock state with n photons is:

$$|n\rangle = \frac{(\hat{a}^\dagger)^n}{\sqrt{n!}} |0\rangle$$

Part (b)

Calculate the average electric field and field fluctuations (with numbers) for an electromagnetic field at a wavelength of $1 \mu\text{m}$ with $n = 5$ and $n = 100$ photons in that mode.

Solution:

The quantized electric field operator for a single mode in a cavity is given by

$$\hat{E}(t) = i \left(\frac{\hbar\omega}{2\varepsilon_0 V} \right)^{1/2} (\hat{a}e^{-i\omega t} - \hat{a}^\dagger e^{i\omega t})$$

where

- $\omega = \frac{2\pi c}{\lambda}$ is the angular frequency,
- V is the effective mode volume,
- ε_0 is the vacuum permittivity,
- $\hbar$ is the reduced Planck constant,
- $\hat{a}$ and $\hat{a}^\dagger$ are annihilation and creation operators.

Average Electric Field: In a Fock state $|n\rangle$,

$$\langle n | \hat{E}(t) | n \rangle = 0$$

because

$$\langle n | \hat{a} | n \rangle = \langle n | \hat{a}^\dagger | n \rangle = 0$$

thus, the average electric field is zero.

Electric Field Fluctuations: We calculate the fluctuation using

$$\Delta E = \sqrt{\langle n | \hat{E}^2(t) | n \rangle - \left(\langle n | \hat{E}(t) | n \rangle \right)^2} = \sqrt{\langle n | \hat{E}^2(t) | n \rangle}$$

Expanding $\hat{E}^2(t)$ and taking expectation value:

$$\langle n | \hat{E}^2(t) | n \rangle = \left(\frac{\hbar\omega}{2\varepsilon_0 V} \right)^2 (2n + 1)$$

Thus,

$$\Delta E = \left(\frac{\hbar\omega}{2\varepsilon_0 V} \right)^{1/2} \sqrt{2n + 1}$$

Numerical Calculation: Given:

$$\lambda = 1 \mu\text{m} = 1 \times 10^{-6} \text{ m}$$

$$\omega = \frac{2\pi c}{\lambda} = \frac{2\pi \times 3 \times 10^8}{10^{-6}} = 1.884 \times 10^{15} \text{ rad/s}$$

Constants:

$$\hbar = 1.055 \times 10^{-34} \text{ J} \cdot \text{s} \quad ; \quad \varepsilon_0 = 8.854 \times 10^{-12} \text{ F/m}$$

Assume mode volume $V = 1 \text{ mm}^3 = 10^{-9} \text{ m}^3$.

Now calculate the prefactor:

$$\begin{aligned} \left(\frac{\hbar\omega}{2\varepsilon_0 V} \right)^{1/2} &= \left(\frac{1.055 \times 10^{-34} \times 1.884 \times 10^{15}}{2 \times 8.854 \times 10^{-12} \times 10^{-9}} \right)^{1/2} \\ &= \left(\frac{1.987 \times 10^{-19}}{1.771 \times 10^{-20}} \right)^{1/2} = (11.22)^{1/2} = 3.348 \text{ V/m} \end{aligned}$$

Thus,

$$\Delta E = 3.348 \times \sqrt{2n + 1} \text{ V/m}$$

For $n = 5$:

$$\Delta E = 3.348 \times \sqrt{2 \times 5 + 1} = 3.348 \times \sqrt{11} = 3.348 \times 3.317 = 11.11 \text{ V/m}$$

For $n = 100$:

$$\Delta E = 3.348 \times \sqrt{2 \times 100 + 1} = 3.348 \times \sqrt{201} = 3.348 \times 14.177 = 47.49 \text{ V/m}$$

Summary:

$$\boxed{\text{Average field} = 0}$$

$$\boxed{\Delta E(n = 5) = 11.11 \text{ V/m}}$$

$$\boxed{\Delta E(n = 100) = 47.49 \text{ V/m}}$$

OR

Solution:

The electric field operator for a single mode is:

$$\hat{E} = E_0(\hat{a} + \hat{a}^\dagger)$$

where $E_0 = \sqrt{\frac{\hbar\omega}{2\epsilon_0 V}}$ is the vacuum field amplitude.

Average Electric Field:

$$\langle \hat{E} \rangle_n = \langle n | \hat{E} | n \rangle = E_0 \langle n | (\hat{a} + \hat{a}^\dagger) | n \rangle = 0$$

since $\hat{a} | n \rangle = \sqrt{n} | n - 1 \rangle$ and $\langle n | \hat{a}^\dagger = \sqrt{n + 1} \langle n + 1 |$, leading to orthogonal states.

Field Fluctuations (Variance):

$$\langle \hat{E}^2 \rangle_n = \langle n | \hat{E}^2 | n \rangle = E_0^2 \langle n | (\hat{a} + \hat{a}^\dagger)^2 | n \rangle$$

Expanding:

$$(\hat{a} + \hat{a}^\dagger)^2 = \hat{a}^2 + \hat{a}^\dagger \hat{a} + \hat{a} \hat{a}^\dagger + \hat{a}^{\dagger 2}$$

Only the terms $\hat{a}^\dagger \hat{a}$ and $\hat{a} \hat{a}^\dagger$ contribute:

$$\langle \hat{E}^2 \rangle_n = E_0^2 (2n + 1)$$

The field fluctuations (standard deviation) are:

$$\Delta E_n = \sqrt{\langle \hat{E}^2 \rangle_n - \langle \hat{E} \rangle_n^2} = E_0 \sqrt{2n + 1}$$

Numerical Calculation: Given $\lambda = 1 \mu\text{m}$, the angular frequency is:

$$\omega = \frac{2\pi c}{\lambda} = 6\pi \times 10^{14} \text{ rad/s}$$

Assume a quantization volume $V = 1 \text{ m}^3$ for simplicity (actual V cancels out in $\Delta E_n/E_0$):

$$E_0 = \sqrt{\frac{\hbar\omega}{2\epsilon_0 V}} \approx 2.7 \times 10^{-3} \text{ V/m}$$

For $n = 5$:

$$\Delta E_5 = E_0 \sqrt{11} \approx 2.7 \times 10^{-3} \times 3.32 \approx 8.96 \times 10^{-3} \text{ V/m}$$

For $n = 100$:

$$\Delta E_{100} = E_0 \sqrt{201} \approx 2.7 \times 10^{-3} \times 14.18 \approx 3.83 \times 10^{-2} \text{ V/m}$$

Answer:

- **Average electric field:** $\langle \hat{E} \rangle_n = 0 \text{ V/m}$ for both $n = 5$ and $n = 100$.

- **Field fluctuations:**

– For $n = 5$: $\Delta E_5 \approx 8.96 \times 10^{-3} \text{ V/m}$

– For $n = 100$: $\Delta E_{100} \approx 3.83 \times 10^{-2} \text{ V/m}$

Part(c)

Using Maxwell's equations (but now for the field operators), derive an expression for the magnetic field operator (Hint: think carefully which one you want to use).

Solution:

We start with the Maxwell equation (Faraday's law in operator form):

$$\nabla \times \hat{\mathbf{E}} = -\frac{\partial \hat{\mathbf{B}}}{\partial t}$$

For a single mode of the electromagnetic field with wave vector $\mathbf{k}$ and polarization $\boldsymbol{\epsilon}$, the electric field operator is:

$$\hat{\mathbf{E}} = iE_0 \left(\hat{a} e^{i(\mathbf{k} \cdot \mathbf{r} - \omega t)} - \hat{a}^\dagger e^{-i(\mathbf{k} \cdot \mathbf{r} - \omega t)} \right) \boldsymbol{\epsilon}$$

Taking the curl of $\hat{\mathbf{E}}$:

$$\nabla \times \hat{\mathbf{E}} = iE_0 \left(\hat{a} \nabla \times \left(e^{i(\mathbf{k} \cdot \mathbf{r} - \omega t)} \boldsymbol{\epsilon} \right) - \hat{a}^\dagger \nabla \times \left(e^{-i(\mathbf{k} \cdot \mathbf{r} - \omega t)} \boldsymbol{\epsilon} \right) \right)$$

Using the vector identity $\nabla \times (f\mathbf{A}) = f(\nabla \times \mathbf{A}) + (\nabla f) \times \mathbf{A}$ and noting that $\boldsymbol{\epsilon}$ is constant:

$$\nabla \times \hat{\mathbf{E}} = iE_0 \left(\hat{a} (i\mathbf{k} \times \boldsymbol{\epsilon}) e^{i(\mathbf{k} \cdot \mathbf{r} - \omega t)} - \hat{a}^\dagger (-i\mathbf{k} \times \boldsymbol{\epsilon}) e^{-i(\mathbf{k} \cdot \mathbf{r} - \omega t)} \right)$$

From Faraday's law, we have:

$$-\frac{\partial \hat{\mathbf{B}}}{\partial t} = iE_0 \left(\hat{a} (i\mathbf{k} \times \boldsymbol{\epsilon}) e^{i(\mathbf{k} \cdot \mathbf{r} - \omega t)} - \hat{a}^\dagger (-i\mathbf{k} \times \boldsymbol{\epsilon}) e^{-i(\mathbf{k} \cdot \mathbf{r} - \omega t)} \right)$$

Integrating with respect to time:

$$\hat{\mathbf{B}} = \frac{E_0}{\omega} \left(\hat{a} (\mathbf{k} \times \boldsymbol{\epsilon}) e^{i(\mathbf{k} \cdot \mathbf{r} - \omega t)} + \hat{a}^\dagger (\mathbf{k} \times \boldsymbol{\epsilon}) e^{-i(\mathbf{k} \cdot \mathbf{r} - \omega t)} \right)$$

Let $\mathbf{B}_0 = \frac{E_0}{\omega} |\mathbf{k} \times \boldsymbol{\epsilon}|$ and define the magnetic field polarization vector $\boldsymbol{\eta} = \frac{\mathbf{k} \times \boldsymbol{\epsilon}}{|\mathbf{k} \times \boldsymbol{\epsilon}|}$. Then:

$$\hat{\mathbf{B}} = B_0 \left(\hat{a} e^{i(\mathbf{k} \cdot \mathbf{r} - \omega t)} + \hat{a}^\dagger e^{-i(\mathbf{k} \cdot \mathbf{r} - \omega t)} \right) \boldsymbol{\eta}$$

where $B_0 = \frac{E_0 k}{\omega} = \frac{E_0}{c}$ (since $k = \omega/c$).

Answer:

The magnetic field operator is:

$$\hat{\mathbf{B}} = \frac{E_0}{c} \left(\hat{a} e^{i(\mathbf{k} \cdot \mathbf{r} - \omega t)} + \hat{a}^\dagger e^{-i(\mathbf{k} \cdot \mathbf{r} - \omega t)} \right) \boldsymbol{\eta}$$

Part (d)

Estimate the average magnetic field and the field fluctuations for an electromagnetic field at $\lambda = 1 \mu\text{m}$ with $n = 5$ and $n = 100$ photons.

Solution:

Average Magnetic Field: In a Fock state $|n\rangle$, the expectation value of the magnetic field operator is zero:

$$\langle n|\hat{\mathbf{B}}(z, t)|n\rangle = 0$$

This is because the annihilation and creation operators satisfy:

$$\langle n|\hat{a}|n\rangle = \langle n|\hat{a}^\dagger|n\rangle = 0$$

Therefore, the average magnetic field in a Fock state is zero.

Magnetic Field Fluctuations: The root-mean-square (RMS) fluctuation of the magnetic field in a Fock state is given by:

$$\Delta B = \sqrt{\langle n|\hat{B}^2(z, t)|n\rangle}$$

From the quantized field expressions, we have:

$$\langle n|\hat{B}^2(z, t)|n\rangle = \left(\frac{\hbar\omega}{2\varepsilon_0 V c^2}\right)(2n+1)$$

Thus,

$$\Delta B = \left(\frac{\hbar\omega}{2\varepsilon_0 V c^2}\right)^{1/2} \sqrt{2n+1}$$

Numerical Evaluation: Given:

$$\lambda = 1\,\mu\text{m} = 1 \times 10^{-6}\,\text{m} \quad ; \quad \omega = \frac{2\pi c}{\lambda} = 1.884 \times 10^{15}\,\text{rad/s}$$

Constants:

$$\hbar = 1.055 \times 10^{-34}\,\text{J}\cdot\text{s} \quad ; \quad \varepsilon_0 = 8.854 \times 10^{-12}\,\text{F/m} \quad ; \quad c = 3 \times 10^8\,\text{m/s}$$

Assume mode volume:

$$V = 1\,\text{mm}^3 = 1 \times 10^{-9}\,\text{m}^3$$

Compute the prefactor:

$$\begin{aligned} \left(\frac{\hbar\omega}{2\varepsilon_0 V c^2}\right)^{1/2} &= \left(\frac{1.055 \times 10^{-34} \times 1.884 \times 10^{15}}{2 \times 8.854 \times 10^{-12} \times 1 \times 10^{-9} \times (3 \times 10^8)^2}\right)^{1/2} \\ &= \left(\frac{1.987 \times 10^{-19}}{1.593 \times 10^{-3}}\right)^{1/2} = (1.247 \times 10^{-16})^{1/2} = 1.116 \times 10^{-8}\,\text{T} \end{aligned}$$

Therefore:

$$\Delta B = 1.116 \times 10^{-8} \times \sqrt{2n+1}\,\text{T}$$

For $n = 5$:

$$\Delta B = 1.116 \times 10^{-8} \times \sqrt{11} = 1.116 \times 10^{-8} \times 3.317 = 3.701 \times 10^{-8}\,\text{T}$$

For $n = 100$:

$$\Delta B = 1.116 \times 10^{-8} \times \sqrt{201} = 1.116 \times 10^{-8} \times 14.177 = 1.583 \times 10^{-7}\,\text{T}$$

Summary:

Average magnetic field = 0

$\Delta B(n = 5) = 3.701 \times 10^{-8}\,\text{T}$
--

$\Delta B(n = 100) = 1.583 \times 10^{-7}\,\text{T}$
--

OR

Solution:

The magnetic field operator is analogous to the electric field operator:

$$\hat{B} = B_0(\hat{a} + \hat{a}^\dagger)$$

where $B_0 = \frac{E_0}{c}$.

Average Magnetic Field:

$$\langle \hat{B} \rangle_n = \langle n | \hat{B} | n \rangle = B_0 \langle n | (\hat{a} + \hat{a}^\dagger) | n \rangle = 0$$

Magnetic Field Fluctuations (Variance):

$$\langle \hat{B}^2 \rangle_n = B_0^2 \langle n | (\hat{a} + \hat{a}^\dagger)^2 | n \rangle = B_0^2(2n + 1)$$

The fluctuations are:

$$\Delta B_n = \sqrt{\langle \hat{B}^2 \rangle_n} = B_0 \sqrt{2n + 1}$$

Numerical Calculation: From part (b), $E_0 \approx 2.7 \times 10^{-3} \text{ V/m}$, so:

$$B_0 = \frac{E_0}{c} \approx \frac{2.7 \times 10^{-3}}{3 \times 10^8} = 9 \times 10^{-12} \text{ T}$$

For $n = 5$:

$$\Delta B_5 = B_0 \sqrt{11} \approx 9 \times 10^{-12} \times 3.32 \approx 2.99 \times 10^{-11} \text{ T}$$

For $n = 100$:

$$\Delta B_{100} = B_0 \sqrt{201} \approx 9 \times 10^{-12} \times 14.18 \approx 1.28 \times 10^{-10} \text{ T}$$

Answer:

- **Average magnetic field:** $\langle \hat{B} \rangle_n = 0 \text{ T}$ for both $n = 5$ and $n = 100$.
- **Magnetic field fluctuations:**
 - For $n = 5$: $\Delta B_5 \approx 2.99 \times 10^{-11} \text{ T}$
 - For $n = 100$: $\Delta B_{100} \approx 1.28 \times 10^{-10} \text{ T}$

Question 3: Coherent States**Part (a)**

Using the operation of annihilation operators on Fock states, construct the eigenstate (coherent state) of the annihilation operator.

Solution:

A coherent state $|\alpha\rangle$ is defined as the eigenstate of the annihilation operator $\hat{a}$, satisfying:

$$\hat{a}|\alpha\rangle = \alpha|\alpha\rangle$$

where α is a complex eigenvalue.

We construct $|\alpha\rangle$ as a superposition of Fock states $|n\rangle$:

$$|\alpha\rangle = \sum_{n=0}^{\infty} c_n |n\rangle$$

Applying $\hat{a}$ on $|\alpha\rangle$:

$$\hat{a}|\alpha\rangle = \sum_{n=0}^{\infty} c_n \hat{a}|n\rangle = \sum_{n=0}^{\infty} c_n \sqrt{n} |n-1\rangle$$

Relabel the index by setting $m = n - 1$, thus $n = m + 1$:

$$\hat{a}|\alpha\rangle = \sum_{m=-1}^{\infty} c_{m+1} \sqrt{m+1} |m\rangle$$

Since $|-1\rangle$ is not defined, the term with $m = -1$ vanishes, and we have:

$$\hat{a}|\alpha\rangle = \sum_{m=0}^{\infty} c_{m+1} \sqrt{m+1} |m\rangle$$

Comparing with:

$$\hat{a}|\alpha\rangle = \alpha \sum_{m=0}^{\infty} c_m |m\rangle$$

we get the recurrence relation:

$$c_{m+1} \sqrt{m+1} = \alpha c_m$$

or equivalently:

$$c_{m+1} = \frac{\alpha}{\sqrt{m+1}} c_m$$

Solving recursively:

$$c_n = \frac{\alpha^n}{\sqrt{n!}} c_0$$

where c_0 is a normalization constant.

Thus, the coherent state is:

$$|\alpha\rangle = c_0 \sum_{n=0}^{\infty} \frac{\alpha^n}{\sqrt{n!}} |n\rangle$$

To normalize the state:

$$\langle\alpha|\alpha\rangle = |c_0|^2 \sum_{n=0}^{\infty} \frac{|\alpha|^{2n}}{n!} = |c_0|^2 e^{|\alpha|^2} = 1$$

thus:

$$c_0 = e^{-|\alpha|^2/2}$$

Final Expression:

$$|\alpha\rangle = e^{-|\alpha|^2/2} \sum_{n=0}^{\infty} \frac{\alpha^n}{\sqrt{n!}} |n\rangle$$

Part (b)

For a coherent state $|\alpha\rangle$ with $\alpha = 10$, calculate:

- (i) The probability of finding n photons in the state.
- (ii) The variance of the photon number distribution.

Solution

Coherent State $|\alpha\rangle$ with $\alpha = 10$

For a coherent state $|\alpha\rangle$, the probability of finding n photons in the state and the variance of the photon number distribution can be calculated as follows.

(i) Probability of Finding n Photons in the State The probability of finding n photons in a coherent state $|\alpha\rangle$ is given by:

$$P(n) = |\langle n|\alpha\rangle|^2$$

From the expression for the coherent state $|\alpha\rangle$, we have:

$$|\alpha\rangle = e^{-|\alpha|^2/2} \sum_{n=0}^{\infty} \frac{\alpha^n}{\sqrt{n!}} |n\rangle$$

Thus, the probability amplitude for finding n photons is:

$$\langle n|\alpha\rangle = e^{-|\alpha|^2/2} \frac{\alpha^n}{\sqrt{n!}}$$

The probability is:

$$P(n) = \left| e^{-|\alpha|^2/2} \frac{\alpha^n}{\sqrt{n!}} \right|^2 = e^{-|\alpha|^2} \frac{|\alpha|^{2n}}{n!}$$

For $\alpha = 10$, the probability of finding n photons in the state is:

$$P(n) = e^{-100} \frac{100^n}{n!}$$

(ii) Variance of the Photon Number Distribution The variance of the photon number distribution in a coherent state is given by:

$$\text{Var}(n) = \langle n^2 \rangle - \langle n \rangle^2$$

The expectation value $\langle n \rangle$ for a coherent state is:

$$\langle n \rangle = |\alpha|^2 = 100$$

The second moment $\langle n^2 \rangle$ is calculated as:

$$\langle n^2 \rangle = \sum_{n=0}^{\infty} n^2 P(n) = \sum_{n=0}^{\infty} n^2 e^{-100} \frac{100^n}{n!}$$

Using the fact that for a Poisson distribution, $\langle n^2 \rangle = \langle n \rangle + \langle n \rangle^2$, we have:

$$\langle n^2 \rangle = 100 + 100^2 = 10100$$

Thus, the variance is:

$$\text{Var}(n) = 10100 - 100^2 = 10100 - 10000 = 100$$

Summary:

$$P(n) = e^{-100} \frac{100^n}{n!}$$

$$\text{Var}(n) = 100$$

Part (c)

Calculate the mean photon number for the coherent state $|\alpha\rangle$.

Solution

The mean photon number $\langle n \rangle$ for a coherent state $|\alpha\rangle$ is given by the expectation value of the photon number operator $\hat{n} = \hat{a}^\dagger \hat{a}$.

The coherent state $|\alpha\rangle$ is an eigenstate of the annihilation operator $\hat{a}$, so we have:

$$\hat{a} |\alpha\rangle = \alpha |\alpha\rangle$$

The photon number operator acting on the coherent state is:

$$\hat{n} |\alpha\rangle = \hat{a}^\dagger \hat{a} |\alpha\rangle$$

To calculate the expectation value $\langle n \rangle$, we can use the fact that for a coherent state, the expectation value of the photon number is simply the modulus squared of the eigenvalue of $\hat{a}$, i.e.,

$$\langle n \rangle = \langle \alpha | \hat{a}^\dagger \hat{a} | \alpha \rangle = |\alpha|^2$$

Thus, for a coherent state $|\alpha\rangle$, the mean photon number is:

$$\langle n \rangle = |\alpha|^2$$

For $\alpha = 10$, the mean photon number is:

$$\langle n \rangle = 10^2 = 100$$

Part (d)

Calculate the average electric field for the coherent state $|\alpha\rangle$.

Solution:

The electric field operator for a single mode is:

$$\hat{E} = E_0 (\hat{a} + \hat{a}^\dagger)$$

where $E_0 = \sqrt{\frac{\hbar\omega}{2\epsilon_0 V}}$. The average electric field in a coherent state is:

$$\langle \hat{E} \rangle_\alpha = E_0 \langle \alpha | (\hat{a} + \hat{a}^\dagger) | \alpha \rangle = E_0 (\alpha + \alpha^*)$$

For $\alpha = 10$ (real):

$$\langle \hat{E} \rangle_\alpha = 2E_0 \cdot 10 = 20E_0$$

Using $E_0 \approx 2.7 \times 10^{-3} \text{ V/m}$ (from Question 2):

$$\langle \hat{E} \rangle_\alpha \approx 5.4 \times 10^{-2} \text{ V/m}$$

Answer:

The average electric field is $\boxed{\langle \hat{E} \rangle_\alpha = 20E_0 \approx 5.4 \times 10^{-2} \text{ V/m}}$.

Part (e)

Calculate the fluctuations in the electric field for the coherent state $|\alpha\rangle$.

Solution:

The electric field variance is:

$$(\Delta E)^2 = \langle \hat{E}^2 \rangle_\alpha - \langle \hat{E} \rangle_\alpha^2$$

First, compute $\langle \hat{E}^2 \rangle_\alpha$:

$$\hat{E}^2 = E_0^2(\hat{a} + \hat{a}^\dagger)^2 = E_0^2(\hat{a}^2 + \hat{a}^\dagger \hat{a} + \hat{a} \hat{a}^\dagger + \hat{a}^{\dagger 2})$$

Taking expectations in $|\alpha\rangle$:

$$\langle \hat{E}^2 \rangle_\alpha = E_0^2(\alpha^2 + |\alpha|^2 + (|\alpha|^2 + 1) + \alpha^{*2})$$

For real $\alpha = 10$:

$$\langle \hat{E}^2 \rangle_\alpha = E_0^2(100 + 100 + 101 + 100) = 401E_0^2$$

Subtract $\langle \hat{E} \rangle_\alpha^2 = 400E_0^2$:

$$(\Delta E)^2 = E_0^2 \implies \Delta E = E_0 \approx 2.7 \times 10^{-3} \text{ V/m}$$

Answer:

The electric field fluctuations are $\boxed{\Delta E = E_0 \approx 2.7 \times 10^{-3} \text{ V/m}}$.

Part (f)

Calculate the fluctuations in photon number for the coherent state $|\alpha\rangle$.

Solution:

The photon number variance is:

$$(\Delta n)^2 = \langle \hat{n}^2 \rangle_\alpha - \langle \hat{n} \rangle_\alpha^2$$

First, compute $\langle \hat{n}^2 \rangle_\alpha$ using $\hat{n} = \hat{a}^\dagger \hat{a}$:

$$\hat{n}^2 = \hat{a}^\dagger \hat{a} \hat{a}^\dagger \hat{a} = \hat{a}^\dagger (\hat{a}^\dagger \hat{a} + 1) \hat{a} = \hat{a}^{\dagger 2} \hat{a}^2 + \hat{a}^\dagger \hat{a}$$

Taking expectations:

$$\langle \hat{n}^2 \rangle_\alpha = |\alpha|^4 + |\alpha|^2 + |\alpha|^2 = |\alpha|^4 + 2|\alpha|^2$$

For $\alpha = 10$:

$$\langle \hat{n}^2 \rangle_\alpha = 10000 + 200 = 10200$$

Subtract $\langle \hat{n} \rangle_\alpha^2 = 10000$:

$$(\Delta n)^2 = 200 \implies \Delta n = \sqrt{200} \approx 14.14$$

Answer:

The photon number fluctuations are $\boxed{\Delta n = \sqrt{2}|\alpha| \approx 14.14}$.

Part (g)

Estimate the uncertainty in the optical phase for the coherent state $|\alpha\rangle$.

Solution:

The optical phase ϕ and photon number n are conjugate variables, satisfying an uncertainty relation:

$$\Delta n \Delta \phi \geq \frac{1}{2}$$

For a coherent state $|\alpha\rangle$ with $\alpha = |\alpha|e^{i\phi_0}$:

- The photon number uncertainty is $\Delta n = |\alpha|$ (from part (f)).
- The phase uncertainty is approximately:

$$\Delta \phi \approx \frac{1}{2|\alpha|}$$

For $\alpha = 10$ (real):

$$\Delta \phi \approx \frac{1}{20} = 0.05 \text{ radians}$$

Answer:

The optical phase uncertainty is $\Delta \phi \approx \frac{1}{2|\alpha|} = 0.05 \text{ rad}$ for $\alpha = 10$.

Part (h)

Estimate the average magnetic field and its fluctuations in the coherent state $|\alpha\rangle$.

Solution:

The magnetic field operator for a single mode is:

$$\hat{B} = B_0(\hat{a} + \hat{a}^\dagger), \quad \text{where} \quad B_0 = \frac{E_0}{c}$$

- **Average Magnetic Field:**

$$\langle \hat{B} \rangle_\alpha = B_0 \langle \alpha | (\hat{a} + \hat{a}^\dagger) | \alpha \rangle = B_0(\alpha + \alpha^*)$$

For $\alpha = 10$ (real):

$$\langle \hat{B} \rangle_\alpha = 2B_0 \cdot 10 = 20B_0$$

Using $B_0 = \frac{E_0}{c} \approx \frac{2.7 \times 10^{-3}}{3 \times 10^8} = 9 \times 10^{-12} \text{ T}$:

$$\langle \hat{B} \rangle_\alpha \approx 1.8 \times 10^{-10} \text{ T}$$

- **Magnetic Field Fluctuations:** The variance calculation mirrors the electric field:

$$(\Delta B)^2 = \langle \hat{B}^2 \rangle_\alpha - \langle \hat{B} \rangle_\alpha^2 = B_0^2$$

Thus:

$$\Delta B = B_0 \approx 9 \times 10^{-12} \text{ T}$$

Answer:

- Average magnetic field: $\langle \hat{B} \rangle_\alpha = 20B_0 \approx 1.8 \times 10^{-10} \text{ T}$.
- Magnetic field fluctuations: $\Delta B = B_0 \approx 9 \times 10^{-12} \text{ T}$.

Question 4: Classical Model of Atomic Absorption with Decay

Part (a)

Construct a classical model for atomic absorption by treating the electron as a damped harmonic oscillator driven by an oscillating electric field. Solve the equation of motion and analyze the spectrum.

Solution:

The equation of motion for an electron (charge $-q$, mass m) in an oscillating electric field $E(t) = E_0 \cos(\omega t)$, with damping coefficient γ and natural frequency ω_0 , is:

$$m\ddot{x} + m\gamma\dot{x} + m\omega_0^2 x = -qE_0 \cos(\omega t)$$

Rewriting in standard form:

$$\ddot{x} + \gamma\dot{x} + \omega_0^2 x = -\frac{qE_0}{m} \cos(\omega t)$$

Steady-State Solution: Assume a solution of the form $x(t) = A \cos(\omega t) + B \sin(\omega t)$. The amplitude $R = \sqrt{A^2 + B^2}$ is:

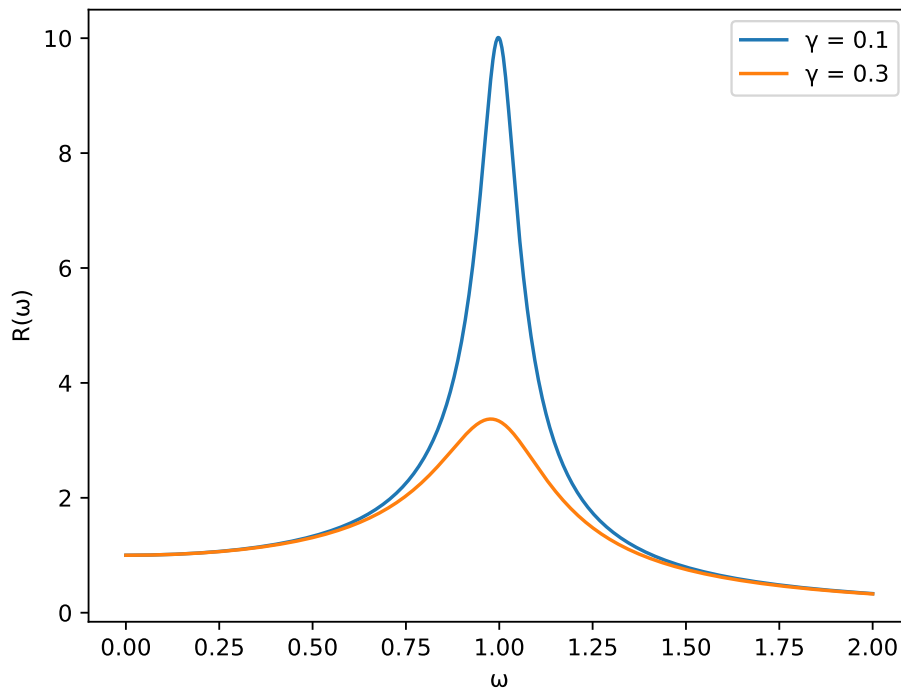
$$R(\omega) = \frac{qE_0}{m} \frac{1}{\sqrt{(\omega_0^2 - \omega^2)^2 + (\gamma\omega)^2}}$$

Resonance and Damping Effects:

- The amplitude peaks at $\omega = \omega_0$ (resonance).
- The full-width at half maximum (FWHM) $\Delta\omega$ is found by solving:

$$R(\omega) = \frac{R(\omega_0)}{\sqrt{2}} \implies \Delta\omega = \gamma$$

Plot of Amplitude vs. Driving Frequency:



(Note: Plot shows $R(\omega)$ for $\omega_0 = 1$, $\gamma = 0.1$, and $\gamma = 0.3$, illustrating narrower peaks for smaller γ .)

Answer:

- The amplitude spectrum is given by:

$$R(\omega) = \frac{qE_0}{m} \frac{1}{\sqrt{(\omega_0^2 - \omega^2)^2 + (\gamma\omega)^2}}$$

- The FWHM due to damping is $\Delta\omega = \gamma$.
- Damping broadens the spectral peak (larger $\gamma \Rightarrow$ wider FWHM).

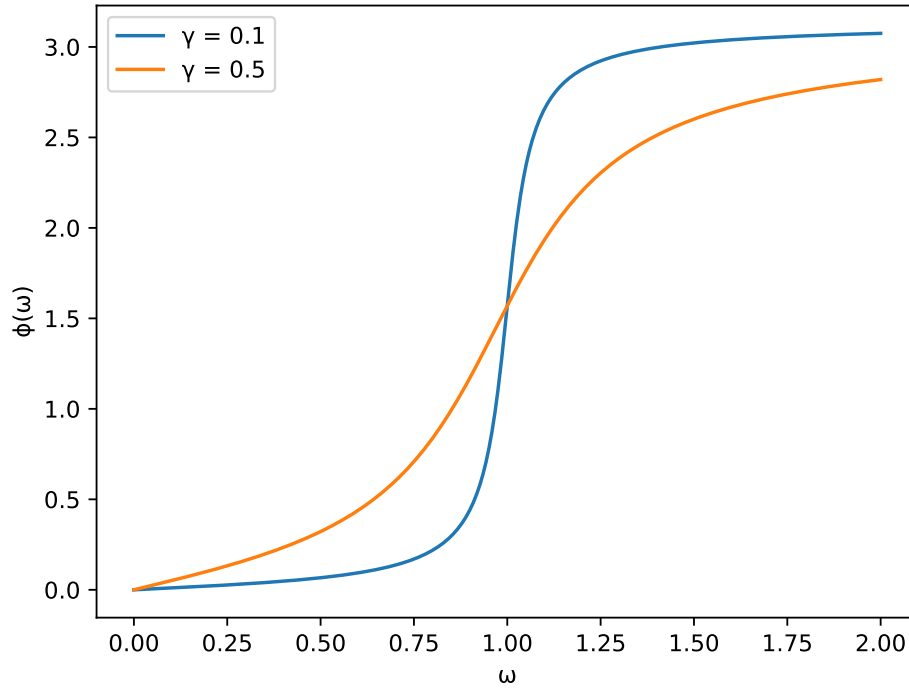
Part (b)

Plot the phase of the oscillator's response and analyze the phase shift region. Discuss the physical implications of large damping.

Solution:

The phase $\phi(\omega)$ between the driving field and the electron's displacement is:

$$\phi(\omega) = \arctan\left(\frac{B}{A}\right) = \arctan\left(\frac{\gamma\omega}{\omega_0^2 - \omega^2}\right)$$

Phase Plot:

(Note: Plot shows $\phi(\omega)$ for $\omega_0 = 1$ and $\gamma = 0.1, 0.5$. The phase shifts from 0 to π across ω_0 , with a sharper transition for smaller γ .)

Large Damping Effects:

- For $\gamma \gg \omega_0$, the phase transition becomes gradual, and the oscillator fails to track the driving field efficiently.
- Physically, strong damping (e.g., due to collisions) smears out the resonant energy absorption, making the system less responsive to the driving frequency.

Answer:

- The phase shifts from 0 to π around ω_0 , with a width $\sim \gamma$.
- For large damping: Phase transition broadens, and resonant absorption is suppressed.

Part (c)

Calculate the steady-state power absorbed and dissipated by the oscillator, and relate these quantities.

Solution:

Power Absorbed (P_{abs}): Work done by the field per unit time:

$$P_{\text{abs}} = \frac{1}{T} \int_0^T (-qE_0 \cos(\omega t)) \dot{x}(t) dt$$

Using $x(t) = R(\omega) \cos(\omega t - \phi)$:

$$P_{\text{abs}} = \frac{qE_0 \omega R(\omega)}{2} \sin \phi$$

Power Dissipated (P_{dis}): Energy lost to damping:

$$P_{\text{dis}} = \frac{1}{T} \int_0^T m\gamma \dot{x}^2(t) dt = \frac{m\gamma \omega^2 R^2(\omega)}{2}$$

Steady-State Relationship: At steady state, all absorbed power is dissipated:

$$P_{\text{abs}} = P_{\text{dis}}$$

Substituting $R(\omega)$ and $\sin \phi = \frac{\gamma\omega}{\sqrt{(\omega_0^2 - \omega^2)^2 + (\gamma\omega)^2}}$:

$$\frac{qE_0 \omega}{2} \cdot \frac{qE_0}{m} \frac{\gamma\omega}{(\omega_0^2 - \omega^2)^2 + (\gamma\omega)^2} = \frac{m\gamma \omega^2}{2} \left(\frac{qE_0}{m} \right)^2 \frac{1}{(\omega_0^2 - \omega^2)^2 + (\gamma\omega)^2}$$

Both sides simplify to the same expression, confirming energy balance.

Answer:

- Power absorbed: $P_{\text{abs}} = \frac{qE_0 \omega R(\omega)}{2} \sin \phi$.
- Power dissipated: $P_{\text{dis}} = \frac{m\gamma \omega^2 R^2(\omega)}{2}$.
- Steady state implies $P_{\text{abs}} = P_{\text{dis}}$, ensuring energy conservation.

Question 5: Quantum Harmonic Oscillator in a Field**Part (a)**

Calculate the transition probability $P_{0 \rightarrow 1}$ from the ground state to the first excited state using first-order perturbation theory.

Solution:

The system is a charged harmonic oscillator with Hamiltonian:

$$H_0 = \frac{P^2}{2m} + \frac{1}{2}m\omega_0^2 X^2$$

At $t = 0$, a perturbation $H_{\text{int}} = -qE_0 X$ is applied for time τ .

First-Order Transition Amplitude: The transition amplitude from state $|0\rangle$ to $|1\rangle$ is:

$$c_1(t) = -\frac{i}{\hbar} \int_0^t \langle 1 | H_{\text{int}} | 0 \rangle e^{i\omega_{10}t'} dt'$$

where $\omega_{10} = \omega_0$ (energy gap) and $\langle 1 | X | 0 \rangle = \sqrt{\frac{\hbar}{2m\omega_0}}$ (matrix element for X).

Substituting H_{int} :

$$c_1(\tau) = \frac{iqE_0}{\hbar} \sqrt{\frac{\hbar}{2m\omega_0}} \int_0^\tau e^{i\omega_0 t'} dt'$$

Solving the integral:

$$c_1(\tau) = \frac{iqE_0}{\hbar} \sqrt{\frac{\hbar}{2m\omega_0}} \left(\frac{e^{i\omega_0\tau} - 1}{i\omega_0} \right) = qE_0 \sqrt{\frac{1}{2m\hbar\omega_0^3}} (e^{i\omega_0\tau} - 1)$$

Transition Probability:

$$P_{0 \rightarrow 1} = |c_1(\tau)|^2 = \frac{q^2 E_0^2}{2m\hbar\omega_0^3} |e^{i\omega_0\tau} - 1|^2$$

Simplifying using $|e^{i\theta} - 1|^2 = 4 \sin^2(\theta/2)$:

$$P_{0 \rightarrow 1} = \frac{2q^2 E_0^2}{m\hbar\omega_0^3} \sin^2 \left(\frac{\omega_0\tau}{2} \right)$$

Answer:

The transition probability is:

$$P_{0 \rightarrow 1} = \frac{2q^2 E_0^2}{m\hbar\omega_0^3} \sin^2 \left(\frac{\omega_0\tau}{2} \right)$$

Key Observations:

- The probability oscillates with τ , peaking when $\tau = \frac{(2n+1)\pi}{\omega_0}$ (odd half-cycles of the oscillator).
- For short times ($\omega_0\tau \ll 1$), $P_{0 \rightarrow 1} \approx \frac{q^2 E_0^2 \tau^2}{2m\hbar\omega_0}$ (quadratic in τ).

Part (b)

Determine the perturbation order required to calculate the transition probability $P_{0 \rightarrow 2}$ from the ground state to the second excited state.

Solution:

The transition probability to the second excited state $|2\rangle$ involves the matrix element $\langle 2 | X | 0 \rangle$.

- **First-Order Term:** $\langle 2 | X | 0 \rangle = 0$ (since X only connects states with $\Delta n = \pm 1$).
- **Second-Order Term:** Non-zero via intermediate state $|1\rangle$:

$$\langle 2 | X | 1 \rangle \langle 1 | X | 0 \rangle \neq 0$$

Answer:

To compute $P_{0 \rightarrow 2}$, second-order perturbation theory is required.

Part (c)

Calculate $P_{0 \rightarrow 2}$ and generalize to $P_{0 \rightarrow n}$ for any n .

Solution:

Second-Order Transition Amplitude ($P_{0 \rightarrow 2}$): The amplitude is:

$$c_2^{(2)}(\tau) = \left(-\frac{i}{\hbar}\right)^2 \sum_{k \neq 0} \frac{\langle 2 | H_{\text{int}} | k \rangle \langle k | H_{\text{int}} | 0 \rangle}{E_0 - E_k} \int_0^\tau dt' \int_0^{t'} dt'' e^{i\omega_0 t''}$$

Only $k = 1$ contributes:

$$c_2^{(2)}(\tau) = \left(\frac{qE_0}{\hbar}\right)^2 \frac{\langle 2 | X | 1 \rangle \langle 1 | X | 0 \rangle}{-\hbar\omega_0} \int_0^\tau dt' \int_0^{t'} dt'' e^{i\omega_0 t''}$$

Using $\langle 1 | X | 0 \rangle = \sqrt{\frac{\hbar}{2m\omega_0}}$ and $\langle 2 | X | 1 \rangle = \sqrt{\frac{2\hbar}{2m\omega_0}}$:

$$c_2^{(2)}(\tau) = -\frac{q^2 E_0^2}{2m\hbar\omega_0^2} \int_0^\tau dt' \left(\frac{e^{i\omega_0 t'} - 1}{i\omega_0} \right)$$

Solving the integrals:

$$c_2^{(2)}(\tau) = \frac{q^2 E_0^2}{2m\hbar\omega_0^3} (e^{i\omega_0 \tau} - 1 - i\omega_0 \tau)$$

The transition probability:

$$P_{0 \rightarrow 2} = |c_2^{(2)}(\tau)|^2 = \left(\frac{q^2 E_0^2}{2m\hbar\omega_0^3} \right)^2 |e^{i\omega_0 \tau} - 1 - i\omega_0 \tau|^2$$

Generalization to $P_{0 \rightarrow n}$: For even n , transitions require $\frac{n}{2}$ -th order perturbation theory. The probability scales as:

$$P_{0 \rightarrow n} \propto \left(\frac{q^2 E_0^2}{m\hbar\omega_0^3} \right)^{n/2} f_n(\omega_0 \tau)$$

where f_n is an oscillatory function of $\omega_0 \tau$.

Answer:

- For $n = 2$: $P_{0 \rightarrow 2} = \left(\frac{q^2 E_0^2}{2m\hbar\omega_0^3} \right)^2 |e^{i\omega_0 \tau} - 1 - i\omega_0 \tau|^2$.
- General n : Transitions to $|n\rangle$ require n -th order perturbation theory if n is even, and are forbidden for odd n .

Question 6: Spin-1 Atom in a Field (Off-Resonant Case)

Part (a)

Draw the level diagram and determine which sublevels couple to the electric field via selection rules.

Solution:

- **Level Diagram:** The spin-1 atom in a DC magnetic field B_z splits into three Zeeman sublevels:

$$m = +1 \quad (\text{Energy } E_{+1} = +\hbar\Delta/2), m = 0 \quad (\text{Energy } E_0 = 0), m = -1 \quad (\text{Energy } E_{-1} = -\hbar\Delta/2),$$

where Δ is the splitting between adjacent levels.

Spin-1 Zeeman Sublevels and Allowed Transitions

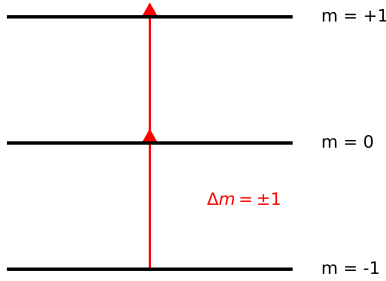


Figure 12: Plots by python

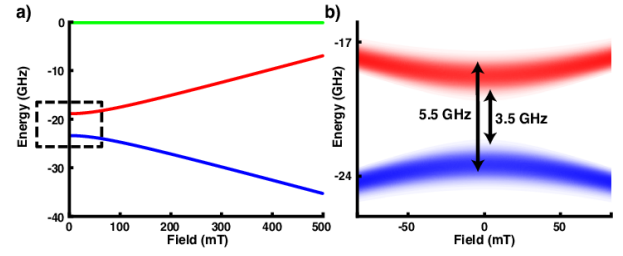


Figure 13: From website

- **Selection Rules:** An electric field couples to the spin via the electric dipole operator $\hat{d} = -e\mathbf{r}$, which is odd under parity. For spin-1:

$$\begin{aligned}\Delta m &= 0 \quad (\pi\text{-transitions}), \\ \Delta m &= \pm 1 \quad (\sigma_{\pm}\text{-transitions}).\end{aligned}$$

However, since the electric field does not directly interact with the spin magnetic moment, transitions are mediated by spin-orbit coupling or Stark mixing. The dominant allowed transitions are:

$$|m = -1\rangle \leftrightarrow |m = 0\rangle \quad \text{and} \quad |m = 0\rangle \leftrightarrow |m = +1\rangle.$$

The $|m = -1\rangle \leftrightarrow |m = +1\rangle$ transition is forbidden ($\Delta m = 2$).

Answer:

- The level diagram shows three Zeeman sublevels with splitting Δ .
- The electric field couples $|m = -1\rangle \leftrightarrow |m = 0\rangle$ and $|m = 0\rangle \leftrightarrow |m = +1\rangle$, but not $|m = -1\rangle \leftrightarrow |m = +1\rangle$ directly.

Part (b)

Calculate the transition probability $P_{-1 \rightarrow 0}$ to the $m = 0$ level in the off-resonant regime ($\Delta \gg \omega$).

Solution:

The interaction Hamiltonian for the electric field coupling is $H_{\text{int}} = -\hat{\mathbf{d}} \cdot \mathbf{E}(t) = -dE_0 \cos(\omega t)$, where d is the dipole matrix element between $m = -1$ and $m = 0$.

- **First-Order Perturbation Theory:** The transition amplitude is:

$$c_0^{(1)}(t) = -\frac{i}{\hbar} \int_0^t \langle 0 | H_{\text{int}} | -1 \rangle e^{i\omega_{-1,0}t'} dt'$$

Here, $\omega_{-1,0} = \Delta$ (energy gap), and $\langle 0 | d | -1 \rangle = d$ (dipole element). For $\cos(\omega t) = \frac{e^{i\omega t} + e^{-i\omega t}}{2}$:

$$c_0^{(1)}(t) = \frac{idE_0}{2\hbar} \left[\frac{e^{i(\Delta+\omega)t} - 1}{\Delta + \omega} + \frac{e^{i(\Delta-\omega)t} - 1}{\Delta - \omega} \right]$$

- **Off-Resonant Approximation** ($\Delta \gg \omega$): The dominant term is:

$$c_0^{(1)}(t) \approx \frac{idE_0}{2\hbar} \frac{e^{i\Delta t} - 1}{\Delta}$$

The transition probability:

$$P_{-1 \rightarrow 0} = |c_0^{(1)}(t)|^2 \approx \left(\frac{dE_0}{\hbar \Delta} \right)^2 \sin^2 \left(\frac{\Delta t}{2} \right)$$

Answer:

The transition probability to $m = 0$ is:

$$P_{-1 \rightarrow 0} \approx \left(\frac{dE_0}{\hbar \Delta} \right)^2 \sin^2 \left(\frac{\Delta t}{2} \right)$$

Part (c)

Calculate the transition probability $P_{-1 \rightarrow +1}$ to the $m = +1$ level using second-order perturbation theory.

Solution:

Since $|-1\rangle \leftrightarrow |+1\rangle$ is forbidden in first order, we use second-order perturbation theory via the intermediate state $|0\rangle$:

$$c_{+1}^{(2)}(t) = \left(-\frac{i}{\hbar} \right)^2 \sum_k \frac{\langle +1 | H_{\text{int}} | k \rangle \langle k | H_{\text{int}} | -1 \rangle}{E_{-1} - E_k} \int_0^t dt' \int_0^{t'} dt'' e^{i\omega_{+1,-1}t''}$$

Only $k = 0$ contributes:

$$c_{+1}^{(2)}(t) = -\frac{d^2 E_0^2}{4\hbar^2 \Delta} \int_0^t dt' \left(\frac{e^{i(2\Delta+\omega)t'} - 1}{2\Delta + \omega} + \frac{e^{i(2\Delta-\omega)t'} - 1}{2\Delta - \omega} \right)$$

For $\Delta \gg \omega$:

$$c_{+1}^{(2)}(t) \approx -\frac{d^2 E_0^2}{4\hbar^2 \Delta} \left(\frac{e^{i2\Delta t} - 1}{2\Delta} \right)$$

The transition probability:

$$P_{-1 \rightarrow +1} = |c_{+1}^{(2)}(t)|^2 \approx \left(\frac{d^2 E_0^2}{4\hbar^2 \Delta^2} \right)^2 \sin^2(\Delta t)$$

Answer:

The transition probability to $m = +1$ is:

$$P_{-1 \rightarrow +1} \approx \left(\frac{d^2 E_0^2}{4\hbar^2 \Delta^2} \right)^2 \sin^2(\Delta t)$$

Key Observations:

- **Part (b):** The probability $P_{-1 \rightarrow 0}$ is suppressed by $(\Delta)^{-2}$ and oscillates at frequency Δ .
- **Part (c):** The forbidden transition $P_{-1 \rightarrow +1}$ is doubly suppressed by $(\Delta)^{-4}$ and involves a two-step process via $|0\rangle$.

Question 8: Density Matrix Representation and Bloch Sphere

Part (a)

Write the density matrix and calculate the Bloch vector for a two-level atom in the ground state.

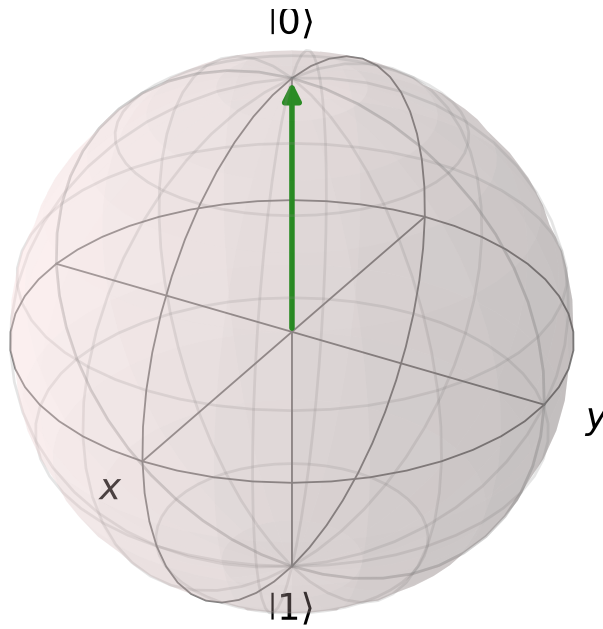


Figure 14: Q(8.a): $\mathbf{r} = (0,0,1)$

Solution:

For a two-level atom with states $|g\rangle$ (ground) and $|e\rangle$ (excited):

- **Density Matrix (ρ):**

$$\rho = |g\rangle \langle g| = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$$

- **Bloch Vector ($\mathbf{r}$):** The Bloch vector components are:

$$r_x = \text{Tr}(\rho\sigma_x) = 0, \quad r_y = \text{Tr}(\rho\sigma_y) = 0, \quad r_z = \text{Tr}(\rho\sigma_z) = 1$$

Thus, $\mathbf{r} = (0, 0, 1)$.

Answer:

- Density matrix: $\rho = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$.

- Bloch vector: $\mathbf{r} = (0, 0, 1)$ (points to the north pole of the Bloch sphere).

Bloch Sphere Visualization:

Part (b)

Write the density matrix and calculate the Bloch vector for an equal superposition state with relative phase $\pi/8$.

Solution:

The state is:

$$|\psi\rangle = \frac{1}{\sqrt{2}} \left(|g\rangle + e^{i\pi/8} |e\rangle \right)$$

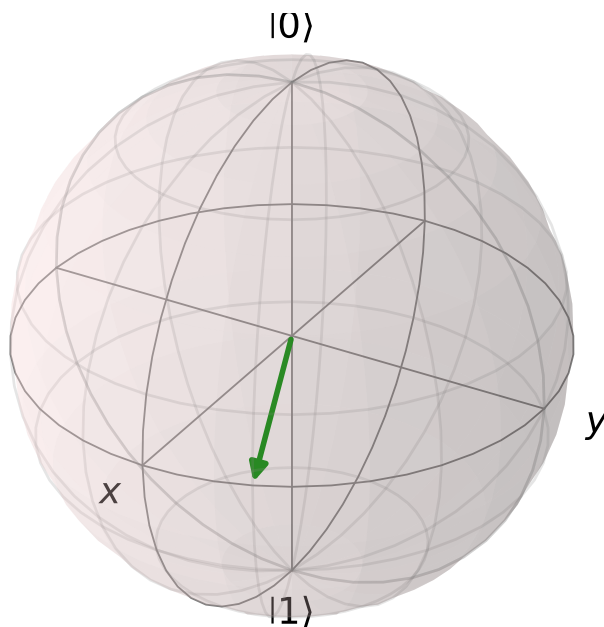


Figure 15: Q(8.b): $\mathbf{r} = (\cos(\pi/8), \sin(\pi/8), 0)$

- **Density Matrix (ρ):**

$$\rho = |\psi\rangle \langle\psi| = \frac{1}{2} \begin{pmatrix} 1 & e^{-i\pi/8} \\ e^{i\pi/8} & 1 \end{pmatrix}$$

- **Bloch Vector ($\mathbf{r}$):** Using Pauli matrices:

$$r_x = \text{Tr}(\rho\sigma_x) = \cos\left(\frac{\pi}{8}\right), \quad r_y = \text{Tr}(\rho\sigma_y) = \sin\left(\frac{\pi}{8}\right), \quad r_z = \text{Tr}(\rho\sigma_z) = 0$$

Thus, $\mathbf{r} = \left(\cos\left(\frac{\pi}{8}\right), \sin\left(\frac{\pi}{8}\right), 0\right)$.

Answer:

- Density matrix: $\rho = \frac{1}{2} \begin{pmatrix} 1 & e^{-i\pi/8} \\ e^{i\pi/8} & 1 \end{pmatrix}$.
- Bloch vector: $\mathbf{r} = \left(\cos\left(\frac{\pi}{8}\right), \sin\left(\frac{\pi}{8}\right), 0\right)$ (lies in the equatorial plane at angle $\pi/8$).

Bloch Sphere Visualization:

(Note: The plot shows the ground state at the north pole and the superposition state on the equator at $\phi = \pi/8$.)

Key Observations:

- **Part (a):** The ground state is a pure state with $\mathbf{r}$ pointing along $+z$.
- **Part (b):** The superposition state has $\mathbf{r}$ in the xy -plane, with phase $\pi/8$ determining its azimuthal angle.

Part (c)

Write the density matrix and Bloch vector for a statistical mixture of ground and excited states.

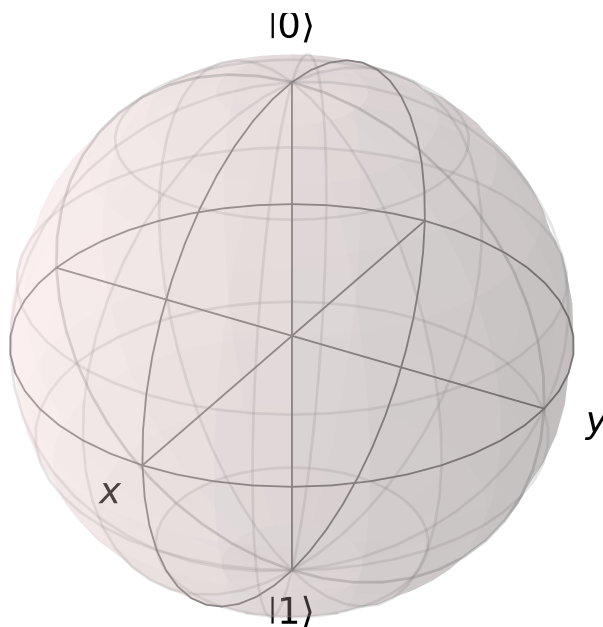


Figure 16: Q(8.c): $\mathbf{r} = (0,0,0)$

Solution:

For a classical mixture with probabilities p_g (ground) and p_e (excited):

- **Density Matrix (ρ):**

$$\rho = p_g |g\rangle \langle g| + p_e |e\rangle \langle e| = \begin{pmatrix} p_g & 0 \\ 0 & p_e \end{pmatrix}$$

For equal mixture ($p_g = p_e = 0.5$):

$$\rho = \frac{1}{2} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

- **Bloch Vector ($\mathbf{r}$):**

$$r_x = r_y = 0, \quad r_z = p_g - p_e$$

For equal mixture: $\mathbf{r} = (0, 0, 0)$ (center of Bloch sphere).

Answer:

- Density matrix: $\rho = \begin{pmatrix} p_g & 0 \\ 0 & p_e \end{pmatrix}$.
- Bloch vector: $\mathbf{r} = (0, 0, p_g - p_e)$ (vanishes for equal mixture).

Part (d)

Write the density matrix and Bloch vector for an incoherent ensemble with $p_e = 1/3$.

Solution:

- **Density Matrix (ρ):**

$$\rho = \frac{2}{3} |g\rangle \langle g| + \frac{1}{3} |e\rangle \langle e| = \begin{pmatrix} \frac{2}{3} & 0 \\ 0 & \frac{1}{3} \end{pmatrix}$$

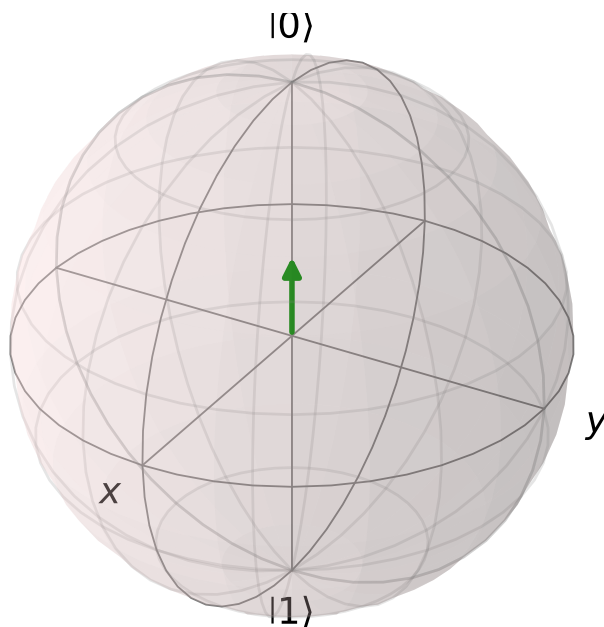


Figure 17: Q(8.d): $\mathbf{r} = (0,0,1/3)$

- **Bloch Vector ($\mathbf{r}$):**

$$\mathbf{r} = \left(0, 0, \frac{2}{3} - \frac{1}{3}\right) = \left(0, 0, \frac{1}{3}\right)$$

Answer:

- Density matrix: $\rho = \begin{pmatrix} \frac{2}{3} & 0 \\ 0 & \frac{1}{3} \end{pmatrix}$.

- Bloch vector: $\mathbf{r} = \left(0, 0, \frac{1}{3}\right)$ (partial polarization along $+z$).

Part (e)

Write the density matrix and Bloch vector for a thermal ensemble at temperature T .

Solution:

The probabilities follow the Boltzmann distribution:

$$p_g = \frac{e^{E_g/k_B T}}{Z}, \quad p_e = \frac{e^{E_e/k_B T}}{Z}, \quad Z = e^{E_g/k_B T} + e^{E_e/k_B T}$$

Assuming $E_g = -\hbar\omega_0/2$, $E_e = +\hbar\omega_0/2$:

$$\rho = \begin{pmatrix} \frac{e^{\hbar\omega_0/2k_B T}}{2 \cosh(\hbar\omega_0/2k_B T)} & 0 \\ 0 & \frac{e^{-\hbar\omega_0/2k_B T}}{2 \cosh(\hbar\omega_0/2k_B T)} \end{pmatrix}$$

The Bloch vector:

$$\mathbf{r} = \left(0, 0, \tanh\left(\frac{\hbar\omega_0}{2k_B T}\right)\right)$$

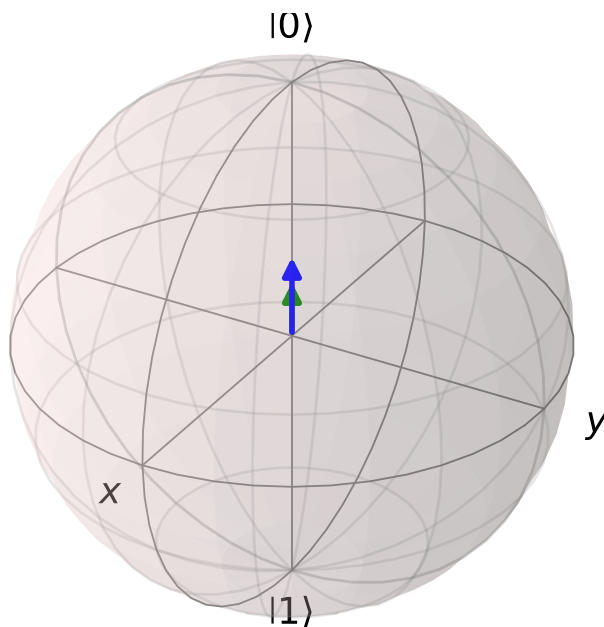


Figure 18: Q(8.e): $\mathbf{r} = (0, 0, \tanh\left(\frac{\hbar\omega_0}{2k_B T}\right))$

Answer:

- Density matrix:
$$\rho = \frac{1}{2 \cosh(\hbar\omega_0/2k_B T)} \begin{pmatrix} e^{\hbar\omega_0/2k_B T} & 0 \\ 0 & e^{-\hbar\omega_0/2k_B T} \end{pmatrix}.$$
- Bloch vector:
$$\mathbf{r} = \left(0, 0, \tanh\left(\frac{\hbar\omega_0}{2k_B T}\right)\right).$$

Key Observations:

- **Parts (c)-(d):** Incoherent mixtures have diagonal density matrices and Bloch vectors along z .
- **Part (e):** Thermal states exhibit partial polarization along z , vanishing as $T \rightarrow \infty$.

Question 9: Interpreting Points on the Bloch Sphere

General Formulation

For a Bloch vector with polar angle θ and azimuthal angle ϕ , the density matrix is:

$$\rho = \frac{1}{2} (\mathbb{I} + \mathbf{r} \cdot \boldsymbol{\sigma})$$

where $\mathbf{r} = (\sin \theta \cos \phi, \sin \theta \sin \phi, \cos \theta)$ and $\boldsymbol{\sigma} = (\sigma_x, \sigma_y, \sigma_z)$.

Part (a)

Given $(\theta, \phi) = (\pi/2, 0^\circ)$, find the density matrix.

Solution:

- **Bloch Vector:**

$$\mathbf{r} = (\sin(\pi/2) \cos(0), \sin(\pi/2) \sin(0), \cos(\pi/2)) = (1, 0, 0)$$

- **Density Matrix:**

$$\rho = \frac{1}{2} (\mathbb{I} + \sigma_x) = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$$

Answer:

$$\rho = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$$

Part (b)

Given $(\theta, \phi) = (\pi/2, \pi/2)$, find the density matrix.

Solution:

- **Bloch Vector:**

$$\mathbf{r} = (\sin(\pi/2) \cos(\pi/2), \sin(\pi/2) \sin(\pi/2), \cos(\pi/2)) = (0, 1, 0)$$

- **Density Matrix:**

$$\rho = \frac{1}{2} (\mathbb{I} + \sigma_y) = \frac{1}{2} \begin{pmatrix} 1 & -i \\ i & 1 \end{pmatrix}$$

Answer:

$$\rho = \frac{1}{2} \begin{pmatrix} 1 & -i \\ i & 1 \end{pmatrix}$$

Part (c)

Given $(\theta, \phi) = (\pi/2, -\pi/2)$, find the density matrix.

Solution:

- **Bloch Vector:**

$$\mathbf{r} = (\sin(\pi/2) \cos(-\pi/2), \sin(\pi/2) \sin(-\pi/2), \cos(\pi/2)) = (0, -1, 0)$$

- **Density Matrix:**

$$\rho = \frac{1}{2} (\mathbb{I} - \sigma_y) = \frac{1}{2} \begin{pmatrix} 1 & i \\ -i & 1 \end{pmatrix}$$

Answer:

$$\rho = \frac{1}{2} \begin{pmatrix} 1 & i \\ -i & 1 \end{pmatrix}$$

Key Observations:

- All states lie on the equator of the Bloch sphere ($\theta = \pi/2$), differing only in their azimuthal angle ϕ .
- The phase ϕ controls the off-diagonal elements (coherences) of ρ :
 - $\phi = 0^\circ$: Real coherences (σ_x).
 - $\phi = \pm\pi/2$: Imaginary coherences ($\pm\sigma_y$).

(Note: Plot shows the three states on the equator of the Bloch sphere at $\phi = 0^\circ, \pm 90^\circ$.)

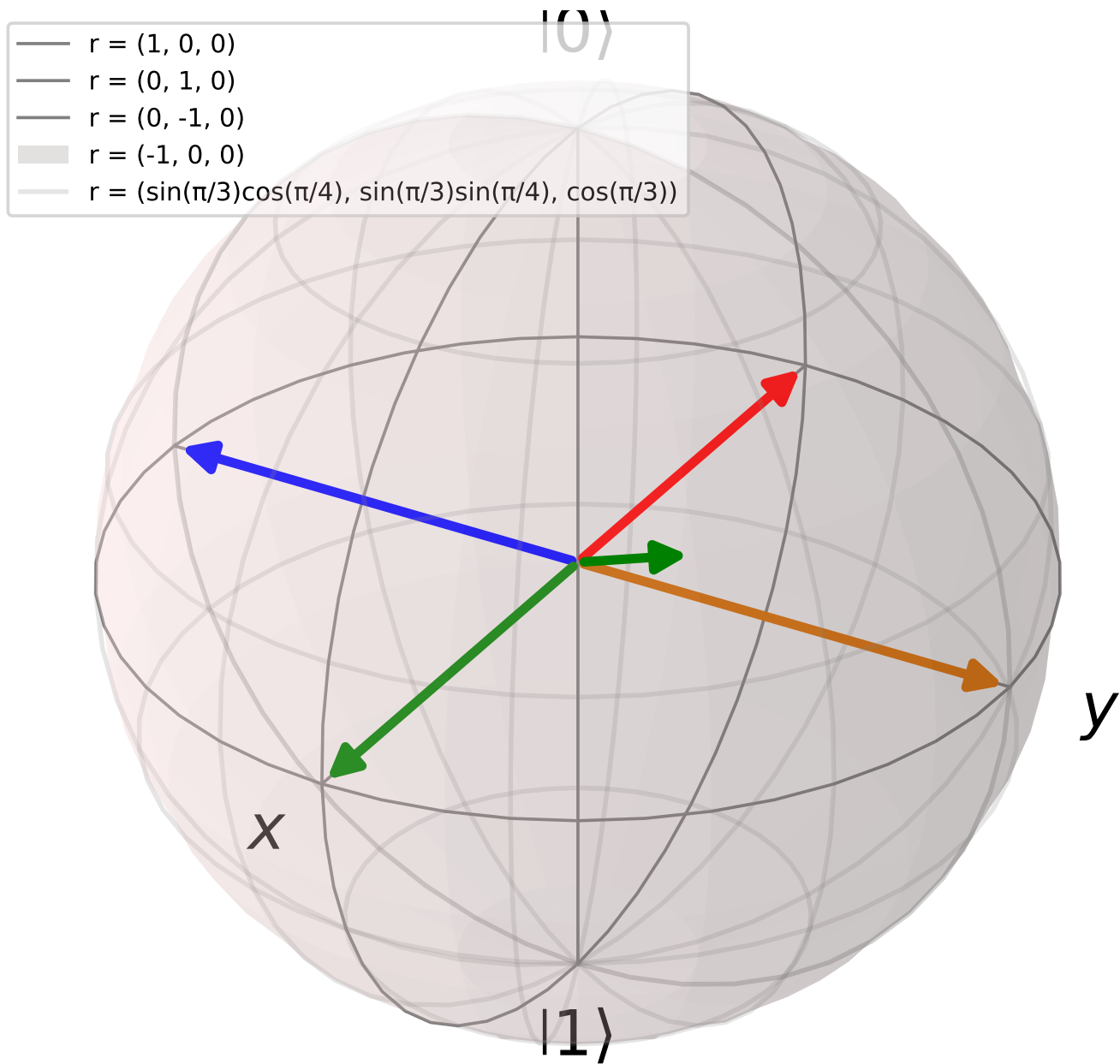


Figure 19: Bloch vectors ($Q9(a,b,c,d,e)$)

Part (d)

Given $(\theta, \phi) = (\pi/2, \pi)$, find the density matrix.

Solution:

- **Bloch Vector:**

$$\mathbf{r} = (\sin(\pi/2) \cos(\pi), \sin(\pi/2) \sin(\pi), \cos(\pi/2)) = (-1, 0, 0)$$

- **Density Matrix:**

$$\rho = \frac{1}{2} (\mathbb{I} - \sigma_x) = \frac{1}{2} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$$

This represents the state $\frac{1}{\sqrt{2}}(|g\rangle - |e\rangle)$.

Answer:

$$\rho = \frac{1}{2} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$$

Part (e)

Given $(\theta, \phi) = (\pi/3, \pi/4)$, find the density matrix.

Solution:

- **Bloch Vector:**

$$\mathbf{r} = (\sin(\pi/3) \cos(\pi/4), \sin(\pi/3) \sin(\pi/4), \cos(\pi/3)) = \left(\frac{\sqrt{3}}{2\sqrt{2}}, \frac{\sqrt{3}}{2\sqrt{2}}, \frac{1}{2} \right)$$

- **Density Matrix:**

$$\rho = \frac{1}{2} \left(\mathbb{I} + \frac{\sqrt{3}}{2\sqrt{2}} \sigma_x + \frac{\sqrt{3}}{2\sqrt{2}} \sigma_y + \frac{1}{2} \sigma_z \right) = \frac{1}{2} \begin{pmatrix} 1 + \frac{1}{2} & \frac{\sqrt{3}}{2\sqrt{2}}(1 - i) \\ \frac{\sqrt{3}}{2\sqrt{2}}(1 + i) & 1 - \frac{1}{2} \end{pmatrix}$$

Simplifying:

$$\rho = \begin{pmatrix} \frac{3}{4} & \frac{\sqrt{6}}{4} e^{-i\pi/4} \\ \frac{\sqrt{6}}{4} e^{i\pi/4} & \frac{1}{4} \end{pmatrix}$$

Answer:

$$\rho = \begin{pmatrix} \frac{3}{4} & \frac{\sqrt{6}}{4} e^{-i\pi/4} \\ \frac{\sqrt{6}}{4} e^{i\pi/4} & \frac{1}{4} \end{pmatrix}$$

Key Observations:

- **Part (d):** The state is an antisymmetric superposition $(|g\rangle - |e\rangle)$ with purely real coherences.
- **Part (e):** The state has unequal populations and complex coherences, reflecting both polar and azimuthal angles.

Question 10: Off-Resonant Rabi Oscillations

Part (a)

Find the generalized Rabi frequency for an atom with detuning $\delta = \omega_0 - \omega$.

Solution:

The Hamiltonian in the rotating frame is:

$$H = \frac{\hbar}{2} \begin{pmatrix} -\delta & \Omega \\ \Omega & \delta \end{pmatrix}$$

where:

- $\delta = \omega_0 - \omega$ is the detuning
- $\Omega = \frac{dE_0}{\hbar}$ is the on-resonance Rabi frequency

The generalized Rabi frequency $\tilde{\Omega}$ is the eigenvalue difference:

$$\tilde{\Omega} = \sqrt{\delta^2 + \Omega^2}$$

Answer:

The generalized Rabi frequency is:

$$\tilde{\Omega} = \sqrt{\delta^2 + \Omega^2}$$

Part (b)

Express the dynamics on the Bloch sphere, identifying the rotation axis and Bloch vector magnitude.

Solution:

The system evolves under the unitary operator:

$$U(t) = e^{-iHt/\hbar} = \cos\left(\frac{\tilde{\Omega}t}{2}\right) \mathbb{I} - i \sin\left(\frac{\tilde{\Omega}t}{2}\right) \frac{H}{\hbar\tilde{\Omega}/2}$$

- **Rotation Axis:** The Hamiltonian defines the rotation axis $\mathbf{n}$:

$$\mathbf{n} = \frac{(\Omega, 0, \delta)}{\tilde{\Omega}}$$

(Tilted in the xz -plane by angle $\theta = \arctan(\Omega/\delta)$ from the z -axis)

- **Bloch Vector Dynamics:** For initial state $|g\rangle$:

$$r_x(t) = \frac{\Omega\delta}{\tilde{\Omega}^2}(1 - \cos\tilde{\Omega}t), \quad r_y(t) = -\frac{\Omega}{\tilde{\Omega}}\sin\tilde{\Omega}t, \quad r_z(t) = \frac{\delta^2}{\tilde{\Omega}^2} + \frac{\Omega^2}{\tilde{\Omega}^2}\cos\tilde{\Omega}t$$

The magnitude $|\mathbf{r}| = 1$ is preserved (pure state).

Answer:

- Rotation axis: $\mathbf{n} = \frac{(\Omega, 0, \delta)}{\sqrt{\delta^2 + \Omega^2}}$ (tilted in xz -plane)

- Bloch vector precesses about $\mathbf{n}$ at frequency $\tilde{\Omega} = \sqrt{\delta^2 + \Omega^2}$